

Thirty characters are not enou

CPC Reference Notebook

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1 Template and Stress

Template.cpp

Use: Starter file: put solution code inside SOLVE(), keep main(), and add snippets above SOLVE() as needed.

Tip: Use this as the wrapper for copied snippets; most snippets in the notebook have duplicate includes/main removed.

Before: Keep common includes, typedefs, and fast I/O here so snippets can stay short.

Pitfall: If a copied snippet needs globals, place them above SOLVE() and reset them for each test case.

```
#include <bits/stdc++.h>
#include <cassert>
#pragma GCC optimize("Ofast,unroll-loops")
#ifdef ONLINE_JUDGE
    #define debug(...) ((void)0)
#else
    #include "Template\debug.h"
#endif
using namespace std;
using ll = long long;
#define endl '\n'

void SOLVE(){

}

signed main(){
    ios_base::sync_with_stdio(false); cin.tie(nullptr);
    int o_o; cin >> o_o; while(o_o--){
        SOLVE(); return 0;
    }
}
```

stress.bat

Use: Put brute.cpp, fast.cpp, and generator.cpp beside it, then run stress.bat to repeatedly compare outputs.

Tip: When it reports a mismatch, rerun with the printed seed/input first, then shrink the generated constraints.

Before: Compile all three C++ files first, or keep the script compile commands in sync with your local flags.

Pitfall: A mismatch can come from nondeterministic output order, so sort outputs when any valid order is accepted.

```
@echo off
setlocal EnableDelayedExpansion

if not exist bin mkdir bin

g++ -std=c++17 -O2 generator.cpp -o bin\gen.exe || exit /b
g++ -std=c++17 -O2 brute.cpp -o bin\brute.exe || exit /b
g++ -std=c++17 -O2 fast.cpp -o bin\fast.exe || exit /b

set /a I=0
set LIM=%1
if "%LIM%"==" " set LIM=0

:loop
set /a I+=1
echo Test !I!

bin\gen.exe !I! > input.txt
bin\brute.exe < input.txt > out_brute.txt
bin\fast.exe < input.txt > out_fast.txt
```

```
fc out_brute.txt out_fast.txt > nul || (
    echo MISMATCH ON SEED !I!
    echo ==== INPUT ==== & type input.txt
    echo ==== BRUTE ==== & type out_brute.txt
    echo ==== FAST ==== & type out_fast.txt
    pause
    exit /b
)

if not "%LIM%"=="0" if !I! GEQ %LIM% exit /b
goto loop
```

brute.cpp

Use: Write the slow trusted solution in solve(); stress.bat feeds it generated tests and compares against fast.cpp.

Care: Keep this solution obvious and trusted; slow exhaustive logic is fine because it is only an oracle for small tests.

In: Use the same input format as fast.cpp, even when constraints are intentionally tiny.

Pitfall: Do not optimize the brute until it becomes hard to trust; clarity matters more than speed here.

```
#include <bits/stdc++.h>
using namespace std;

using i64 = long long;

void solve() {
    int n;
    cin >> n;
    vector<i64> a(n);
    for (i64 &x : a) cin >> x;

    i64 best = LLONG_MIN;
    for (int l = 0; l < n; l++) {
        i64 sum = 0;
        for (int r = l; r < n; r++) {
            sum += a[r];
            best = max(best, sum);
        }
    }
    cout << best << '\n';
}

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    int t;
    cin >> t;
    while (t--) solve();
}
```

fast.cpp

Use: Write the optimized solution in solve(); stress.bat checks its output against brute.cpp.

Care: Match brute.cpp output exactly, including spaces and newlines, so the diff points to algorithm bugs only.

In: Keep the generator limits visible, then paste the contest solution logic inside solve().

Pitfall: Undefined behavior may pass samples but usually appears quickly under repeated random stress.

```
#include <bits/stdc++.h>
```

```

using namespace std;

using i64 = long long;

void solve() {
    int n;
    cin >> n;
    vector<i64> a(n);
    for (i64 &x : a) cin >> x;

    i64 best = a[0], current = a[0];
    for (int i = 1; i < n; i++) {
        current = max(a[i], current + a[i]);
        best = max(best, current);
    }
    cout << best << '\n';
}

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    int t;
    cin >> t;
    while (t--) solve();
}

```

generator.cpp

Use: Edit main()/helpers to print one random test; optional argv seed lets stress.bat reproduce failing tests.

Tip: Mix random cases with edge cases such as minimum sizes, duplicates, sorted data, and dense/sparse graphs.

Out: Print exactly one complete test file, including the test count if the solution expects one.

Pitfall: Use the argv seed while debugging so a failing random case can be reproduced exactly.

```

#include <bits/stdc++.h>
using namespace std;

using ll = long long;

mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());

ll rnd(ll l, ll r) {
    return uniform_int_distribution<ll>(l, r)(rng);
}

int rnd(int l, int r) {
    return uniform_int_distribution<int>(l, r)(rng);
}

bool coin(int p = 50) {
    return rnd(1, 100) <= p;
}

template <class T>
void print(vector<T> a) {
    for (int i = 0; i < (int)a.size(); i++) cout << a[i] << " \n"[i + 1 == (int)a.size()];
}

vector<ll> arr(int n, ll l, ll r) {
    vector<ll> a(n);
    for (auto &x : a) x = rnd(l, r);
}

```

```

    return a;
}

vector<int> perm(int n) {
    vector<int> p(n);
    iota(p.begin(), p.end(), 1);
    shuffle(p.begin(), p.end(), rng);
    return p;
}

string str(int n, char l = 'a', char r = 'z') {
    string s;
    while (n--) s += char(rnd(l, r));
    return s;
}

vector<string> grid(int n, int m, string chars = ".#") {
    vector<string> g(n, string(m, '.'));
    for (auto &row : g) {
        for (char &c : row) c = chars[rnd(0, (int)chars.size() - 1)];
    }
    return g;
}

vector<pair<int, int>> tree(int n) {
    vector<pair<int, int>> e;
    for (int i = 2; i <= n; i++) e.push_back({rnd(1, i - 1), i});
    shuffle(e.begin(), e.end(), rng);
    return e;
}

vector<pair<int, int>> path_tree(int n) {
    vector<pair<int, int>> e;
    for (int i = 2; i <= n; i++) e.push_back({i - 1, i});
    return e;
}

vector<pair<int, int>> star_tree(int n) {
    vector<pair<int, int>> e;
    int c = rnd(1, n);
    for (int i = 1; i <= n; i++) {
        if (i != c) e.push_back({c, i});
    }
    shuffle(e.begin(), e.end(), rng);
    return e;
}

vector<pair<int, int>> graph(int n, int m) {
    vector<pair<int, int>> all, e;
    for (int i = 1; i <= n; i++) {
        for (int j = i + 1; j <= n; j++) all.push_back({i, j});
    }
    shuffle(all.begin(), all.end(), rng);
    m = min(m, (int)all.size());
    for (int i = 0; i < m; i++) e.push_back(all[i]);
    return e;
}

vector<pair<int, int>> digraph(int n, int m) {
}

```

```

vector<pair<int, int>> all, e;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        if (i != j) all.push_back({i, j});
    }
}
shuffle(all.begin(), all.end(), rng);
m = min(m, (int)all.size());
for (int i = 0; i < m; i++) e.push_back(all[i]);
return e;
}

vector<pair<int, int>> dag(int n, int m) {
    vector<int> p = perm(n);
    vector<pair<int, int>> all, e;
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) all.push_back({p[i], p[j]});
    }
    shuffle(all.begin(), all.end(), rng);
    m = min(m, (int)all.size());
    for (int i = 0; i < m; i++) e.push_back(all[i]);
    return e;
}

vector<pair<int, int>> connected_graph(int n, int m) {
    vector<pair<int, int>> e = tree(n);
    set<pair<int, int>> used;
    for (auto [u, v] : e) {
        if (u > v) swap(u, v);
        used.insert({u, v});
    }
    m = min(m, n * (n - 1) / 2);
    while ((int)e.size() < m) {
        int u = rnd(1, n), v = rnd(1, n);
        if (u == v) continue;
        if (u > v) swap(u, v);
        if (used.insert({u, v}).second) e.push_back({u, v});
    }
    shuffle(e.begin(), e.end(), rng);
    return e;
}

vector<pair<int, int>> queries(int q, int n) {
    vector<pair<int, int>> res;
    while (q--) {
        int l = rnd(1, n), r = rnd(1, n);
        if (l > r) swap(l, r);
        res.push_back({l, r});
    }
    return res;
}

void print_edges(vector<pair<int, int>> e) {
    for (auto [u, v] : e) cout << u << ' ' << v << '\n';
}

void print_weighted_edges(vector<pair<int, int>> e, int l, int r) {
    for (auto [u, v] : e) cout << u << ' ' << v << ' ' << rnd(l, r) << '\n';
}

```

```

void solve() {
    int n = rnd(1, 20);
    cout << n << '\n';
    print(arr(n, -20, 20));
}

int main(int argc, char **argv) {
    ios::sync_with_stdio(0);
    cin.tie(0);
    if (argc > 1) rng.seed(stoull(argv[1]));
    int t = 1; // TESTCASES
    if (argc > 2) t = stoi(argv[2]);
    cout << t << '\n';
    while (t--) solve();
}

```

2 Data structures

2D sparse table.cpp

Use: SparseTable2D<T> st(a, merge); query(r1,c1,r2,c2) returns the idempotent aggregate on an inclusive 0-indexed rectangle.

Care: The merge function must be idempotent, such as min, max, or gcd; do not use it for sum without changing the query logic.

Complexity: Build and memory are $O(n*m*\log n*\log m)$, query is $O(1)$, and the matrix must stay static.

Pitfall: Input rows must all have the same length; coordinates are inclusive and may be swapped if passed reversed.

```

template<class T>
struct SparseTable2D{
    int n, m, LOGN, LOGM;
    vector<int> lgN, lgM;
    vector<vector<vector<T>>> table;
    function<T(const T&, const T&> f;
    SparseTable2D(){
        SparseTable2D(const vector<vector<T>> &a, function<T(const T&, const T&> func){
            init(a, func);
        }
        int rows(int kx) const{
            return n - (1 << kx) + 1;
        }
        int cols(int ky) const{
            return m - (1 << ky) + 1;
        }
        int id(int ky, int i, int j) const{
            return i * cols(ky) + j;
        }
        T& at(int kx, int ky, int i, int j){
            return table[kx][ky][id(ky, i, j)];
        }
        const T& at(int kx, int ky, int i, int j) const{
            return table[kx][ky][id(ky, i, j)];
        }
        void init(const vector<vector<T>> &a, function<T(const T&, const T&> func){
            n = a.size();
            m = n ? a[0].size() : 0;
            assert(n > 0 && m > 0);
            for (int i = 0; i < n; i++){
                assert((int)a[i].size() == m);
            }
            f = func;
            lgN.assign(n + 1, 0);
            lgM.assign(m + 1, 0);
            for (int i = 2; i <= n; i++){
                lgN[i] = lgN[i >> 1] + 1;
            }
            for (int i = 2; i <= m; i++){
                lgM[i] = lgM[i >> 1] + 1;
            }
            LOGN = lgN[n] + 1;
            LOGM = lgM[m] + 1;
            table.assign(LOGN, vector<vector<T>>(LOGM));
            table[0][0].resize(n * m);
            for (int i = 0; i < n; i++){
                for (int j = 0; j < m; j++){
                    at(0, 0, i, j) = a[i][j];
                }
            }
        }
    };
};

```

```

    }
    for (int kx = 0; kx < LOGN; kx++){
        for (int ky = 0; ky < LOGM; ky++){
            if (kx == 0 && ky == 0) continue;
            int R = rows(kx), C = cols(ky);
            table[kx][ky].resize(R * C);
            if (kx == 0){
                for (int i = 0; i < R; i++){
                    for (int j = 0; j < C; j++){
                        at(kx, ky, i, j) = f(at(kx, ky - 1, i, j), at(kx, ky - 1, i,
                            ↪ j + (1 << (ky - 1))));
                    }
                }
            }
            }else{
                for (int i = 0; i < R; i++){
                    for (int j = 0; j < C; j++){
                        at(kx, ky, i, j) = f(at(kx - 1, ky, i, j), at(kx - 1, ky, i +
                            ↪ (1 << (kx - 1)), j));
                    }
                }
            }
        }
    }
}

T query(int r1, int c1, int r2, int c2) const{
    if (r1 > r2) swap(r1, r2);
    if (c1 > c2) swap(c1, c2);
    assert(0 <= r1 && r2 < n && 0 <= c1 && c2 < m);
    int kx = lgN[r2 - r1 + 1];
    int ky = lgM[c2 - c1 + 1];
    int nr = r2 - (1 << kx) + 1;
    int nc = c2 - (1 << ky) + 1;
    T a = f(at(kx, ky, r1, c1), at(kx, ky, nr, c1));
    T b = f(at(kx, ky, r1, nc), at(kx, ky, nr, nc));
    return f(a, b);
}
};

```

bin_Trie.cpp

Use: Trie $t(\text{bits})$; call $\text{insert}(x)$, $\text{del}(x)$, $\text{cnt}(x)$, $\text{query}(x, k)$, $\text{mnXor}(x)$, or $\text{mxXor}(x)$ for nonnegative integers with that bit width.

Care: Values are stored with multiplicity; delete only inserted values and choose bits high enough for the maximum number.

Complexity: insert, del, cnt, and xor queries are $O(\text{bits})$, independent of how many numbers are stored.

Pitfall: `query(x,k)` counts stored `y` with $(x \text{ xor } y) < k$, so confirm strict inequality before using it.

```
struct Trie{
    struct Node{
        int child[2], freq;
        Node(){
            child[0] = child[1] = -1;
            freq = 0;
        }
    };
    vector<Node> trie;
    int bits;
    Trie(int x){
```

```

    bits = x;
    trie.emplace_back();
}
int newNode(){
    trie.emplace_back();
    return (int)trie.size() - 1;
}
void insert(ll val){
    int cur = 0;
    trie[cur].freq++;
    for (int i = bits; i >= 0; i--){
        bool bit = (val >> i) & 1;
        if (trie[cur].child[bit] == -1){
            trie[cur].child[bit] = newNode();
        }
        cur = trie[cur].child[bit];
        trie[cur].freq++;
    }
}
void del(ll val){
    if (!cnt(val)) return;
    int cur = 0;
    trie[cur].freq--;
    for (int i = bits; i >= 0; i--){
        bool bit = (val >> i) & 1;
        cur = trie[cur].child[bit];
        trie[cur].freq--;
    }
}
int cnt(ll val){
    int cur = 0;
    for (int i = bits; i >= 0; i--){
        bool bit = (val >> i) & 1;
        cur = trie[cur].child[bit];
        if (cur == -1) return 0;
    }
    return trie[cur].freq;
}
ll query(ll val, ll k){ // num ^ val < k
    int cur = 0;
    ll answer = 0;
    for (int i = bits; i >= 0 && cur != -1; i--){
        bool b1 = (val >> i) & 1;
        bool b2 = (k >> i) & 1;
        if (b2){
            int same = trie[cur].child[b1];
            if (same != -1) answer += trie[same].freq;
            cur = trie[cur].child[b1 ^ 1];
        }else{
            cur = trie[cur].child[b1];
        }
    }
    return answer;
}
ll mxXor(ll val){
    int cur = 0;
    ll ret = 0;
    for (int i = bits; i >= 0 && cur != -1; i--){
        bool bit = (val >> i) & 1;

```

```

        int nxt = trie[cur].child[bit ^ 1];
        if (nxt != -1 && trie[nxt].freq){
            ret |= (1LL << i);
            cur = nxt;
        }else{
            cur = trie[cur].child[bit];
        }
    }
    return ret;
}
ll mnXor(ll val){
    int cur = 0;
    ll ret = 0;
    for (int i = bits; i >= 0 && cur != -1; i--){
        bool bit = (val >> i) & 1;
        int nxt = trie[cur].child[bit];
        if (nxt != -1 && trie[nxt].freq){
            cur = nxt;
        }else{
            ret |= (1LL << i);
            cur = trie[cur].child[bit ^ 1];
        }
    }
    return ret;
}
};

```

Cartesian Tree.cpp

Use: Build MinCartesianTree<T> or MaxCartesianTree<T> ct(a); query_index/query_value answer RMQ and get_tree(directed) returns adjacency with root.

Out: Queries are 0-indexed inclusive; L[v], R[v] give the subtree interval, and tie behavior follows the better() comparator.

Complexity: Construction is O(n), build_info adds binary lifting, and range queries are O(log n).

When: Use it when RMQ index/value queries are static and you also want the Cartesian tree structure.

```

template<class T, bool is_min = true>
struct CartesianTree{
    int n, root, LOG;
    vector<T> a;
    vector<int> parent, left, right, L, R, depth;
    vector<vector<int>>> up;
    CartesianTree(){}
    CartesianTree(const vector<T> &v){ init(v); }
    bool better(int i, int j) const{
        if (a[i] != a[j]) return is_min ? a[i] < a[j] : a[i] > a[j];
        return is_min ? i < j : i > j; // pair {a[i], i}
    }
    void init(const vector<T> &v){
        a = v;
        n = a.size();
        root = -1;
        parent.assign(n, -1);
        left.assign(n, -1); right.assign(n, -1);
        L.resize(n); R.resize(n);
        depth.assign(n, 0);
        vector<int> st;
        for (int i = 0; i < n; i++){
            int last = -1;

```

```

while (!st.empty() && better(i, st.back())){
    last = st.back();
    st.pop_back();
}
if (!st.empty()){
    right[st.back()] = i;
    parent[i] = st.back();
}
if (last != -1){
    left[i] = last;
    parent[last] = i;
}
st.push_back(i);
}
for (int i = 0; i < n; i++){
    if (parent[i] == -1) root = i;
}
build_info();
}
void build_info(){
    if (!n) return;
    LOG = __lg(n) + 1;
    up.assign(LOG, vector<int>(n, root));
    vector<int> order = {root};
    up[0][root] = root;
    for (int ptr = 0; ptr < (int)order.size(); ptr++){
        int v = order[ptr];
        for (int i = 1; i < LOG; i++) up[i][v] = up[i - 1][up[i - 1][v]];
        for (int u : {left[v], right[v]}){
            if (u == -1) continue;
            depth[u] = depth[v] + 1;
            up[0][u] = v;
            order.push_back(u);
        }
    }
    reverse(order.begin(), order.end());
    for (auto v : order){
        L[v] = R[v] = v;
        if (left[v] != -1) L[v] = L[left[v]];
        if (right[v] != -1) R[v] = R[right[v]];
    }
}
int lca(int u, int v) const{
    if (depth[u] < depth[v]) swap(u, v);
    int diff = depth[u] - depth[v];
    for (int i = 0; i < LOG; i++){
        if ((diff >> i) & 1) u = up[i][u];
    }
    if (u == v) return u;
    for (int i = LOG - 1; i >= 0; i--){
        if (up[i][u] != up[i][v]){
            u = up[i][u];
            v = up[i][v];
        }
    }
    return up[0][u];
}
int query_index(int l, int r) const{
    if (l > r) swap(l, r);

```

```

return lca(l, r);
}
T query_value(int l, int r) const{
    return a[query_index(l, r)];
}
pair<vector<vector<int>>, int> get_tree(bool directed = true) const {
    vector<vector<int>> adj(n);
    for (int i = 0; i < n; i++) {
        if (left[i] != -1) {
            adj[i].push_back(left[i]);
            if (!directed) adj[left[i]].push_back(i);
        }
        if (right[i] != -1) {
            adj[i].push_back(right[i]);
            if (!directed) adj[right[i]].push_back(i);
        }
    }
    return {adj, root};
}
};
template<class T>
using MinCartesianTree = CartesianTree<T, true>;
template<class T>
using MaxCartesianTree = CartesianTree<T, false>;

```

CHT.cpp

Use: CHT<true> for min or CHT<false> for max; add(k,m) inserts $y=kx+m$ and query(x) returns the best value.

Care: This is a dynamic integer LineContainer; queries may be arbitrary, but slopes/intersections use floored $\parallel$ division.

Complexity: Each add/query is $O(\log n)$; it handles arbitrary insertion order and arbitrary x queries.

Pitfall: For minimum queries the template negates lines internally, so do not negate answers yourself.

```

struct Line {
    mutable ll k, m, p;
    bool operator<(const Line &o) const { return k < o.k; }
    bool operator<(ll x) const { return p < x; }
};

// CHT<true> for min, CHT<false> for max. add(k, m) adds  $y = kx + m$ 
template<bool is_min = false> struct CHT : multiset<Line, less<>> {
    // (for doubles, use inf = 1/.0, div(a,b) = a/b)
    static const ll inf = LLONG_MAX;
    ll div(ll a, ll b) { // floored division
        return a / b - ((a ^ b) < 0 && a % b);
    }
    bool isect(iterator x, iterator y) {
        if (y == end())
            return x->p = inf, 0;
        if (x->k == y->k)
            x->p = x->m > y->m ? inf : -inf;
        else
            x->p = div(y->m - x->m, x->k - y->k);
        return x->p >= y->p;
    }
    void add(ll k, ll m) {
        if constexpr (is_min) {
            k = -k;

```

```

    m = -m;
}
auto z = insert({k, m, 0}), y = z++, x = y;
while (isect(y, z))
    z = erase(z);
if (x != begin() && isect(--x, y))
    isect(x, y = erase(y));
while ((y = x) != begin() && (--x)->p >= y->p)
    isect(x, erase(y));
}
ll query(ll x) {
    if (empty())
        return is_min ? LLONG_MAX : LLONG_MIN;
    auto l = *lower_bound(x);
    ll res = l.k * x + l.m;
    if constexpr (is_min) {
        return -res;
    }
    return res;
}
};

```

DSU rollbacks.cpp

Use: DSU d(n); save int t = d.time(), call unite(a,b), then rollback(t) to undo to that state.

Care: Rollback DSU intentionally has no path compression, so find() stays undoable but costs logarithmic-ish recursion.

Complexity: unite and rollback are undo-friendly without path compression; rollback is proportional to undone changes.

When: Use it for offline dynamic connectivity, divide-and-conquer on time, or backtracking over graph states.

```

struct DSU{
    int n, components;
    vector<int> parent, sz;
    vector<pair<int&, int>> history;
    DSU(){
        DSU(int n){
            init(n);
        }
        void init(int _n){
            n = _n;
            components = n;
            parent.resize(n + 1);
            sz.assign(n + 1, 1);
            history.clear();
            for (int i = 1; i <= n; i++){
                parent[i] = i;
            }
        }
        int find(int v){
            return parent[v] == v ? v : find(parent[v]);
        }
        bool same(int u, int v){
            return find(u) == find(v);
        }
        bool unite(int a, int b){
            a = find(a);
            b = find(b);
            if (a == b) return false;

```

```

            if (sz[a] < sz[b]) swap(a, b);
            history.push_back({sz[a], sz[a]});
            history.push_back({parent[b], parent[b]});
            history.push_back({components, components});
            parent[b] = a;
            sz[a] += sz[b];
            components--;
            return true;
        }
        bool union_sets(int a, int b){
            return unite(a, b);
        }
        int time(){
            return (int)history.size();
        }
        int get_size(int v){
            return sz[find(v)];
        }
        void rollback(int until){
            while ((int)history.size() > until){
                history.back().first = history.back().second;
                history.pop_back();
            }
        }
    };

```

DSU.cpp

Use: DSU d(n); use unite(a,b), same(a,b), get_size(v), and the components variable on 1-indexed nodes.

Care: Nodes are 1-indexed and components is a public variable, not a function call.

Complexity: find/unite are effectively $O(\alpha n)$ with path compression and union by size.

Pitfall: The structure is 1-indexed, so allocate with n and avoid node 0 unless you adapt init.

```

struct DSU{
    int n;
    vector<int> parent, size;
    int components;
    DSU(int n) : n(n), parent(n + 1), size(n + 1, 1), components(n) {
        iota(parent.begin(), parent.end(), 0);
    }
    int find(int v){
        return parent[v] == (parent[v] == v ? v : find(parent[v]));
    }
    bool same(int u, int v) {
        return find(u) == find(v);
    }
    bool union_sets(int a, int b){
        a = find(a);
        b = find(b);
        if (a == b) return false;
        if (size[a] < size[b]) swap(a, b);
        parent[b] = a;
        size[a] += size[b];
        components--;
        return true;
    }
    int get_size(int v){
        return size[find(v)];
    }
}

```

```
};
```

Fenwick.cpp

Use: Fenwick<T> fw(n, ID, op); add(i, val), query(i), query(l, r), and lower_bound(sum) use 1-indexed positions.

Care: Range query uses subtraction, so op should behave like sum/xor with an inverse if you use query(l, r).

Complexity: add, prefix query, range query, and lower_bound are $O(\log n)$.

Pitfall: lower_bound assumes prefix values are monotone increasing, usually nonnegative point updates.

```
template <class T>
struct Fenwick { // 1-indexed
    int n, LOG; T ID;
    function<T(T, T)> op;
    vector<T> tree;
    Fenwick(int _n, T _ID, function<T(T, T)> _op) : n(_n + 5), ID(_ID), op(_op),
        LOG(log2(n) + 1) {
        tree.assign(n + 1, ID);
    }
    void add(int i, T val) {
        if (i <= 0) return;
        for (; i <= n; i += (i & -i)) tree[i] = op(tree[i], val);
    }
    T query(int i) {
        T res = ID;
        for (; i >= 1; i -= (i & -i)) res = op(res, tree[i]);
        return res;
    }
    T query(int l, int r) {
        return query(r) - query(l - 1);
        // return query(r) ^ query(l - 1);
    }
    int lower_bound(ll val) {
        T sum = 0;
        int pos = 0;
        for(int i = LOG; i >= 0; i--) {
            if(pos + (1 << i) <= n && sum + tree[pos + (1 << i)] < val) {
                pos += (1 << i);
                sum += tree[pos];
            }
        }
        return pos + 1;
    }
};
```

lichao segment with rollbacks.cpp

Use: LiChaoSegment<is_min> lc(L, R); add_line(a, b), add_segment(a, b, l, r), query(x), time(), and rollback(t).

Care: The domain is the integer interval [L, R]; use add_segment only when a line is valid on a subrange.

Complexity: Line insertion and query are $O(\log \text{coordinate_range})$; segment insertion adds another log factor.

When: Use the segment version when each line is valid only on [ql, qr], not on the whole domain.

```
template<bool is_min = true>
struct LiChaoSegment {
    const ll INF = 4e18;
    struct Line {
        ll a, b;
        Line(ll _a = 0, ll _b = 4e18) : a(_a), b(_b) {}
    };
};
```

```
vector<Line> tree;
vector<pair<int, Line>> history;
ll L, R;
LiChaoSegment() {}
LiChaoSegment(ll _L, ll _R) {
    init(_L, _R);
}
void init(ll _L, ll _R) {
    L = _L;
    R = _R;
    tree.assign(4, Line());
    history.clear();
}
ll eval(Line line, ll x) {
    __int128 ret = (__int128)line.a * x + line.b;
    if (ret > INF) return INF;
    if (ret < -INF) return -INF;
    return (ll)ret;
}
bool better(Line x, Line y, ll at) {
    return eval(x, at) < eval(y, at);
}
void save(int node) {
    if (node >= (int)tree.size()) tree.resize(node + 1, Line());
    history.push_back({node, tree[node]});
}
void add_line(ll a, ll b) {
    if constexpr (!is_min) a = -a, b = -b;
    add_line(Line(a, b), 1, L, R);
}
void add_line(Line cur, int node, ll l, ll r) {
    if (node >= (int)tree.size()) tree.resize(node + 1, Line());
    ll mid = l + (r - l) / 2;
    bool lef = better(cur, tree[node], l);
    bool m = better(cur, tree[node], mid);
    if (m) {
        save(node);
        swap(cur, tree[node]);
    }
    if (l == r) return;
    if (lef != m) add_line(cur, node * 2, l, mid);
    else add_line(cur, node * 2 + 1, mid + 1, r);
}
void add_segment(ll a, ll b, ll ql, ll qr) {
    if constexpr (!is_min) a = -a, b = -b;
    add_segment(Line(a, b), ql, qr, 1, L, R);
}
void add_segment(Line line, ll ql, ll qr, int node, ll l, ll r) {
    if (qr < l || r < ql) return;
    if (ql <= l && r <= qr) {
        add_line(line, node, l, r);
        return;
    }
    if (node >= (int)tree.size()) tree.resize(node + 1, Line());
    ll mid = l + (r - l) / 2;
    add_segment(line, ql, qr, node * 2, l, mid);
    add_segment(line, ql, qr, node * 2 + 1, mid + 1, r);
}
ll query(ll x) {
```

```

    ll ret = query(x, 1, L, R);
    if constexpr (!is_min) ret = -ret;
    return ret;
}
ll query(ll x, int node, ll l, ll r){
    if (node >= (int)tree.size()) return INF;
    ll ret = eval(tree[node], x);
    if (l == r) return ret;
    ll mid = l + (r - l) / 2;
    if (x <= mid) ret = min(ret, query(x, node * 2, l, mid));
    else ret = min(ret, query(x, node * 2 + 1, mid + 1, r));
    return ret;
}
int time(){
    return (int)history.size();
}
void rollback(int t){
    while ((int)history.size() > t){
        auto [node, line] = history.back();
        history.pop_back();
        tree[node] = line;
    }
}
};

```

lichao with rollbacks.cpp

Use: LiChao<is_min> lc(L,R); add_line(a,b) inserts $y=ax+b$, query(x) asks best value, and rollback(snapshot()) undoes.

Care: Use snapshot() before tentative insertions; rollback(t) restores line nodes but not any variables outside the structure.

Complexity: add_line and query are $O(\log \text{coordinate_range})$; rollback cost is proportional to saved node changes.

Pitfall: Pick L and R to cover every x query exactly; outside-domain queries are not protected.

```

template<bool is_min = true>
struct LiChao{
    const ll INF = 4e18;
    struct Line{
        ll a, b;
        Line(ll _a = 0, ll _b = 4e18) : a(_a), b(_b){}
    };
    vector<Line> tree;
    vector<pair<int, Line>> history;
    ll L, R;
    LiChao(){
        LiChao(ll _L, ll _R){
            init(_L, _R);
        }
    }
    void init(ll _L, ll _R){
        L = _L;
        R = _R;
        tree.assign(4, Line());
        history.clear();
    }
    ll eval(Line line, ll x){
        __int128 ret = (__int128)line.a * x + line.b;
        if (ret > INF) return INF;
        if (ret < -INF) return -INF;
    }
};

```

```

    return (ll)ret;
}
bool better(Line x, Line y, ll at){
    return eval(x, at) < eval(y, at);
}
void save(int node){
    if (node >= (int)tree.size()) tree.resize(node + 1, Line());
    history.push_back({node, tree[node]});
}
void add_line(ll a, ll b){
    if constexpr (!is_min) a = -a, b = -b;
    add_line(Line(a, b), 1, L, R);
}
void add_line(Line cur, int node, ll l, ll r){
    if (node >= (int)tree.size()) tree.resize(node + 1, Line());
    ll mid = l + (r - l) / 2;
    bool lef = better(cur, tree[node], l);
    bool m = better(cur, tree[node], mid);
    if (m){
        save(node);
        swap(cur, tree[node]);
    }
    if (l == r) return;
    if (lef != m) add_line(cur, node * 2, l, mid);
    else add_line(cur, node * 2 + 1, mid + 1, r);
}
ll query(ll x){
    ll ret = query(x, 1, L, R);
    if constexpr (!is_min) ret = -ret;
    return ret;
}
ll query(ll x, int node, ll l, ll r){
    if (node >= (int)tree.size()) return INF;
    ll ret = eval(tree[node], x);
    if (l == r) return ret;
    ll mid = l + (r - l) / 2;
    if (x <= mid) ret = min(ret, query(x, node * 2, l, mid));
    else ret = min(ret, query(x, node * 2 + 1, mid + 1, r));
    return ret;
}
int snapshot(){
    return (int)history.size();
}
void rollback(int t){
    while ((int)history.size() > t){
        auto [node, line] = history.back();
        history.pop_back();
        tree[node] = line;
    }
}
};

```

merge sort tree.cpp

Use: merge_sort_tree mt(a); count_less/count_leq/count_greater(l,r,x) and sum_less/sum_leq use 0-indexed inclusive ranges.

Out: Counts and sums are for values compared to x on $[l,r]$; memory is $O(n \log n)$, so reserve it for static arrays.

Complexity: Build is $O(n \log n)$; each count/sum query is $O(\log^2 n)$.

Pitfall: The array is static; point updates require rebuilding or a different structure.

```

struct merge_sort_tree{
    int n;
    vector<vector<ll>> tree, pref;
    merge_sort_tree(){}
    merge_sort_tree(vector<ll> &a){
        init(a);
    }
    void init(vector<ll> &a){
        n = 1;
        while (n < (int)a.size()) n <= 1;
        tree.assign(n << 1, {});
        pref.assign(n << 1, {});
        build(0, 0, n, a);
    }
    void pull(int node){
        auto &l = tree[node * 2 + 1];
        auto &r = tree[node * 2 + 2];
        tree[node].resize(l.size() + r.size());
        merge(l.begin(), l.end(), r.begin(), r.end(), tree[node].begin());
        pref[node].resize(tree[node].size() + 1);
        for (int i = 0; i < (int)tree[node].size(); i++){
            pref[node][i + 1] = pref[node][i] + tree[node][i];
        }
    }
    void build(int node, int l, int r, vector<ll> &a){
        if (r - l == 1){
            if (l < (int)a.size()){
                tree[node] = {a[l]};
                pref[node] = {0, a[l]};
            }
            return;
        }
        int mid = (l + r) >> 1;
        build(node * 2 + 1, l, mid, a);
        build(node * 2 + 2, mid, r, a);
        pull(node);
    }
    int cnt_less(int node, ll x){
        return lower_bound(tree[node].begin(), tree[node].end(), x) - tree[node].begin();
    }
    int cnt_leq(int node, ll x){
        return upper_bound(tree[node].begin(), tree[node].end(), x) - tree[node].begin();
    }
    ll sum_first(int node, int cnt){
        return pref[node][cnt];
    }
    int count_less(int l, int r, ll x){
        return count_less(l, r + 1, x, 0, 0, n);
    }
    int count_less(int ql, int qr, ll x, int node, int l, int r){
        if (qr <= l || r <= ql) return 0;
        if (ql <= l && r <= qr) return cnt_less(node, x);
        int mid = (l + r) >> 1;
        return count_less(ql, qr, x, node * 2 + 1, l, mid) + count_less(ql, qr, x, node * 2 + 2, mid, r);
    }
    int count_leq(int l, int r, ll x){
        return count_leq(l, r + 1, x, 0, 0, n);
    }
};

```

```

}
int count_leq(int ql, int qr, ll x, int node, int l, int r){
    if (qr <= l || r <= ql) return 0;
    if (ql <= l && r <= qr) return cnt_leq(node, x);
    int mid = (l + r) >> 1;
    return count_leq(ql, qr, x, node * 2 + 1, l, mid) + count_leq(ql, qr, x, node * 2 + 2, mid, r);
}
int count_greater(int l, int r, ll x){
    return r - l + 1 - count_leq(l, r, x);
}
ll sum_less(int l, int r, ll x){
    return sum_less(l, r + 1, x, 0, 0, n);
}
ll sum_less(int ql, int qr, ll x, int node, int l, int r){
    if (qr <= l || r <= ql) return 0;
    if (ql <= l && r <= qr) return sum_first(node, cnt_less(node, x));
    int mid = (l + r) >> 1;
    return sum_less(ql, qr, x, node * 2 + 1, l, mid) + sum_less(ql, qr, x, node * 2 + 2, mid, r);
}
ll sum_leq(int l, int r, ll x){
    return sum_leq(l, r + 1, x, 0, 0, n);
}
ll sum_leq(int ql, int qr, ll x, int node, int l, int r){
    if (qr <= l || r <= ql) return 0;
    if (ql <= l && r <= qr) return sum_first(node, cnt_leq(node, x));
    int mid = (l + r) >> 1;
    return sum_leq(ql, qr, x, node * 2 + 1, l, mid) + sum_leq(ql, qr, x, node * 2 + 2, mid, r);
}
};

```

Mo's on tree.cpp

Use: Fill add(v), del(v), get_answer(); add edges, build(root), add_query(u,v,id), then process() fills answer[id].
Care: Call clear(n) between tests; vertex toggling means add/del should be symmetric and based on node contribution.

Complexity: Processing is roughly $O((n+q) \sqrt{n})$ after Euler tour and LCA preprocessing.

Pitfall: Path queries need special LCA handling; verify whether the LCA node is toggled separately.

```

const int N = 2e5 + 5;
int n, q, timer, LOG, block;
vector<int> graph[N];
int tin[N], tout[N], euler[2 * N], depth[N], up[20][N];
bool active[N];
ll curAns, answer[N];

struct Query{
    int l, r, lca, id;
    bool operator<(const Query &other) const{
        if (l / block != other.l / block) return l < other.l;
        return (l / block & 1 ? r < other.r : other.r < r);
    }
};
vector<Query> queries;

void add(int v){

```

```

}
void del(int v){
}
ll get_answer(){
    return curAns;
}
void toggle(int idx){
    int v = euler[idx];
    if (active[v]){
        del(v);
    }else{
        add(v);
    }
    active[v] ^= 1;
}
void dfs(int v, int p){
    tin[v] = ++timer;
    euler[timer] = v;
    up[0][v] = p;
    for (int i = 1; i < LOG; i++) up[i][v] = up[i - 1][up[i - 1][v]];
    for (auto u : graph[v]){
        if (u == p) continue;
        depth[u] = depth[v] + 1;
        dfs(u, v);
    }
    tout[v] = ++timer;
    euler[timer] = v;
}
bool is_ancestor(int u, int v){
    return tin[u] <= tin[v] && tout[v] <= tout[u];
}
int lca(int u, int v){
    if (is_ancestor(u, v)) return u;
    if (is_ancestor(v, u)) return v;
    for (int i = LOG - 1; i >= 0; i--){
        if (!is_ancestor(up[i][u], v)) u = up[i][u];
    }
    return up[0][u];
}
void add_query(int u, int v, int id){
    if (tin[u] > tin[v]) swap(u, v);
    int w = lca(u, v);
    if (w == u){
        queries.push_back({tin[u], tin[v], 0, id});
    }else{
        queries.push_back({tout[u], tin[v], w, id});
    }
}
void build(int root = 1){
    timer = 0;
    LOG = __lg(max(1, n)) + 1;
    depth[root] = 0;
    dfs(root, root);
    block = max(1, (int)sqrt(timer));
}
void process(){
    sort(queries.begin(), queries.end());
    int l = 1, r = 0;

```

```

    for (auto query : queries){
        while (l > query.l) toggle(--l);
        while (r < query.r) toggle(++r);
        while (l < query.l) toggle(l++);
        while (r > query.r) toggle(r--);
        if (query.lca) toggle(tin[query.lca]);
        answer[query.id] = get_answer();
        if (query.lca) toggle(tin[query.lca]);
    }
}
void clear(int n){
    for (int i = 1; i <= n; i++){
        graph[i].clear();
        active[i] = false;
    }
    queries.clear();
    curAns = 0;
    timer = 0;
}

```

Mo_s with updates.cpp

Use: Fill add(i), del(i), and maintain curAns; store queries/updates, call process(), then read answer[id].

Care: The kept SOLVE() shows the required input handling for updates; do not remove its current-array bookkeeping.

Complexity: Typical cost depends on block sizes, often around $O((n+q)^{2/3})$ pointer/update movement.

Pitfall: apply/undo must change both the array and maintained answer consistently for active indices.

```

const int N = 2e5 + 5;
int n, q, block;
int a[N];
ll curAns, answer[N];

struct Query{
    int l, r, t, id;
    bool operator<(const Query &other) const{
        if (l / block != other.l / block) return l < other.l;
        if (r / block != other.r / block) return r < other.r;
        return t < other.t;
    }
};
struct Update{
    int i, old, val;
};

vector<Query> queries;
vector<Update> updates;
void add(int i){
}
void del(int i){
}
void apply(int t, int l, int r){
    auto &u = updates[t];
    if (l <= u.i && u.i <= r) del(u.i);
    a[u.i] = u.val;
    if (l <= u.i && u.i <= r) add(u.i);
}

```

```

void undo(int t, int l, int r){
    auto &u = updates[t];
    if (l <= u.i && u.i <= r) del(u.i);
    a[u.i] = u.old;
    if (l <= u.i && u.i <= r) add(u.i);
}
void process(){
    block = max(1, (int)pow(n, 2.0 / 3.0));
    sort(queries.begin(), queries.end());
    int l = 1, r = 0, t = 0;
    for (auto query : queries){
        while(t < query.t) apply(t++, l, r);
        while(t > query.t) undo(--t, l, r);
        while(r < query.r) add(++r);
        while(l > query.l) add(--l);
        while(r > query.r) del(r--);
        while(l < query.l) del(l++);
        answer[query.id] = curAns;
    }
}
void clear(){
    queries.clear();
    updates.clear();
    curAns = 0;
}
void SOLVE(){
    cin >> n >> q;
    for (int i = 1; i <= n; i++) cin >> a[i];
    vector<int> cur(a, a + n + 1);
    int id = 0;
    for (int i = 0; i < q; i++){
        char type;
        cin >> type;
        if (type == 'Q'){
            int l, r;
            cin >> l >> r;
            queries.push_back({l, r, (int)updates.size(), id++});
        }else{
            int idx, val;
            cin >> idx >> val;
            updates.push_back({idx, cur[idx], val});
            cur[idx] = val;
        }
    }
    process();
    for (int i = 0; i < id; i++){
        cout << answer[i] << endl;
    }
    clear();
}

```

Mo_s_algorithm.cpp

Use: Define add/remove logic around sorted query{*l,r,id*}; move current [*L,R*] while processing and write ans[id].

Care: The file is only the ordering skeleton; keep ans[id] outside and update the current range while iterating queries.

Complexity: After sorting, pointer moves dominate; total cost is about $O((n+q) \sqrt{n})$ for $O(1)$ add/remove.

Before: Store queries as inclusive [*l,r*], choose block size, and initialize current L/R before processing.

```

const int N = 1e5 + 5, sq = 317;
struct query{
    int l, r, i;
    bool operator<(const query &other) const {
        if (l / sq != other.l / sq)
            return l / sq < other.l / sq;
        return (l / sq & 1 ? r < other.r : other.r < r);
    }
};

```

Multiset lazy deletion.cpp

Use: MS<T> ms; insert(*x*), erase(*x*), top(), pop(), size(), empty(); normalize() lazily removes deleted elements.

Care: top()/pop() assume the structure is nonempty after normalize(); compare parameter makes min-heap or max-heap behavior.

Complexity: insert, erase, top, and pop are $O(\log n)$ amortized because stale deleted values are removed lazily.

Pitfall: If you erase a value never inserted, a future equal value may be accidentally removed.

```

template<typename T, typename Compare = less<T>>
struct MS {
    priority_queue<T, vector<T>, Compare> pq, del;
    void normalize(){
        while(!pq.empty() && !del.empty() && pq.top() == del.top()){
            pq.pop();
            del.pop();
        }
    }
    bool empty() { normalize(); return size() == 0; }
    int size() { return (int)pq.size() - (int)del.size(); }
    void insert(T x) { pq.push(x); }
    void erase(T x) { del.push(x); }
    T top() { normalize(); return pq.top(); }
    void pop() { normalize(); pq.pop(); }
    void clear() {
        while(!pq.empty()) pq.pop();
        while(!del.empty()) del.pop();
    }
};

```

Offline 2D BIT.cpp

Use: OfflineBIT2D<T> bit(*n*, points); add(*r,c,val*) and rect_sum(*r1,c1,r2,c2*) answer offline 2D point-add rectangle-sum queries.

Before: Pass every coordinate that can ever receive add() into todo during init, otherwise that point cannot be updated later.

Complexity: Memory is $O(\text{total compressed coordinates} \log n)$; add and rectangle sum are $O(\log^2 n)$.

When: Good for all queries known beforehand with point updates and rectangle sums on sparse coordinates.

```

template<typename T>
struct OfflineBIT2D{
    int n;
    vector<int> head, vals;
    vector<T> bit;
    OfflineBIT2D(){
        OfflineBIT2D(int _n, vector<array<int, 2>> todo){
            init(_n, todo);
        }
        int ind(int r, int c){

```

```

    int start = head[r], end = head[r + 1];
    return upper_bound(vals.begin() + start, vals.begin() + end, c) - vals.begin() -
        ↪ start - 1;
}

void init(int _n, vector<array<int, 2>> todo){
    n = _n;
    sort(todo.begin(), todo.end(), [](array<int, 2> &a, array<int, 2> &b){
        return a[1] < b[1];
    });
    vector<int> last(n + 1, -1), sz(n + 1, 1);
    for (auto [r, c] : todo){
        r++, c++;
        for (; r <= n; r += r & -r){
            if (last[r] != c){
                sz[r]++;
                last[r] = c;
            }
        }
    }
    head.assign(n + 2, 0);
    for (int i = 1; i <= n; i++){
        head[i + 1] = head[i] + sz[i];
    }
    vals.assign(head[n + 1], 0);
    bit.assign(head[n + 1], 0);
    last.assign(n + 1, -1);
    vector<int> cur = head;
    for (int i = 1; i <= n; i++){
        vals[cur[i]++] = 0;
    }
    for (auto [r, c] : todo){
        r++, c++;
        for (; r <= n; r += r & -r){
            if (last[r] != c){
                vals[cur[r]++] = c;
                last[r] = c;
            }
        }
    }
}

void add(int r, int c, T val){
    r++, c++;
    for (; r <= n; r += r & -r){
        int i = ind(r, c);
        int sz = head[r + 1] - head[r];
        for (; i < sz; i += i & -i){
            bit[head[r] + i] += val;
        }
    }
}

T rect_sum(int r, int c){
    r++, c++;
    T ret = 0;
    for (; r > 0; r -= r & -r){
        int i = ind(r, c);
        for (; i > 0; i -= i & -i){
            ret += bit[head[r] + i];
        }
    }
}

```

```

        return ret;
    }
    T rect_sum(int r1, int c1, int r2, int c2){
        if (r1 > r2 || c1 > c2) return 0;
        return rect_sum(r2, c2) - rect_sum(r2, c1 - 1) - rect_sum(r1 - 1, c2) +
            ↪ rect_sum(r1 - 1, c1 - 1);
    }
    T sum(int r, int c){
        return rect_sum(r, c);
    }
    T sum(int r1, int c1, int r2, int c2){
        return rect_sum(r1, c1, r2, c2);
    }
};

```

ordered_set.cpp

Use: ordered_set<T> supports find_by_order(k) for kth element and order_of_key(x) for count of elements < x.

Care: For duplicates, store pairs like {value, unique_id}; plain ordered_set<T> behaves like a mathematical set.

Complexity: insert, erase, order_of_key, and find_by_order are O(log n).

Pitfall: find_by_order(k) is 0-indexed and invalid when k is outside [0,size).

```

template<class T>
using ordered_set = tree<T, null_type, std::less<T>, rb_tree_tag,
    ↪ tree_order_statistics_node_update>;

```

Persistent DSU.cpp

Use: PersistentDSU d(n); add_edge/unite(a,b,t), same(a,b,t), get_size(v,t), components(t), and when_connected(u,v).

Care: Times are edge insertion versions; same/get_size/components answer the state at time t without mutating history.

Complexity: Queries are logarithmic over history vectors; when_connected does a binary search over time.

Before: Add edges in nondecreasing time so parent_time and component history stay meaningful.

// Add edges in nondecreasing time. Query with time 0 for the initial graph.

```

struct PersistentDSU{
    const int INF = 1e9;
    int n, timer;
    vector<int> parent, parent_time;
    vector<vector<pair<int, int>>> sz;
    vector<pair<int, int>> components_hist;
    PersistentDSU(){
        PersistentDSU(int _n){
            init(_n);
        }
        void init(int _n){
            n = _n;
            timer = 0;
            parent.resize(n + 1);
            parent_time.assign(n + 1, INF);
            sz.assign(n + 1, {});
            components_hist = {{0, n}};
            for (int i = 1; i <= n; i++){
                parent[i] = i;
                sz[i].push_back({0, 1});
            }
        }
    }
}

```

```

}
int find(int v, int t){
    return parent_time[v] > t ? v : find(parent[v], t);
}
int get_size(int v, int t){
    v = find(v, t);
    auto &h = sz[v];
    int p = upper_bound(h.begin(), h.end(), make_pair(t, INF)) - h.begin() - 1;
    return h[p].second;
}
int components(int t){
    int p = upper_bound(components_hist.begin(), components_hist.end(), make_pair(t,
    ↪ INF)) - components_hist.begin() - 1;
    return components_hist[p].second;
}
bool same(int u, int v, int t){
    return find(u, t) == find(v, t);
}
bool unite(int a, int b, int t){
    timer = max(timer, t);
    a = find(a, t);
    b = find(b, t);
    if (a == b) return false;
    int sa = get_size(a, t), sb = get_size(b, t);
    if (sa < sb){
        swap(a, b);
        swap(sa, sb);
    }
    parent[b] = a;
    parent_time[b] = t;
    sz[a].push_back({t, sa + sb});
    components_hist.push_back({t, components(t) - 1});
    return true;
}
bool add_edge(int a, int b, int t){
    return unite(a, b, t);
}
bool unite(int a, int b){
    return unite(a, b, timer + 1);
}
bool add_edge(int a, int b){
    return unite(a, b);
}
int when_connected(int u, int v, int hi){
    if (!same(u, v, hi)) return -1;
    int l = 0, r = hi;
    while (l < r){
        int mid = (l + r) >> 1;
        if (same(u, v, mid)) r = mid;
        else l = mid + 1;
    }
    return l;
}
int when_connected(int u, int v){
    return when_connected(u, v, timer);
}
};

```

PST.cpp

Use: PST pst(n); keep root 0 as empty, newRoot=pst.update(oldRoot,idx,val), query(root,l,r), and kth(L,R,k).

Care: Updates add val to one position and return a new root index; kth(L,R,k) compares two version roots.

Complexity: Each point update creates $O(\log n)$ nodes; each query and kth call is $O(\log n)$.

When: Use two roots for prefix versions to answer kth/order-statistics on subarrays.

```

struct PST{
    struct Node{ int l, r; ll val; };
    int n;
    vector<Node> tree;
    PST(){}
    PST(int _n, int m = 0){
        init(_n, m);
    }
    void init(int _n, int m = 0){
        n = _n;
        tree.clear();
        if(m) tree.reserve(m);
        tree.push_back({0, 0, 0});
    }
    int clone(int node){
        tree.push_back(tree[node]);
        return (int)tree.size() - 1;
    }
    int update(int node, int idx, ll val){
        return update(node, idx, val, 0, n - 1);
    }
    int update(int node, int idx, ll val, int l, int r){
        int cur = clone(node);
        tree[cur].val += val;
        if (l == r) return cur;
        int mid = (l + r) >> 1;
        if (idx <= mid) tree[cur].l = update(tree[cur].l, idx, val, l, mid);
        else tree[cur].r = update(tree[cur].r, idx, val, mid + 1, r);
        return cur;
    }
    ll query(int node, int ql, int qr){
        return query(node, ql, qr, 0, n - 1);
    }
    ll query(int node, int ql, int qr, int l, int r){
        if (!node || qr < l || r < ql) return 0;
        if (ql <= l && r <= qr) return tree[node].val;
        int mid = (l + r) >> 1;
        return query(tree[node].l, ql, qr, l, mid) + query(tree[node].r, ql, qr, mid + 1,
        ↪ r);
    }
    int kth(int L, int R, ll k){
        return kth(L, R, k, 0, n - 1);
    }
    int kth(int L, int R, ll k, int l, int r){
        if (l == r) return l;
        int mid = (l + r) >> 1;
        ll left = tree[tree[R].l].val - tree[tree[L].l].val;
        if (k <= left) return kth(tree[L].l, tree[R].l, k, l, mid);
        return kth(tree[L].r, tree[R].r, k - left, mid + 1, r);
    }
};

```

seg_tree (lazy generic).cpp

Use: Edit node::add and merge for the problem, then build(v), update(l,r,val), and query(l,r) on inclusive ranges.

Care: Public update/query convert [l,r] to internal half-open ranges, so call them with inclusive endpoints.

Complexity: Build is $O(n)$, range update/query are $O(\log n)$ when merge and lazy application are $O(1)$.

Before: Change neutral node, merge, and lazy apply together; mismatching them gives silent wrong answers.

```
struct node {
    ll lazy = 0, val = 0;
    bool isLazy = 0;
    node() {}
    node(ll x) : val(x) {}
    void add(ll x, int lx, int rx) {
        val += x; // for sum use: val += x * (rx - lx)
        lazy += x;
        isLazy = 1;
    }
};

#define lnode (x * 2 + 1)
#define rnode (x * 2 + 2)
#define md (lx + (rx - lx) / 2)
struct segment_tree {
    int n;
    vector<node> seg;
    segment_tree(int sz) {
        n = 1;
        while (n < sz) n *= 2;
        seg.assign(n * 2, node());
    }
    node merge(node &l, node &r) {
        node ret;
        ret.val = l.val + r.val; // min/max: change this
        return ret;
    }
    void build(vector<int> &v, int x, int lx, int rx) {
        if (rx - lx == 1) {
            if (lx < v.size()) seg[x] = node(v[lx]);
            return;
        }
        build(v, lnode, lx, md);
        build(v, rnode, md, rx);
        seg[x] = merge(seg[lnode], seg[rnode]);
    }
    void build(vector<int> &v) { build(v, 0, 0, n); }
    void propagate(int x, int lx, int rx) {
        if (rx - lx == 1 || !seg[x].isLazy) return;
        seg[lnode].add(seg[x].lazy, lx, md);
        seg[rnode].add(seg[x].lazy, md, rx);
        seg[x].isLazy = seg[x].lazy = 0;
    }
    void update(int l, int r, ll val, int x, int lx, int rx) {
        propagate(x, lx, rx);
        if (lx >= r || rx <= l) return;
        if (lx >= l && rx <= r) return seg[x].add(val, lx, rx);
        update(l, r, val, lnode, lx, md);
        update(l, r, val, rnode, md, rx);
        seg[x] = merge(seg[lnode], seg[rnode]);
    }
};
```

```
void update(int l, int r, ll val) { update(l, r + 1, val, 0, 0, n); }
node query(int l, int r, int x, int lx, int rx) {
    propagate(x, lx, rx);
    if (lx >= l && rx <= r) return seg[x];
    if (lx >= r || rx <= l) return node();
    node L = query(l, r, lnode, lx, md);
    node R = query(l, r, rnode, md, rx);
    return merge(L, R);
}
node query(int l, int r) {
    return query(l, r + 1, 0, 0, n);
}
};
```

seg_tree (lazy propagation).cpp

Use: Build with vector, then add/multiply/set/add_ap(l,r,...) and query(l,r) on inclusive 0-indexed ranges.

Care: Default value is range sum with affine index update $a*i+b$; change node if the problem needs another aggregate.

Complexity: Each add, multiply, set, add_ap, and query is $O(\log n)$.

Pitfall: The operations compose through lazyMul/lazyA/lazyB, so new update types need careful algebra.

```
struct node {
    ll lazyMul = 1, lazyA = 0, lazyB = 0, val = 0;
    bool isLazy = 0;
    node() {}
    node(ll x) : val(x) {}
    ll sumIdx(ll lx, ll rx) {
        return (lx + rx - 1) * (rx - lx) / 2;
    }
    void apply(ll mul, ll a, ll b, int lx, int rx) {
        val = val * mul + a * sumIdx(lx, rx) + b * (rx - lx);
        lazyMul *= mul;
        lazyA = lazyA * mul + a;
        lazyB = lazyB * mul + b;
        isLazy = 1;
    }
    void add(ll x, int lx, int rx) { apply(1, 0, x, lx, rx); }
    void multiply(ll x, int lx, int rx) { apply(x, 0, 0, lx, rx); }
    void set(ll x, int lx, int rx) { apply(0, 0, x, lx, rx); }
    void add_ap(ll first, ll diff, int l, int lx, int rx) {
        apply(1, diff, first - diff * l, lx, rx);
    }
};

#define lnode (x * 2 + 1)
#define rnode (x * 2 + 2)
#define md (lx + (rx - lx) / 2)
struct segment_tree {
    int n;
    vector<node> seg;
    segment_tree(int sz) {
        n = 1;
        while (n < sz) n *= 2;
        seg.assign(n * 2, node());
    }
    node merge(node &l, node &r) {
        node ret;
        ret.val = l.val + r.val;
    }
};
```

```

    return ret;
}
void build(vector<int> &v, int x, int lx, int rx) {
    if (rx - lx == 1) {
        if (lx < v.size()) seg[x] = node(v[lx]);
        return;
    }
    build(v, lnode, lx, md);
    build(v, rnode, md, rx);
    seg[x] = merge(seg[lnode], seg[rnode]);
}
void build(vector<int> &v) { build(v, 0, 0, n); }
void propagate(int x, int lx, int rx) {
    if (rx - lx == 1 || !seg[x].isLazy) return;
    seg[lnode].apply(seg[x].lazyMul, seg[x].lazyA, seg[x].lazyB, lx, md);
    seg[rnode].apply(seg[x].lazyMul, seg[x].lazyA, seg[x].lazyB, md, rx);
    seg[x].lazyMul = 1;
    seg[x].lazyA = seg[x].lazyB = 0;
    seg[x].isLazy = 0;
}
void update(int l, int r, ll mul, ll a, ll b, int x, int lx, int rx) {
    propagate(x, lx, rx);
    if (lx >= r || rx <= l) return;
    if (lx >= l && rx <= r) return seg[x].apply(mul, a, b, lx, rx);
    update(l, r, mul, a, b, lnode, lx, md);
    update(l, r, mul, a, b, rnode, md, rx);
    seg[x] = merge(seg[lnode], seg[rnode]);
}
void update(int l, int r, ll val) { add(l, r, val); }
void add(int l, int r, ll val) { update(l, r + 1, 1, 0, val, 0, 0, n); }
void multiply(int l, int r, ll val) { update(l, r + 1, val, 0, 0, 0, 0, n); }
void set(int l, int r, ll val) { update(l, r + 1, 0, 0, val, 0, 0, n); }
void add_ap(int l, int r, ll first, ll diff) {
    update(l, r + 1, 1, diff, first - diff * l, 0, 0, n);
}
}
node query(int l, int r, int x, int lx, int rx) {
    propagate(x, lx, rx);
    if (lx >= l && rx <= r) return seg[x];
    if (lx >= r || rx <= l) return node();
    node L = query(l, r, lnode, lx, md);
    node R = query(l, r, rnode, md, rx);
    return merge(L, R);
}
node query(int l, int r) {
    return query(l, r + 1, 0, 0, n);
}
};

```

seg_tree (point, range query).cpp

Use: segment_tree st(n, neutral, merge); set(i,val) point-updates and query(l,r) answers inclusive range queries.

Care: Give a true neutral node for empty segments; public query(l,r) is inclusive even though the recursion is half-open.

Complexity: Point set and range query are $O(\log n)$; memory is $O(\text{power_of_two } n)$.

When: Use this for associative range queries with point updates and no lazy propagation.

```

struct node{
    ll val;
    node(){

```

```

        node(ll v): val(v) {}
    };

struct segment_tree{
    int n;
    vector<node> tree;
    node NEUTRAL;
    function<node(const node&, const node&>> merge;
    segment_tree(int n_, node neutral,
        function<node(const node&, const node&>> merge_)
        : NEUTRAL(neutral), merge(merge_){
        n = 1;
        while (n < n_) n <<= 1;
        tree.assign(n << 1, NEUTRAL);
    }
    void set(int i, ll val, int pos, int l, int r){
        if (r - l == 1) { tree[pos] = node(val); return; }
        int mid = (l + r) >> 1;
        if (i < mid) set(i, val, 2 * pos + 1, l, mid);
        else set(i, val, 2 * pos + 2, mid, r);
        tree[pos] = merge(tree[2 * pos + 1], tree[2 * pos + 2]);
    }
    void set(int i, ll val){ set(i, val, 0, 0, n); }
    node query(int l, int r, int pos, int nl, int nr){
        if (nr <= l || r <= nl) return NEUTRAL;
        if (l <= nl && nr <= r) return tree[pos];
        int mid = (nl + nr) >> 1;
        return merge(query(l, r, 2 * pos + 1, nl, mid), query(l, r, 2 * pos + 2, mid,
            nr));
    }
    node query(int l, int r){ return query(l, r + 1, 0, 0, n); }
};

```

sparse_table.cpp

Use: SparseTable<T> st(a, merge_func); query(l,r) answers inclusive idempotent range queries in $O(1)$.

Care: merge_func must be idempotent, like min/max/gcd, because query overlaps two blocks.

Complexity: Preprocessing is $O(n \log n)$, query is $O(1)$, and memory is $O(n \log n)$.

Pitfall: It cannot support updates; rebuild if the array changes.

```

template<class T>
struct SparseTable{
    int n, LOG;
    vector<int> lg;
    vector<vector<T>> table;
    function<T(T, T)> f;
    SparseTable(){
        SparseTable(vector<T> &v, function<T(T, T)> func){
            init(v, func);
        }
        void init(vector<T> &v, function<T(T, T)> func){
            n = v.size();
            f = func;
            LOG = __lg(n) + 1;
            lg.assign(n + 1, 0);
            for (int i = 2; i <= n; i++){
                lg[i] = lg[i >> 1] + 1;
            }
            table.assign(LOG, vector<T>(n));

```

```

        table[0] = v;
        for (int j = 1; j < LOG; j++){
            for (int i = 0; i + (1 << j) <= n; i++){
                table[j][i] = f(table[j - 1][i], table[j - 1][i + (1 << (j - 1))]);
            }
        }
    }
    T query(int l, int r){
        if (l > r) swap(l, r);
        int j = lg[r - l + 1];
        return f(table[j][l], table[j][r - (1 << j) + 1]);
    }
};

```

topK.cpp

Use: TopKSum ds(k); insert/erase values, setK(k), and get() returns the sum of the current largest k elements.

Care: It supports duplicate values; erase(x) removes one copy if present, then rebalances the top-k set.

Complexity: insert, erase, and setK are $O(\log n)$ plus rebalancing moved elements.

When: Use it for sliding windows or dynamic sets where you need the sum of the largest K values.

```

struct TopKSum{
    int K;
    multiset<ll> top_k, rest;
    ll sum;
    TopKSum(){
        TopKSum(int k){
            init(k);
        }
        void init(int k){
            K = k;
            sum = 0;
            top_k.clear();
            rest.clear();
        }
        void fix(){
            while ((int)top_k.size() > K){
                auto it = top_k.begin();
                sum -= *it;
                rest.insert(*it);
                top_k.erase(it);
            }
            while ((int)top_k.size() < K && !rest.empty()){
                auto it = prev(rest.end());
                sum += *it;
                top_k.insert(*it);
                rest.erase(it);
            }
            while (!top_k.empty() && !rest.empty() && *top_k.begin() < *prev(rest.end())){
                auto small = top_k.begin();
                auto big = prev(rest.end());
                ll x = *small, y = *big;
                sum += y - x;
                top_k.erase(small);
                rest.erase(big);
                top_k.insert(y);
                rest.insert(x);
            }
        }
    }
};

```

```

void setK(int k){
    K = k;
    fix();
}
void insert(ll val){
    if (K == 0){
        rest.insert(val);
        return;
    }
    top_k.insert(val);
    sum += val;
    fix();
}
void erase(ll val){
    auto it = rest.find(val);
    if (it != rest.end()){
        rest.erase(it);
    }else{
        it = top_k.find(val);
        if (it != top_k.end()){
            sum -= val;
            top_k.erase(it);
        }
    }
    fix();
}
int size(){
    return (int)top_k.size() + (int)rest.size();
}
void clear(){
    top_k.clear();
    rest.clear();
    sum = 0;
}
ll get(){
    return sum;
}
};

```

TwoStackQ.cpp

Use: TwoStackQ q; edit Node and mrg for your aggregate, then push(x), pop(), and get() for the whole queue.

Care: Current Node tracks min and max; add more fields to Node and update mrg() before using other aggregates.

Complexity: push and pop are $O(1)$ amortized; get is $O(1)$.

Pitfall: This only supports queue order, not arbitrary deletions from the middle.

```

const ll inf = 1e18;
struct TwoStackQ {
    struct Node {
        ll mx = -inf, mn = inf, val;
        Node(): val(0) { }
        Node(ll x): mx(x), mn(x), val(x) { }
    };
    stack<Node> a, b;
    int size() { return a.size() + b.size(); }
    bool empty() { return size() == 0; }
    void clear() {
        while (!a.empty()) a.pop();
    }
};

```

```
        while (!b.empty()) b.pop();
    }
    void mrg(Node& a, Node& b) {
        a.mn = min(a.mn, b.mn);
        a.mx = max(a.mx, b.mx);
    }
    void push(int val) {
        auto nd = Node(val);
        if (!a.empty()) mrg(nd, a.top());
        a.push(nd);
    }
    void move() {
        while (!a.empty()) {
            auto nd = Node(a.top().val);
            if (!b.empty()) mrg(nd, b.top());
            b.push(nd), a.pop();
        }
    }
    Node get() {
        Node res;
        if (!b.empty()) mrg(res, b.top());
        if (!a.empty()) mrg(res, a.top());
        return res;
    }
    void pop() {
        if (b.empty()) move();
        if (!b.empty()) b.pop();
    }
};
```

3 graphs

2SAT.cpp

Use: two_sat ts(n); add implications/clauses, then satisfiable() returns bool and fills answer[i].

Care: Variables are 0-indexed; add_or(i,val,j,val) represents literal choices directly through var(i,val).

Complexity: satisfiable() is $O(n + \text{clauses})$ after all implication edges are added.

Pitfall: Literal true/false convention is handled by var(i,val); avoid manual offsets unless adding raw edges.

```
struct two_sat{
    int n;
    vector<vector<int>> g, gr;
    vector<int> comp, top, answer;
    vector<bool> vis;
    two_sat(){
        two_sat(int _n){
            init(_n);
        }
        void init(int _n){
            n = _n;
            g.assign(2 * n, {});
            gr.assign(2 * n, {});
            comp.assign(2 * n, 0);
            vis.assign(2 * n, false);
            answer.assign(n, 0);
            top.clear();
        }
        int var(int i, bool val){
            return i + (val ? 0 : n);
        }
        void add_edge(int u, int v){
            g[u].push_back(v);
            gr[v].push_back(u);
        }
        void add_imply(int i, bool f, int j, bool k){
            add_edge(var(i, f), var(j, k));
        }
        void set_true(int i){
            add_imply(i, false, i, true);
        }
        void set_false(int i){
            add_imply(i, true, i, false);
        }
        void add_or(int i, bool f, int j, bool k){
            add_imply(i, !f, j, k);
            add_imply(j, !k, i, f);
        }
        void add_xor(int i, bool f, int j, bool k){
            add_or(i, f, j, k);
            add_or(i, !f, j, !k);
        }
        void add_xnor(int i, bool f, int j, bool k){
            add_or(i, f, j, !k);
            add_or(i, !f, j, k);
        }
        void add_and(int i, bool f, int j, bool k){
            add_imply(i, !f, i, f);
            add_imply(j, !k, j, k);
        }
        void dfs(int u){
```

```
            vis[u] = true;
            for (auto v : g[u]){
                if (!vis[v]) dfs(v);
            }
            top.push_back(u);
        }
        void scc(int u, int id){
            vis[u] = true;
            comp[u] = id;
            for (auto v : gr[u]){
                if (!vis[v]) scc(v, id);
            }
        }
        bool satisfiable(){
            top.clear();
            fill(vis.begin(), vis.end(), false);
            for (int i = 0; i < 2 * n; i++){
                if (!vis[i]) dfs(i);
            }
            fill(vis.begin(), vis.end(), false);
            reverse(top.begin(), top.end());
            int id = 0;
            for (auto v : top){
                if (!vis[v]) scc(v, id++);
            }
            for (int i = 0; i < n; i++){
                if (comp[i] == comp[i + n]) return false;
                answer[i] = comp[i] > comp[i + n];
            }
            return true;
        }
    };
```

Articulation points.cpp

Use: ArticulationPoints ap(n); add_edge(u,v), build() returns cut vertices and also fills isCut[v].

Care: The graph is undirected and 1-indexed; build() resets tin/low/isCut but keeps all added edges.

Complexity: build() is $O(n+m)$ and handles disconnected undirected graphs.

Pitfall: Parallel edges are safe because edge ids are tracked; self-loops are usually ignored before adding.

```
struct ArticulationPoints{
    int n, timer;
    vector<vector<pair<int, int>>> graph;
    vector<pair<int, int>> edges;
    vector<int> tin, low, isCut;
    ArticulationPoints(){
        ArticulationPoints(int _n){
            init(_n);
        }
        void init(int _n){
            n = _n;
            timer = 0;
            graph.assign(n + 1, {});
            edges.clear();
            tin.assign(n + 1, 0);
            low.assign(n + 1, 0);
            isCut.assign(n + 1, 0);
        }
        void add_edge(int u, int v){
```

```

    int id = edges.size();
    edges.push_back({u, v});
    graph[u].push_back({v, id});
    graph[v].push_back({u, id});
}
void dfs(int v, int pe = -1){
    tin[v] = low[v] = ++timer;
    int children = 0;
    for (auto [u, id] : graph[v]){
        if (id == pe) continue;
        if (tin[u]){
            low[v] = min(low[v], tin[u]);
        }else{
            dfs(u, id);
            low[v] = min(low[v], low[u]);
            if (pe != -1 && low[u] >= tin[v]){
                isCut[v] = true;
            }
            children++;
        }
    }
    if (pe == -1 && children > 1){
        isCut[v] = true;
    }
}
vector<int> build(){
    timer = 0;
    fill(tin.begin(), tin.end(), 0);
    fill(low.begin(), low.end(), 0);
    fill(isCut.begin(), isCut.end(), 0);
    for (int i = 1; i <= n; i++){
        if (!tin[i]) dfs(i);
    }
    vector<int> ret;
    for (int i = 1; i <= n; i++){
        if (isCut[i]) ret.push_back(i);
    }
    return ret;
}
};

```

Biconnected components.cpp

Use: BiconnectedComponents bc(n); add_edge(u,v), build(), then components and comp_edges store each biconnected block.

Out: components stores vertices and comp_edges stores original edge ids for each block; isolated vertices become singleton blocks.

Complexity: build() is $O(n+m)$; each edge enters exactly one component stack interval.

When: Use it for block-cut tree construction, articulation decomposition, or vertex-biconnected blocks.

```

struct BiconnectedComponents{
    int n, timer;
    vector<vector<pair<int, int>>> graph;
    vector<pair<int, int>> edges;
    vector<vector<int>> components, comp_edges;
    vector<int> tin, low, isCut, st;
    BiconnectedComponents(){
        BiconnectedComponents(int _n){
            init(_n);
        }
    }
};

```

```

}
void init(int _n){
    n = _n;
    timer = 0;
    graph.assign(n + 1, {});
    edges.clear();
    components.clear();
    comp_edges.clear();
    tin.assign(n + 1, 0);
    low.assign(n + 1, 0);
    isCut.assign(n + 1, 0);
    st.clear();
}
void add_edge(int u, int v){
    int id = edges.size();
    edges.push_back({u, v});
    graph[u].push_back({v, id});
    graph[v].push_back({u, id});
}
void add_component(int until){
    vector<int> verts, eids;
    while (true){
        int id = st.back();
        st.pop_back();
        eids.push_back(id);
        verts.push_back(edges[id].first);
        verts.push_back(edges[id].second);
        if (id == until) break;
    }
    sort(verts.begin(), verts.end());
    verts.erase(unique(verts.begin(), verts.end()), verts.end());
    components.push_back(verts);
    comp_edges.push_back(eids);
}
void dfs(int v, int pe = -1){
    tin[v] = low[v] = ++timer;
    int children = 0;
    for (auto [u, id] : graph[v]){
        if (id == pe) continue;
        if (!tin[u]){
            st.push_back(id);
            dfs(u, id);
            low[v] = min(low[v], low[u]);
            if (low[u] >= tin[v]){
                if (pe != -1 || children > 0) isCut[v] = true;
                add_component(id);
            }
            children++;
        }else if (tin[u] < tin[v]){
            st.push_back(id);
            low[v] = min(low[v], tin[u]);
        }
    }
    if (pe == -1 && children > 1){
        isCut[v] = true;
    }
}
void build(){
    timer = 0;
}

```

```

components.clear();
comp_edges.clear();
st.clear();
fill(tin.begin(), tin.end(), 0);
fill(low.begin(), low.end(), 0);
fill(isCut.begin(), isCut.end(), 0);
for (int i = 1; i <= n; i++){
    if (tin[i]) continue;
    if (graph[i].empty()){
        components.push_back({i});
        comp_edges.push_back({});
    }else{
        dfs(i);
    }
}
}
};

```

Bipartite check.cpp

Use: Bipartite bp(n); add_edge(u,v), build(), then ok says bipartite and color[v] gives the side.

Out: build() also stores ok and color; get_side(0/1) is meaningful after a successful build().

Complexity: build() is $O(n+m)$ over all connected components.

Pitfall: If ok is false, color is still filled but sides are not a valid bipartition.

```

struct Bipartite{
    int n;
    bool ok;
    vector<vector<int>> graph;
    vector<int> color;
    Bipartite(){}
    Bipartite(int _n){
        init(_n);
    }
    void init(int _n){
        n = _n;
        ok = true;
        graph.assign(n + 1, {});
        color.assign(n + 1, -1);
    }
    void add_edge(int u, int v){
        graph[u].push_back(v);
        graph[v].push_back(u);
    }
    bool build(){
        fill(color.begin(), color.end(), -1);
        ok = true;
        queue<int> q;
        for (int i = 1; i <= n; i++){
            if (color[i] != -1) continue;
            color[i] = 0;
            q.push(i);
            while (!q.empty()){
                int v = q.front();
                q.pop();
                for (auto u : graph[v]){
                    if (color[u] == -1){
                        color[u] = color[v] ^ 1;
                        q.push(u);
                    }
                }
            }
        }
    }
};

```

```

        }else if (color[u] == color[v]){
            ok = false;
        }
    }
}
return ok;
}
vector<int> get_side(int c){
    vector<int> ret;
    for (int i = 1; i <= n; i++){
        if (color[i] == c) ret.push_back(i);
    }
    return ret;
}
};

```

Blossom.cpp

Use: blossom bm(n); add_edge(u,v), then solve() returns maximum matching size and mate[v] gives partner or -1.

Care: Nodes are 0-indexed and edges are unweighted; use only for general graph maximum cardinality matching.

Complexity: This is polynomial but heavy; use Hopcroft-Karp instead when the graph is bipartite.

Pitfall: The implementation is for maximum cardinality matching, not weighted matching.

```

struct blossom {
    int n, m;
    vector<int> mate;
    vector<vector<int>> b;
    vector<int> p, d, bl;
    vector<vector<int>> g;
    blossom(int n) : n(n) {
        m = n + n / 2;
        mate.assign(n, -1);
        b.resize(m);
        p.resize(m);
        d.resize(m);
        bl.resize(m);
        g.assign(m, vector<int>(m, -1));
    }
    void add_edge(int u, int v) {
        g[u][v] = u;
        g[v][u] = v;
    }
    void match(int u, int v) {
        g[u][v] = g[v][u] = -1;
        mate[u] = v;
        mate[v] = u;
    }
    vector<int> trace(int x) {
        vector<int> vx;
        while(true) {
            while(bl[x] != x) x = bl[x];
            if(!vx.empty() && vx.back() == x) break;
            vx.push_back(x);
            x = p[x];
        }
        return vx;
    }
};

```

```

void contract(int c, int x, int y, vector<int> &vx, vector<int> &vy) {
    b[c].clear();
    int r = vx.back();
    while(!vx.empty() && !vy.empty() && vx.back() == vy.back()) {
        r = vx.back();
        vx.pop_back();
        vy.pop_back();
    }
    b[c].push_back(r);
    b[c].insert(b[c].end(), vx.rbegin(), vx.rend());
    b[c].insert(b[c].end(), vy.begin(), vy.end());
    for(int i = 0; i <= c; i++) {
        g[c][i] = g[i][c] = -1;
    }
    for(int z : b[c]) {
        bl[z] = c;
        for(int i = 0; i < c; i++) {
            if(g[z][i] != -1) {
                g[c][i] = z;
                g[i][c] = g[i][z];
            }
        }
    }
}

vector<int> lift(vector<int> &vx) {
    vector<int> A;
    while(vx.size() >= 2) {
        int z = vx.back(); vx.pop_back();
        if(z < n) {
            A.push_back(z);
            continue;
        }
        int w = vx.back();
        int i = (A.size() % 2 == 0 ? find(b[z].begin(), b[z].end(), g[z][w]) -
        ↪ b[z].begin() : 0);
        int j = (A.size() % 2 == 1 ? find(b[z].begin(), b[z].end(), g[z][A.back()]) -
        ↪ b[z].begin() : 0);
        int k = b[z].size();
        int dif = (A.size() % 2 == 0 ? i % 2 == 1 : j % 2 == 0) ? 1 : k - 1;
        while(i != j) {
            vx.push_back(b[z][i]);
            i = (i + dif) % k;
        }
        vx.push_back(b[z][i]);
    }
    return A;
}

int solve() {
    for(int ans = 0; ; ans++) {
        fill(d.begin(), d.end(), 0);
        queue<int> Q;
        for(int i = 0; i < m; i++) bl[i] = i;
        for(int i = 0; i < n; i++) {
            if(mate[i] == -1) {
                Q.push(i);
                p[i] = i;
                d[i] = 1;
            }
        }
    }
}

```

```

int c = n;
bool aug = false;
while(!Q.empty() && !aug) {
    int x = Q.front(); Q.pop();
    if(bl[x] != x) continue;
    for(int y = 0; y < c; y++) {
        if(bl[y] == y && g[x][y] != -1) {
            if(d[y] == 0) {
                p[y] = x;
                d[y] = 2;
                p[mate[y]] = y;
                d[mate[y]] = 1;
                Q.push(mate[y]);
            } else if(d[y] == 1) {
                vector<int> vx = trace(x);
                vector<int> vy = trace(y);
                if(vx.back() == vy.back()) {
                    contract(c, x, y, vx, vy);
                    Q.push(c);
                    p[c] = p[b[c][0]];
                    d[c] = 1;
                    c++;
                } else {
                    aug = true;
                    vx.insert(vx.begin(), y);
                    vy.insert(vy.begin(), x);
                    vector<int> A = lift(vx);
                    vector<int> B = lift(vy);
                    A.insert(A.end(), B.rbegin(), B.rend());
                    for(int i = 0; i < (int) A.size(); i += 2) {
                        match(A[i], A[i + 1]);
                        if(i + 2 < (int) A.size()) add_edge(A[i + 1], A[i +
                        ↪ 2]);
                    }
                }
            }
        }
    }
    break;
}

if(!aug) return ans;
}
}
};

```

Bridges.cpp

Use: clear(n), add_edge(u,v), build(n), then bridges contains bridge endpoints and isBridge[id] marks bridge ids.

Care: Call clear(n) before rebuilding; bridge edge ids follow the order in which add_edge was called.

Complexity: build(n) is O(n+m) and supports disconnected graphs.

Pitfall: Global arrays have fixed N; increase N before copying if constraints are larger.

```

const int N = 2e5 + 5;
vector<pair<int, int>> graph[N];
vector<pair<int, int>> edges, bridges;
vector<bool> isBridge;
int tin[N], low[N], timer;

```

```

void add_edge(int u, int v){
    int id = edges.size();
    edges.push_back({u, v});
    graph[u].push_back({v, id});
    graph[v].push_back({u, id});
}

void dfs(int v, int pe = -1){
    tin[v] = low[v] = ++timer;
    for (auto [u, id] : graph[v]){
        if (id == pe) continue;
        if (tin[u]){
            low[v] = min(low[v], tin[u]);
        }else{
            dfs(u, id);
            low[v] = min(low[v], low[u]);
            if (low[u] > tin[v]){
                isBridge[id] = true;
                bridges.push_back(edges[id]);
            }
        }
    }
}

void build(int n){
    timer = 0;
    bridges.clear();
    isBridge.assign(edges.size(), false);
    for (int i = 1; i <= n; i++){
        tin[i] = low[i] = 0;
    }
    for (int i = 1; i <= n; i++){
        if (!tin[i]) dfs(i);
    }
}

void clear(int n){
    for (int i = 1; i <= n; i++){
        graph[i].clear();
        tin[i] = low[i] = 0;
    }
    edges.clear();
    bridges.clear();
    isBridge.clear();
    timer = 0;
}

```

Centroid decomposition buckets.cpp

Use: Set graph/limit k, build(1), toggle(v) active nodes, and countWith(v) counts active nodes satisfying the bucket condition.

Out: paths is the maintained answer for active pairs with distance < k; set global k before toggles.

Complexity: build is $O(n \log n)$; each toggle/countWith touches $O(\log n)$ centroid ancestors with ordered_set logs.

Pitfall: Distances are weighted; make sure global k uses the same weight units as edge weights.

```

#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;

template<class T>
using ordered_set = tree<T, null_type, less<T>, rb_tree_tag,
    tree_order_statistics_node_update>;

```

```

const int N = 1e5 + 5;
vector<pair<int, ll>> graph[N];
vector<pair<int, ll>> up[N], side[N]; // {centroid / side id, distance}
ordered_set<pair<ll, int>> bucket[N];
vector<ordered_set<pair<ll, int>>> sideBucket(1);
bool rem[N], active[N];
int sz[N], par[N], sideId;
ll k, paths;

```

```

int pre(int v, int p = -1){
    sz[v] = 1;
    for (auto [u, w] : graph[v]){
        if (u != p && !rem[u]) sz[v] += pre(u, v);
    }
    return sz[v];
}

int cen(int v, int tree_size, int p = -1){
    for (auto [u, w] : graph[v]){
        if (u != p && !rem[u] && sz[u] * 2 > tree_size){
            return cen(u, tree_size, v);
        }
    }
    return v;
}

void dfs(int v, int p, int c, ll d, int id){
    up[v].push_back({c, d});
    side[v].push_back({id, d});
    for (auto [u, w] : graph[v]){
        if (u != p && !rem[u]) dfs(u, v, c, d + w, id);
    }
}

void build(int v = 1, int p = 0){
    int c = cen(v, pre(v));
    par[c] = p;
    rem[c] = true;
    up[c].push_back({c, 0});
    side[c].push_back({0, 0});
    for (auto [u, w] : graph[c]){
        if (rem[u]) continue;
        sideBucket.emplace_back();
        dfs(u, c, c, w, ++sideId);
    }
    for (auto [u, w] : graph[c]){
        if (!rem[u]) build(u, c);
    }
}

int get(ordered_set<pair<ll, int>> &st, ll x){
    if (x < 0) return 0;
    return st.order_of_key({x, INT_MAX});
}

ll countWith(int v){
    ll ret = 0;
    for (int i = 0; i < (int)up[v].size(); i++){
        auto [c, d] = up[v][i];
        auto [id, _] = side[v][i];
        ret += get(bucket[c], k - d);
        if (id) ret -= get(sideBucket[id], k - d);
    }
}

```

```

    return ret;
}
void add(int v){
    if (active[v]) return;
    paths += countWith(v);
    active[v] = true;
    for (int i = 0; i < (int)up[v].size(); i++){
        auto [c, d] = up[v][i];
        auto [id, _] = side[v][i];
        bucket[c].insert({d, v});
        if (id) sideBucket[id].insert({d, v});
    }
}
void erase(int v){
    if (!active[v]) return;
    active[v] = false;
    for (int i = 0; i < (int)up[v].size(); i++){
        auto [c, d] = up[v][i];
        auto [id, _] = side[v][i];
        bucket[c].erase({d, v});
        if (id) sideBucket[id].erase({d, v});
    }
    paths -= countWith(v);
}
void toggle(int v){
    active[v] ? erase(v) : add(v);
}
void clear(int n){
    for (int i = 1; i <= n; i++){
        graph[i].clear();
        up[i].clear();
        side[i].clear();
        bucket[i].clear();
        rem[i] = active[i] = false;
        sz[i] = par[i] = 0;
    }
    sideBucket.assign(1, ordered_set<pair<ll, int>>());
    sideId = 0;
    paths = 0;
}

```

Centroid decomposition with updates.cpp

Use: Build after filling tree; add/erase/toggle active nodes and query(v) returns the best distance-style answer for v.

Out: query(v) returns the minimum distance from v to an active node, or INF when no active node exists.

Complexity: build is $O(n \log n)$; add, erase, toggle, and query are $O(\log^2 n)$ with multisets.

Pitfall: When erasing, remove only one matching distance from each multiset.

```

const int N = 1e5 + 5;
const ll INF = 4e18;
vector<pair<int, ll>> graph[N], path[N];
multiset<ll> st[N];
bool rem[N], active[N];
int sz[N], par[N];

int pre(int v, int p = -1){
    sz[v] = 1;
    for (auto [u, w] : graph[v]){

```

```

        if (u != p && !rem[u]) sz[v] += pre(u, v);
    }
    return sz[v];
}
int cen(int v, int tree_size, int p = -1){
    for (auto [u, w] : graph[v]){
        if (u != p && !rem[u] && sz[u] * 2 > tree_size){
            return cen(u, tree_size, v);
        }
    }
    return v;
}
void dfs(int v, int p, int c, ll d){
    path[v].push_back({c, d});
    for (auto [u, w] : graph[v]){
        if (u != p && !rem[u]) dfs(u, v, c, d + w);
    }
}
void build(int v = 1, int p = 0){
    int c = cen(v, pre(v));
    par[c] = p;
    rem[c] = true;
    dfs(c, 0, c, 0);
    for (auto [u, w] : graph[c]){
        if (!rem[u]) build(u, c);
    }
}
void add(int v){
    if (active[v]) return;
    active[v] = true;
    for (auto [c, d] : path[v]) st[c].insert(d);
}
void erase(int v){
    if (!active[v]) return;
    active[v] = false;
    for (auto [c, d] : path[v]){
        auto it = st[c].find(d);
        if (it != st[c].end()) st[c].erase(it);
    }
}
void toggle(int v){
    active[v] ? erase(v) : add(v);
}
ll query(int v){
    ll ans = INF;
    for (auto [c, d] : path[v]){
        if (!st[c].empty()) ans = min(ans, d + *st[c].begin());
    }
    return ans;
}
void clear(int n){
    for (int i = 1; i <= n; i++){
        graph[i].clear();
        path[i].clear();
        st[i].clear();
        rem[i] = active[i] = false;
        sz[i] = par[i] = 0;
    }
}

```

```
}
```

Centroid decomposition.cpp

Use: Fill process_centroid/dfs_calc/dfs_add for your counting task, then build(1) decomposes and processes the tree.

Care: This is a fill-in skeleton: put your counting logic in dfs_calc, dfs_add, and process_centroid.

Complexity: Centroid decomposition visits each level once, usually $O(n \log n)$ plus your custom work.

Before: Clear per-centroid temporary arrays after process_centroid to avoid state leaking between subtrees.

```
const int N = 2e5 + 5;
vector<int> graph[N];
bool rem[N];
int sz[N];
int answer;

int pre(int v, int p = -1){
    sz[v] = 1;
    for(auto &u : graph[v]){
        if(u == p || rem[u]) continue;
        sz[v] += pre(u, v);
    }
    return sz[v];
}

int cen(int v, int tree_size, int p = -1){
    for(auto &u : graph[v]){
        if(u == p || rem[u]) continue;
        if(sz[u] * 2 > tree_size) return cen(u, tree_size, v);
    }
    return v;
}

void process_centroid(int c);
void build(int v = 1){
    int centroid = cen(v, pre(v));
    process_centroid(centroid);
    rem[centroid] = true;
    for(auto &u : graph[centroid]){
        if(rem[u]) continue;
        build(u);
    }
}

void dfs_calc(int &v, int &p) {
    for(int &u : graph[v]){
        if(u == p || rem[u]) continue;
        dfs_calc(u, v);
    }
}

void dfs_add(int &v, int &p){
    for(int &u : graph[v]){
        if(u == p || rem[u]) continue;
        dfs_add(u, v);
    }
}

void process_centroid(int c){
    for(int &u : graph[c]) {
        if(rem[u]) continue;
        dfs_calc(u, c);
        dfs_add(u, c);
    }
}
```

```
for(int &u : graph[c]){
    if(rem[u]) continue;
    dfs_add(u, c);
}

void clear(int n){
    for(int i = 1; i <= n; i++){
        rem[i] = sz[i] = 0;
    }
}
```

dijkstra.cpp

Use: Fill graph adjacency with (to,w); dijkstra(src,parent) returns distances and parents, or dijkstra(src) returns distances only.

Care: Set global n and graph first; edge weights must be nonnegative or Dijkstra is invalid.

Complexity: Runs in $O((n+m) \log n)$ with the priority queue.

Out: Unreachable vertices keep INF distance and parent -1.

```
const ll INF = 4e18;
int n;
vector<vector<pair<int, ll>>> graph;

vector<ll> dijkstra(int src, vector<int> &parent){
    vector<ll> dist(n + 1, INF);
    parent.assign(n + 1, -1);
    priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<pair<ll, int>>> pq;
    dist[src] = 0;
    pq.push({0, src});
    while (!pq.empty()){
        auto [d, v] = pq.top();
        pq.pop();
        if (d != dist[v]) continue;
        for (auto [u, w] : graph[v]){
            if (dist[u] > d + w){
                dist[u] = d + w;
                parent[u] = v;
                pq.push({dist[u], u});
            }
        }
    }
    return dist;
}

vector<ll> dijkstra(int src){
    vector<int> parent;
    return dijkstra(src, parent);
}
```

Dinic.cpp

Use: Dinic flow(n,s,t); add_edge(u,v,cap), then flow() returns max flow and residual data stays in E.

Out: getEdges() returns a min-cut edge set after flow(); getMatchingEdges(n,m) extracts bipartite matching edges.

Complexity: Fast in practice; standard bound is $O(V^2 E)$, often better on unit networks.

Pitfall: Constructor stores s and t, so create a new Dinic or reset flows before another max-flow run.

```
const ll inf = 1e18;
struct Edge {
    int u, v;
```

```

    ll cap, flow = 0;
    Edge(int u, int v, ll cap) : u(u), v(v), cap(cap) {}
};

struct Dinic {
    vector<Edge> E;
    vector<vector<int>>> adj;
    vector<int> level, ptr;
    queue<int> q;
    int n, m = 0, s, t;
    Dinic(int n, int s, int t) : n(n), s(s), t(t) {
        adj.resize(n);
        level.resize(n);
        ptr.resize(n);
    }
    void add_edge(int u, int v, ll cap) {
        E.push_back(Edge(u, v, cap));
        E.push_back(Edge(v, u, 0));
        adj[u].push_back(m);
        adj[v].push_back(m + 1);
        m += 2;
    }
    bool bfs() {
        while (q.size()) {
            auto u = q.front();
            q.pop();
            for (auto &id : adj[u]) {
                if (E[id].cap == E[id].flow) continue;
                if (level[E[id].v] != -1) continue;
                level[E[id].v] = level[u] + 1;
                q.push(E[id].v);
            }
        }
        return level[t] != -1;
    }
    ll dfs(int u, ll pushed) {
        if (pushed == 0) return 0;
        if (u == t) return pushed;
        for (int &cid = ptr[u]; cid < (int)adj[u].size(); ++cid) {
            int id = adj[u][cid];
            ll v = E[id].v;
            if (level[u] + 1 != level[v]) continue;
            ll tr = dfs(v, min(pushed, E[id].cap - E[id].flow));
            if (tr == 0) continue;
            E[id].flow += tr;
            E[id ^ 1].flow -= tr;
            return tr;
        }
        return 0;
    }
    ll flow() {
        ll ret = 0;
        while (true) {
            fill(level.begin(), level.end(), -1);
            level[s] = 0;
            q.push(s);
            if (!bfs()) break;
            fill(ptr.begin(), ptr.end(), 0);
            while (ll pushed = dfs(s, inf)) {
                ret += pushed;
            }
        }
    }
};

```

```

    }
    return ret;
}

// get any set of E to achieve the min cut (max flow)
vector<pair<int, int>> getEdges() {
    vector<int> srcSide(n, 0);
    queue<int> q;
    q.push(s);
    srcSide[s] = 1;
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        for (auto &id : adj[u]) {
            if (!srcSide[E[id].v] && E[id].flow < E[id].cap) {
                srcSide[E[id].v] = 1;
                q.push(E[id].v);
            }
        }
    }
    vector<pair<int, int>> res;
    for (int u = 0; u < n; ++u) {
        if (!srcSide[u]) continue;
        for (auto &id : adj[u]) {
            if (!srcSide[E[id].v] && E[id].cap > 0)
                res.push_back({u, E[id].v});
        }
    }
    return res;
}

// Get The Edges Used In Max Bipartite Matching : s(0), t(n + m + 1)
vector<pair<int, int>> getMatchingEdges(int n, int m) {
    vector<pair<int, int>> ret;
    for (auto &e : E) {
        if (1 <= e.u && e.u <= n && n + 1 <= e.v && e.v <= n + m) {
            if (e.flow > 0) {
                ret.push_back({e.u, e.v - n});
            }
        }
    }
    return ret;
}
};

```

DSU on trees.cpp

Use: Fill add/remove/query logic inside the DFS lambdas; run dfs(root,parent,true) to keep the root subtree data.

Care: The kept SOLVE() contains the actual DSU-on-tree traversal; fill add/remove before dfs(1,-1,1).

Complexity: The small-to-large traversal is $O(n \log n)$ or $O(n)$ depending on add/remove work.

Pitfall: Only the heavy child data is kept; light subtrees are removed unless keep is true.

```

void SOLVE(){
    int n; cin >> n;
    vector<vector<int>>> graph(n + 1);
    for(int i = 0, u, v; i < n - 1; i++){
        cin >> u >> v;
        graph[u].push_back(v);
        graph[v].push_back(u);
    }
}

```

```

}
vector<int> tin(n + 1), tout(n + 1), vertex(n + 1), sz(n + 1, 1), d(n + 1, 1);
int timer = 0;
function<int(int, int)> euler = [&](int u, int p){
    tin[u] = ++timer;
    vertex[tin[u]] = u;
    for(auto v : graph[u]) if(v != p){
        d[v] = d[u] + 1;
        sz[u] += euler(v, u);
    }
    tout[u] = timer;
    return sz[u];
};
euler(1, -1);

auto add = [&](int u){};
auto remove = [&](int u){};

function<void(int, int, bool)> dfs = [&](int u, int p, bool keep){
    int big = -1;
    for(auto v : graph[u]){
        if(v == p) continue;
        if(big == -1 || sz[v] > sz[big]) big = v;
    }
    for(auto v : graph[u])
        if(v != p && v != big) dfs(v, u, 0);
    if(big != -1) dfs(big, u, 1);
    for(auto v : graph[u]){
        if(v == p || v == big) continue;
        for(int i = tin[v]; i <= tout[v]; i++){
            // here
            add(vertex[i]);
        }
    }
    add(u);
    if(!keep){
        for(int i = tin[u]; i <= tout[u]; i++) remove(vertex[i]);
    }
};
dfs(1, -1, 1);
}

```

eulerWalk.cpp

Use: Pass adjacency with (to, edge_id), edge count, and source; eulerWalk returns the Euler vertex order or empty if impossible.

Care: Use unique edge ids from 0 to nedges-1; for undirected graphs add both directions with the same id.

Complexity: Runs in $O(n+m)$ and consumes each edge id once.

Pitfall: The function checks degree feasibility through D; still choose a valid source for paths.

```

/*
undirected/cycle: every vertex is even
undirected/path: 0/2 vertices are odd.
directed/cycle: every vertex in == out
directed/path: exactly one (out - in == 1) -> src
                exactly one (in - out == 1) -> sink
                all other in == out.
*/
vector<int> eulerWalk(vector<vector<pair<int, int>>> &gr, int nedges, int src = 0) {

```

```

int n = gr.size();
vector<int> D(n), its(n), eu(nedges), ret, s = {src};
D[src]++; // to allow Euler paths, not just cycles
while (!s.empty()) {
    int x = s.back(), y, e, &it = its[x], end = gr[x].size();
    if (it == end) {
        ret.push_back(x);
        s.pop_back();
        continue;
    }
    tie(y, e) = gr[x][it++];
    if (!eu[e]) {
        D[x]--, D[y]++;
        eu[e] = 1;
        s.push_back(y);
    }
}
for (int x : D) if (x < 0 || ret.size() != nedges + 1) return {};
return {ret.rbegin(), ret.rend()};
}

```

HLD.cpp

Use: HLD h(n); add edges, init(root), then tin/tout/head/par/d/path_sorted split path and subtree ranges.

Out: path_sorted(u,v) returns tin ranges plus a reversed flag, which is useful for ordered path queries.

Complexity: init is $O(n)$; each path is split into $O(\log n)$ tin intervals.

Pitfall: dfs removes parent edges from child adjacency, so do not reuse the original undirected adjacency after init.

```

struct HLD {
    int root = 0;
    vector<vector<int>> g;
    explicit HLD(int n) : g(n) {}
    void add(int u, int v) {
        g[u].push_back(v);
        g[v].push_back(u);
    }
    vector<int> &operator[](int u) {
        return g[u];
    }
    int timer = 0;
    vector<int> tin, tout, d, sz, head, par, vertex;
    void init(int rt = 0) {
        root = rt;
        tin = tout = d = head = par = vertex = vector<int>(g.size() + 1);
        sz.resize(g.size(), 1);
        par[root] = head[root] = root;
        dfs(root); dfs2(root);
    }
    void dfs(int u){
        for(int & v : g[u]){
            d[v] = d[u] + 1;
            par[v] = u;
            g[v].erase(find(g[v].begin(), g[v].end(), u));
            dfs(v);
            sz[u] += sz[v];
            if(sz[v] > sz[g[u][0]]) swap(v, g[u][0]);
        }
    }
}

```

```

void dfs2(int u){
    tin[u] = ++timer;
    vertex[tin[u]] = u;
    for(int v : g[u]){
        head[v] = v == g[u][0] ? head[u] : v;
        dfs2(v);
    }
    tout[u] = timer;
}

int jump(int u, int k) {
    if(k > d[u]) return -1;
    int depth = d[u] - k;
    while (d[head[u]] > depth) { u = par[head[u]]; }
    return vertex[tin[u] - d[u] + depth];
}

bool isAnc(int u, int v) {
    return tin[u] <= tin[v] && tin[v] <= tout[u];
}

int lca(int u, int v) {
    if(d[u] > d[v]) swap(u, v);
    if(isAnc(u, v)) return u;
    while(head[u] != head[v]){
        if (d[head[u]] > d[head[v]]) u = par[head[u]];
        else v = par[head[v]];
    }
    return d[u] < d[v] ? u : v;
}

int dis(int u, int v) {
    return d[u] + d[v] - 2 * d[lca(u, v)];
}

auto path_sorted(int u, int v) {
    vector<tuple<int, int, bool>> U, V; // from, to, reversed
    while(head[u] != head[v]){
        if(d[head[u]] > d[head[v]]){
            U.emplace_back(tin[head[u]], tin[u], true);
            u = par[head[u]];
        }else{
            V.emplace_back(tin[head[v]], tin[v], false);
            v = par[head[v]];
        }
    }
    if(d[u] < d[v]) V.emplace_back(tin[u], tin[v], false);
    else U.emplace_back(tin[v], tin[u], true);
    U.insert(U.end(), V.rbegin(), V.rend());
    return U;
}
};

```

Hopcroftkarp.cpp

Use: HopcroftKarp hk(nLeft,nRight); add_edge(l,r), then maximum_matching() returns size and l/r store matches.

Care: Left nodes are 1..nLeft and right nodes are passed as 1..nRight; add_edge shifts right nodes internally.

Complexity: maximum_matching is $O(E \sqrt{V})$, much faster than flow for large bipartite matching.

Out: If l[u] is nonzero, left node u is matched to right node l[u]-n.

```

#define ll long long
#define endl "\n"

```

```

struct HopcroftKarp {
    static const int inf = 1e9;
    int n;
    vector<int> l, r, d;
    vector<vector<int>> g;
    HopcroftKarp(int _n, int _m) {
        n = _n;
        int p = _n + _m + 1;
        g.resize(p); l.resize(p, 0); r.resize(p, 0); d.resize(p, 0);
    }
    void add_edge(int u, int v) {
        g[u].push_back(v + n);
    }
    bool bfs() {
        queue<int> q;
        for (int u = 1; u <= n; u++) {
            if (!l[u]) d[u] = 0, q.push(u);
            else d[u] = inf;
        }
        d[0] = inf;
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (auto v : g[u]) {
                if (d[r[v]] == inf) { d[r[v]] = d[u] + 1; q.push(r[v]); }
            }
        }
        return d[0] != inf;
    }
    bool dfs(int u) {
        if (!u) return true;
        for (auto v : g[u]) {
            if (d[r[v]] == d[u] + 1 && dfs(r[v])) {
                l[u] = v; r[v] = u; return true;
            }
        }
        d[u] = inf;
        return false;
    }
    int maximum_matching() {
        int ans = 0;
        while (bfs()) {
            for(int u = 1; u <= n; u++) if (!l[u] && dfs(u)) ans++;
        }
        return ans;
    }
};

```

LCA and Binary lifting.cpp

Use: LCA lca(g,root); use lca(u,v), dist(u,v), distance(u,v), kth(u,k), jump(u,k), and go(u,v,k).

Care: The tree is 0-indexed and unweighted by default; distance() uses distRoot, which currently adds 1 per edge.

Complexity: Preprocessing is $O(n \log n)$; lca, kth, and go are $O(\log n)$.

Pitfall: go(u,v,k) asserts k is on the path length; pass 0 for u and dist(u,v) for v.

```

struct LCA{
    int n, Log, timer;
    vector<vector<int>> up;
    vector<int> depth, tin, tout;

```

```

vector<ll> distRoot;
LCA(){
LCA(vector<vector<int>> &g, int root = 0){
    init(g, root);
}
void init(vector<vector<int>> &g, int root = 0){
    n = g.size();
    Log = __lg(n) + 1;
    timer = 0;
    up.assign(n, vector<int>(Log, root));
    depth.assign(n, 0);
    tin.assign(n, 0);
    tout.assign(n, 0);
    distRoot.assign(n, 0);
    dfs(root, root, g);
}
void dfs(int u, int p, vector<vector<int>> &g){
    tin[u] = ++timer;
    up[u][0] = p;
    for (int l = 1; l < Log; l++){
        up[u][l] = up[up[u][l-1]][l-1];
    }
    for (int v : g[u]){
        if (v == p) continue;
        depth[v] = depth[u] + 1;
        distRoot[v] = distRoot[u] + 1;
        dfs(v, u, g);
    }
    tout[u] = timer;
}
int kth(int u, int k){
    while (k){
        u = up[u][__builtin_ctz(k)];
        k &= k - 1;
    }
    return u;
}
int jump(int u, int k){
    return kth(u, k);
}
bool isAnc(int u, int v){
    return tin[u] <= tin[v] && tout[v] <= tout[u];
}
int lca(int u, int v){
    if (depth[u] < depth[v]) swap(u, v);
    u = kth(u, depth[u] - depth[v]);
    if (u == v) return u;
    for (int l = Log - 1; l >= 0; l--){
        if (up[u][l] != up[v][l]){
            u = up[u][l];
            v = up[v][l];
        }
    }
    return up[u][0];
}
int dist(int u, int v, int l = -1){
    if (l == -1) l = lca(u, v);
    return depth[u] + depth[v] - 2 * depth[l];
}
}

```

```

ll distance(int u, int v, int l = -1){
    if (l == -1) l = lca(u, v);
    return distRoot[u] + distRoot[v] - 2 * distRoot[l];
}
int go(int u, int v, int k){
    int l = lca(u, v);
    int d = dist(u, v, l);
    assert(k <= d);
    if (depth[l] + k <= depth[u]) return kth(u, k);
    k -= depth[u] - depth[l];
    return kth(v, depth[v] - depth[l] - k);
}
}

```

MCMF.cpp

Use: MCMF mf(n,s,t); addEdge(u,v,cost,cap), then minCostMaxFlow(K) returns {flow,cost}.

Care: addEdge parameter order is cost then cap; get_paths(k) assumes cost==1 edges mark the wanted path edges.

Complexity: Each augmentation uses SPFA-style shortest path, so use carefully on large dense graphs.

Out: minCostMaxFlow(K) may return flow < K if the sink cannot receive enough capacity.

```

const int inf = 1000000010;
struct Edge {
    int to, cost, cap, flow, backEdge;
};
struct MCMF {
    int s, t, n;
    vector<vector<Edge>> g;
    MCMF(int _n, int _s, int _t){
        s = _s, t = _t, n = _n + 1;
        g = vector<vector<Edge>>(n);
    }
    void addEdge(int u, int v, int cost, int cap) {
        g[u].push_back({ v, cost, cap, 0, (int)g[v].size() });
        g[v].push_back({ u, -cost, 0, 0, (int)g[u].size() - 1});
    }
    pair<int, int> minCostMaxFlow(int K) {
        int flow = 0, cost = 0;
        vector<int> state(n), from(n), from_edge(n), d(n);
        deque<int> q;

        while(flow < K){
            for (int i = 0; i < n; i++)
                state[i] = 2, d[i] = inf, from[i] = -1;
            state[s] = 1; q.clear(); q.push_back(s); d[s] = 0;
            while (!q.empty()) {
                int v = q.front(); q.pop_front(); state[v] = 0;
                for (int i = 0; i < (int) g[v].size(); i++) {
                    Edge e = g[v][i];
                    if (e.flow >= e.cap || d[e.to] <= d[v] + e.cost)
                        continue;
                    int to = e.to; d[to] = d[v] + e.cost;
                    from[to] = v; from_edge[to] = i;
                    if (state[to] == 1) continue;
                    q.push_back(to);
                    state[to] = 1;
                }
            }
        }
    }
}

```

```

        if (d[t] == inf) break;
        int it = t, addflow = inf;
        while (it != s) {
            addflow = min(addflow, g[from[it]][from_edge[it]].cap
                          - g[from[it]][from_edge[it]].flow);
            it = from[it];
        }
        addflow = min(addflow, K - flow);
        it = t;
        while (it != s) {
            g[from[it]][from_edge[it]].flow += addflow;
            g[it][g[from[it]][from_edge[it]].backEdge].flow -= addflow;
            cost += g[from[it]][from_edge[it]].cost * addflow;
            it = from[it];
        }
        flow += addflow;
    }
    return {flow, cost};
}

vector<vector<int>> get_paths(int k) {
    vector<vector<int>> paths;
    for(int i = 0; i < k; i++){
        vector<int> path;
        int curr = s;
        path.push_back(curr);
        while(curr != t){
            for(auto &e : g[curr]){
                if(e.cost == 1 && e.flow > 0){
                    e.flow--;
                    curr = e.to;
                    path.push_back(curr);
                    break;
                }
            }
        }
        paths.push_back(path);
    }
    return paths;
}
};

```

Rerooting.cpp

Use: Edit base(), merge_into(), finalize_merge(); fill graph and call solve_reroot() to get dp for every root.

Care: Edit Agg, Val, base, merge_into, and finalize_merge; graph must be 0-indexed and connected.

Complexity: The generic reroot pass is $O(n \log \text{degree})$ due the stack split, usually near $O(n)$.

Out: Returns root_dp plus edge_dp/redge_dp, useful when answers also need directed edge contributions.

```

const int N = 2e5 + 5;
using Agg = ll; // in-progress dp of a node
using Val = ll; // final dp of a node
vector<vector<int>> graph; // 0-based nodes
int a[N];
// Initialize the DP state for a node
Agg base(int node) {
    return 0;
}
// Merge a child's finalized value into the current node's aggregate
Agg merge_into(Agg cur_dp, Val ch_val, int node, int edge_idx) {

```

```

    return cur_dp + ch_val;
}
// Finalize the aggregate before passing it up (add node/edge weights)
Val finalize_merge(Agg cur_dp, int node, int par_edge_idx) {
    return cur_dp;
}
// root_dp[u]: dp of the tree when rooted at u
// edge_dp[u][i]: dp of the subtree of node graph[u][i] excluding the branch of u
// redge_dp[u][i]: dp of the tree rooted at u excluding the branch of graph[u][i]
auto solve_reroot() {
    int n = graph.size();
    vector<Val> root_dp(n), dp(n);
    vector<vector<Val>> edge_dp(n), redge_dp(n);
    vector<int> bfs, par(n, -1);
    bfs.reserve(n);
    bfs.push_back(0);
    for(int i = 0; i < n; i++){
        int u = bfs[i];
        for(int v : graph[u]){
            if(v == par[u]) continue;
            par[v] = u;
            bfs.push_back(v);
        }
    }
    for(int i = n - 1; i >= 0; i--){
        int u = bfs[i], p_idx = -1;
        Agg agg = base(u);
        for(int j = 0; j < (int)graph[u].size(); j++){
            int v = graph[u][j];
            if(v == par[u]) p_idx = j;
            else agg = merge_into(agg, dp[v], u, j);
        }
        dp[u] = finalize_merge(agg, u, p_idx);
    }
    for(int u : bfs){
        int deg = graph[u].size();
        edge_dp[u].resize(deg);
        redge_dp[u].resize(deg);
        Agg tot = base(u);
        for (int j = 0; j < deg; j++){
            int v = graph[u][j];
            edge_dp[u][j] = (v == par[u] ? dp[u] : dp[v]);
            tot = merge_into(tot, edge_dp[u][j], u, j);
        }
        root_dp[u] = finalize_merge(tot, u, -1);
    }
    if(deg) {
        struct Frame { int l, r; Agg out; };
        vector<Frame> st;
        st.reserve(deg * 2);
        st.push_back({0, deg - 1, base(u)});
        while(!st.empty()) {
            auto [l, r, out] = st.back();
            st.pop_back();
            if(l == r){
                redge_dp[u][l] = finalize_merge(out, u, l);
                if(graph[u][l] != par[u]) dp[graph[u][l]] = redge_dp[u][l];
                continue;
            }
            int mid = l + ((r - l) >> 1);

```

```

        Agg L_out = out, R_out = out;
        for(int i = mid + 1; i <= r; i++) L_out = merge_into(L_out,
            ↪ edge_dp[u][i], u, i);
        for(int i = 1; i <= mid; i++) R_out = merge_into(R_out, edge_dp[u][i], u,
            ↪ i);
        st.push_back({mid + 1, r, R_out});
        st.push_back({1, mid, L_out});
    }
}
return make_tuple(root_dp, edge_dp, redge_dp);
}

```

SCC.cpp

Use: SCC scc(n); add_edge(u,v), build(), then comp[v] is component id and comps/condensation describe the DAG.

Out: After build(), comp[v] is the component id, groups lists vertices, and dag is the condensed graph.

Complexity: Kosaraju build is $O(n+m)$, then condensation DAG is deduplicated by sorting edges.

Pitfall: Nodes are 1-indexed; component ids are 0..components-1 in the second-pass order.

```

struct SCC{
    int n, components;
    vector<vector<int>> graph, rev, groups, dag;
    vector<int> comp, order, vis;
    SCC(){
        SCC(int _n){
            init(_n);
        }
        void init(int _n){
            n = _n;
            components = 0;
            graph.assign(n + 1, {});
            rev.assign(n + 1, {});
            comp.assign(n + 1, -1);
            vis.assign(n + 1, 0);
            order.clear();
            groups.clear();
            dag.clear();
        }
        void add_edge(int u, int v){
            graph[u].push_back(v);
            rev[v].push_back(u);
        }
        void dfs1(int v){
            vis[v] = 1;
            for (auto u : graph[v]){
                if (!vis[u]) dfs1(u);
            }
            order.push_back(v);
        }
        void dfs2(int v, int id){
            comp[v] = id;
            groups[id].push_back(v);
            for (auto u : rev[v]){
                if (comp[u] == -1) dfs2(u, id);
            }
        }
        void build(){

```

```

            fill(vis.begin(), vis.end(), 0);
            fill(comp.begin(), comp.end(), -1);
            order.clear();
            for (int i = 1; i <= n; i++){
                if (!vis[i]) dfs1(i);
            }
            reverse(order.begin(), order.end());
            components = 0;
            groups.clear();
            for (auto v : order){
                if (comp[v] == -1){
                    groups.push_back({});
                    dfs2(v, components++);
                }
            }
            dag.assign(components, {});
            vector<pair<int, int>> edges;
            for (int v = 1; v <= n; v++){
                for (auto u : graph[v]){
                    if (comp[v] != comp[u]) edges.push_back({comp[v], comp[u]});
                }
            }
            sort(edges.begin(), edges.end());
            edges.erase(unique(edges.begin(), edges.end()), edges.end());
            for (auto [u, v] : edges){
                dag[u].push_back(v);
            }
        }
    };

```

tree isomorphism.cpp

Use: Call tree_hash(graph) for canonical center hashes or isomorphic(a,b) to compare two unrooted trees.

Care: Global ids are shared across calls; that is intended for comparison, but clear ids/ID if you need a fresh hash universe.

Complexity: Hashing is $O(n \log n)$ mainly from sorting child hashes.

Pitfall: The hash ids are deterministic inside one run but not canonical across different programs unless built in same order.

```

map<vector<int>, int> ids;
int ID = 1;

int hash_node(vector<int> ch){
    sort(ch.begin(), ch.end());
    if (!ids.count(ch)) ids[ch] = ID++;
    return ids[ch];
}

int rooted_hash(int v, int p, vector<vector<int>> &graph){
    vector<int> ch;
    for (auto u : graph[v]){
        if (u != p) ch.push_back(rooted_hash(u, v, graph));
    }
    return hash_node(ch);
}

vector<int> centers(vector<vector<int>> &graph){
    int n = graph.size() - 1;
    vector<int> deg(n + 1);
    queue<int> q;
    for (int i = 1; i <= n; i++){

```

```

        deg[i] = graph[i].size();
        if (deg[i] <= 1) q.push(i);
    }
    int left = n;
    while (left > 2){
        int sz = q.size();
        left -= sz;
        while (sz--){
            int v = q.front();
            q.pop();
            for (auto u : graph[v]){
                if (--deg[u] == 1) q.push(u);
            }
        }
    }
    vector<int> ret;
    while (!q.empty()){
        ret.push_back(q.front());
        q.pop();
    }
    return ret;
}
vector<int> tree_hash(vector<vector<int>> &graph){
    vector<int> ret;
    for (auto c : centers(graph)){
        ret.push_back(rooted_hash(c, 0, graph));
    }
    sort(ret.begin(), ret.end());
    return ret;
}
bool isomorphic(vector<vector<int>> &a, vector<vector<int>> &b){
    if (a.size() != b.size()) return false;
    return tree_hash(a) == tree_hash(b);
}

```

Virtual Tree.cpp

Use: Build LCA on the 0-indexed tree, then VirtualTree vt(nodes,lca) fills vt.nodes and weighted vt.edges.

Care: Input nodes may be unsorted; constructor adds needed LCAs, sorts by tin, and stores edges as parent-child-distance triples.

Complexity: Build is $O(k \log k)$ for k marked nodes after LCA preprocessing.

When: Use it when DP only cares about a subset of important nodes and their connecting LCAs.

```

struct LCA{
    int n, Log;
    vector<vector<int>> up;
    vector<int> depth, tin, tout;
    LCA(vector<vector<int>> &g, int root = 0)
        : n((int)g.size()), Log(__lg(n) + 1), up(n, vector<int>(Log, root)), depth(n),
          tin(n), tout(n) {
        int timer = 0;
        function<void(int)> dfs = [&](int u){
            tin[u] = ++timer;
            for (int v : g[u]){
                if (v == up[u][0]) continue;
                depth[v] = depth[u] + 1;
                up[v][0] = u;
                for (int l = 1; l < Log; l++){
                    up[v][l] = up[up[v][l - 1]][l - 1];
                }
            }
            tout[u] = timer;
        };
        dfs(root);
    }
};

```

```

    }
    dfs(v);
}
tout[u] = timer;
};
dfs(root);
}
int kth(int u, int k){
    while (k){
        u = up[u][__builtin_ctz(k)];
        k &= k - 1;
    }
    return u;
}
bool isAnc(int u, int v){
    return tin[u] <= tin[v] && tout[v] <= tout[u];
}
int lca(int u, int v){
    if (depth[u] < depth[v]) swap(u, v);
    u = kth(u, depth[u] - depth[v]);
    if (u == v) return u;
    for (int l = Log - 1; l >= 0; l--){
        if (up[u][l] != up[v][l]){
            u = up[u][l];
            v = up[v][l];
        }
    }
    return up[u][0];
}
int dist(int u, int v){
    int l = lca(u, v);
    return depth[u] + depth[v] - 2 * depth[l];
}
};

struct VirtualTree{
    vector<int> nodes;
    vector<tuple<int, int, int>> edges; // u, v, distance
    VirtualTree(vector<int> subset, LCA &lca){
        if (subset.empty()) return;
        auto cmp = [&](int u, int v){
            return lca.tin[u] < lca.tin[v];
        };
        sort(subset.begin(), subset.end(), cmp);
        int k = subset.size();
        for (int i = 0; i + 1 < k; i++){
            subset.push_back(lca.lca(subset[i], subset[i + 1]));
        }
        sort(subset.begin(), subset.end(), cmp);
        subset.erase(unique(subset.begin(), subset.end()), subset.end());
        nodes = subset;
        vector<int> st;
        st.push_back(subset[0]);
        for (int i = 1; i < (int)subset.size(); i++){
            int u = subset[i];
            while (!st.empty() && !lca.isAnc(st.back(), u)) st.pop_back();
            edges.push_back({st.back(), u, lca.dist(st.back(), u)});
            st.push_back(u);
        }
    }
};

```

```
}  
};
```

4 Math temps

modnum helper

Use: Shared Mint / modnum helper; templates marked with Needs use this MOD/mint code before their snippet.

Before: Change MOD and the mint alias once here; all dependent snippets should use the same modulus.

Pitfall: inv() uses Fermat, so MOD must be prime unless you replace the inverse logic.

```
template<int MOD> struct modnum {
    int v;
    modnum(ll x = 0) : v(x % MOD) { if(v < 0) v += MOD; }
    explicit operator int() const { return v; }
    friend ostream& operator<<(ostream& out, const modnum& n) { return out << n.v; }
    friend istream& operator>>(istream& in, modnum& n) { ll x; in >> x; n = modnum(x);
        ↪ return in; }
    friend bool operator==(const modnum& a, const modnum& b) { return a.v == b.v; }
    friend bool operator!=(const modnum& a, const modnum& b) { return a.v != b.v; }
    modnum pow(ll b) const {
        modnum a = *this, r = 1; while(b) { if(b & 1) r *= a; a *= a; b >>= 1; }
        return r;
    }
    modnum inv() const { return pow(MOD - 2); }
    modnum operator-() const { return modnum(v ? MOD - v : 0); }
    modnum& operator+=(const modnum& o) { v += o.v; if(v >= MOD) v -= MOD; return *this; }
    modnum& operator-=(const modnum& o) { v -= o.v; if(v < 0) v += MOD; return *this; }
    modnum& operator*=(const modnum& o) { v = int(ll(v) * o.v % MOD); return *this; }
    modnum& operator/=(const modnum& o) { return *this *= o.inv(); }
    friend modnum operator+(modnum a, const modnum& b) { return a += b; }
    friend modnum operator-(modnum a, const modnum& b) { return a -= b; }
    friend modnum operator*(modnum a, const modnum& b) { return a *= b; }
    friend modnum operator/(modnum a, const modnum& b) { return a /= b; }
};
```

2D NTT.cpp

Use: conv2d(a,b) pads both dimensions and returns the 2D convolution modulo 998244353; inputs may be rectangular.

Care: The modulus/root are fixed to 998244353 and 3; the 1D dft/idft helpers cache roots and bit-reversal.

Complexity: Build/evaluate cost is $O(H*W*(\log H + \log W))$, where H,W are padded powers of two.

Pitfall: Both dimensions must fit the NTT modulus limit; negative inputs are normalized modulo MOD before transforming.

```
const int MOD = 998244353;
const int root = 3;
using Mat = vector<vector<int>>;

int mod_norm(ll x){ x %= MOD; if (x < 0) x += MOD; return x; }
int addmod(int a, int b){ a += b; if (a >= MOD) a -= MOD; return a; }
int submod(int a, int b){ a -= b; if (a < 0) a += MOD; return a; }
int mulmod(int a, int b){ return 1LL * a * b % MOD; }
int fp(int b, ll p){
    int res = 1;
    while (p){
        if (p & 1) res = mulmod(res, b);
        b = mulmod(b, b);
        p >>= 1;
    }
    return res;
}
```

```
int inv(int x){ return fp(x, MOD - 2); }
bool is_power_two(int n){ return n > 0 && (n & (n - 1)) == 0; }
int next_power_two(int n){
    int p = 1;
    while (p < n) p <<= 1;
    return p;
}
vector<int> roots{0, 1};
void prepare_roots(int n){
    if ((int)roots.size() >= n) return;
    int sz = roots.size();
    roots.resize(n);
    for (int len = sz; len < n; len <<= 1){
        int w = fp(root, (MOD - 1) / (len << 1));
        for (int i = len / 2; i < len; i++){
            roots[2 * i] = roots[i];
            roots[2 * i + 1] = mulmod(roots[i], w);
        }
    }
}
void ntt(vector<int> &a){
    int n = a.size();
    if (n <= 1) return;
    assert(is_power_two(n));
    static vector<int> revs;
    if ((int)revs.size() != n){
        int lg = __builtin_ctz(n);
        revs.assign(n, 0);
        for (int i = 0; i < n; i++){
            revs[i] = (revs[i >> 1] >> 1) | ((i & 1) << (lg - 1));
        }
    }
    prepare_roots(n);
    for (int i = 0; i < n; i++){
        if (i < revs[i]) swap(a[i], a[revs[i]]);
    }
    for (int len = 1; len < n; len <<= 1){
        for (int i = 0; i < n; i += 2 * len){
            for (int j = 0; j < len; j++){
                int u = a[i + j], v = mulmod(a[i + j + len], roots[len + j]);
                a[i + j] = addmod(u, v);
                a[i + j + len] = submod(u, v);
            }
        }
    }
}
void intt(vector<int> &a){
    if ((int)a.size() <= 1) return;
    reverse(a.begin() + 1, a.end());
    ntt(a);
    int iv = inv(a.size());
    for (auto &x : a) x = mulmod(x, iv);
}
void dft(vector<int> &a){
    for (int &x : a) x = mod_norm(x);
    ntt(a);
}
void idft(vector<int> &a){
    intt(a);
```

```

}
void ntt2d(Mat &a, bool inverse = false){
    int n = a.size(), m = a[0].size();
    assert(is_power_two(n) && is_power_two(m));
    assert(n <= (1 << 23) && m <= (1 << 23));
    for (int i = 0; i < n; i++){
        if (inverse) idft(a[i]);
        else dft(a[i]);
    }
    vector<int> col(n);
    for (int j = 0; j < m; j++){
        for (int i = 0; i < n; i++) col[i] = a[i][j];
        if (inverse) idft(col);
        else dft(col);
        for (int i = 0; i < n; i++) a[i][j] = col[i];
    }
}
Mat conv2d(const Mat &a, const Mat &b){
    if (a.empty() || b.empty() || a[0].empty() || b[0].empty()) return {};
    int n = a.size(), m = a[0].size();
    int r = b.size(), c = b[0].size();
    for (int i = 0; i < n; i++) assert((int)a[i].size() == m);
    for (int i = 0; i < r; i++) assert((int)b[i].size() == c);
    int H = next_power_two(n + r - 1);
    int W = next_power_two(m + c - 1);
    Mat A(H, vector<int>(W)), B(H, vector<int>(W));
    for (int i = 0; i < n; i++){
        for (int j = 0; j < m; j++){
            A[i][j] = mod_norm(a[i][j]);
        }
    }
    for (int i = 0; i < r; i++){
        for (int j = 0; j < c; j++){
            B[i][j] = mod_norm(b[i][j]);
        }
    }
    ntt2d(A);
    ntt2d(B);
    for (int i = 0; i < H; i++){
        for (int j = 0; j < W; j++){
            A[i][j] = mulmod(A[i][j], B[i][j]);
        }
    }
    ntt2d(A, true);
    A.resize(n + r - 1);
    for (auto &row : A) row.resize(m + c - 1);
    return A;
}

```

BerlekampMassey.cpp

Use: BerlekampMassey(S) returns recurrence tr; LinearRecurrence(init,tr,k) returns $S[k]$ using the first tr.size() terms.

Care: Provide enough initial terms, usually at least twice the recurrence degree; guess(S,k,ans) validates before answering.

Complexity: Berlekamp-Massey is $O(n^2)$; LinearRecurrence is $O(d^2 \log k)$ for recurrence degree d.

Pitfall: If the sampled prefix is too short, the guessed recurrence may fit samples but fail later.

Needs: Mint / modnum helper above.

```

const int MOD = 998244353;
using mint = modnum<MOD>;
template<typename T>
mint binpow(T aa, ll b){
    assert(b >= 0);
    mint a = aa, r = 1;
    while (b){ if (b & 1) r *= a; a *= a; b >>= 1; }
    return r;
}

vector<mint> BerlekampMassey(vector<mint> S){
    int n = (int)S.size(), L = 0, m = 0;
    vector<mint> C(n), B(n), T;
    C[0] = B[0] = 1;
    mint b = 1;
    for (int i = 0; i < n; i++){
        ++m;
        mint d = S[i];
        for (int j = 1; j <= L; j++) d += C[j] * S[i - j];
        if (d == 0) continue;
        T = C;
        mint coef = d / b;
        for (int j = m; j < n; j++) C[j] -= coef * B[j - m];
        if (2 * L > i) continue;
        L = i + 1 - L; B = T; b = d; m = 0;
    }
    C.resize(L + 1);
    C.erase(C.begin());
    for (auto &x : C) x = -x;
    return C;
}

vector<mint> combine(int n, vector<mint> &a, vector<mint> &b, vector<mint> &tr){
    vector<mint> res(n * 2 + 1, 0);
    for (int i = 0; i < n + 1; i++){
        for (int j = 0; j < n + 1; j++) res[i + j] += a[i] * b[j];
    }
    for (int i = 2 * n; i > n; i--){
        for (int j = 0; j < n; j++) res[i - 1 - j] += res[i] * tr[j];
    }
    res.resize(n + 1);
    return res;
}

mint LinearRecurrence(vector<mint> &S, vector<mint> &tr, ll k){
    int n = S.size();
    assert(n == (int)tr.size());
    if (n == 0) return 0;
    if (k < n) return S[k];
    vector<mint> pol(n + 1), e(n + 1);
    pol[0] = e[1] = 1;
    for (++k; k; k >>= 1){
        if (k & 1) pol = combine(n, pol, e, tr);
        e = combine(n, e, e, tr);
    }
    mint res = 0;
    for (int i = 0; i < n; i++) res += pol[i + 1] * S[i];
    return res;
}

bool guess(vector<mint> S, ll k, mint &ans){

```

```

vector<mint> tr = BerlekampMassey(S);
int d = tr.size();
if (d == 0){
    ans = 0;
    return true;
}
if ((int)S.size() < 2 * d) return false;
for (int i = d; i < (int)S.size(); i++){
    mint cur = 0;
    for (int j = 0; j < d; j++){
        cur += tr[j] * S[i - j - 1];
    }
    if (cur != S[i]) return false;
}
vector<mint> init(S.begin(), S.begin() + d);
ans = LinearRecurrence(init, tr, k);
return true;
}

```

comb.cpp

Use: counting<N> C; call choose(n,k), stars_and_bars(n,k), P(n,k), catalan(n), fact, inv, and invfact under mint MOD.

Care: Set MOD and N before copying; all n used in combinations/catalan must be less than N.

Complexity: Precomputation is O(N), while choose, P, and catalan queries are O(1).

Pitfall: stars_and_bars(n,k) assumes positive boxes; handle k=0 separately if needed.

Needs: Mint / modnum helper above.

```

const int MOD = 998244353;
using mint = modnum<MOD>;
template<typename T>
mint binpow(T aa, ll b){
    assert(b >= 0);
    mint a = aa, r = 1;
    while (b){ if (b & 1) r *= a; a *= a; b >>= 1; }
    return r;
}

template<int N>
struct counting{
    mint fact[N], inv[N], invfact[N];
    counting(){
        fact[0] = invfact[0] = 1;
        for (int i = 1; i < N; i++) fact[i] = fact[i - 1] * i;
        invfact[N - 1] = fact[N - 1].inv();
        for (int i = N - 2; i >= 0; i--) invfact[i] = invfact[i + 1] * (i + 1);
        for (int i = 1; i < N; i++) inv[i] = fact[i - 1] * invfact[i];
    }
    mint choose(int n, int r){
        if (n < 0 || r < 0 || n < r) return 0;
        return fact[n] * invfact[n - r] * invfact[r];
    }
    mint stars_and_bars(int n, int k){
        return choose(n + k - 1, k - 1);
    }
    mint P(int n, int r){
        if (n < 0 || r < 0 || r > n) return 0;
        return fact[n] * invfact[n - r];
    }
}

```

```

mint catalan(int n){
    return fact[2 * n] * invfact[n] * invfact[n + 1];
}
};
const int N = 1e6 + 5;
counting<N> comb;

```

CRT.cpp

Use: CRT({{a,m},...}) returns {x,lcm} for x = a mod m or {0,-1}; mod_inv requires gcd(a,m)=1.

Care: Works for non-coprime moduli using __int128; print results carefully because T is not directly stream-friendly everywhere.

Complexity: Combining equations is O(number_of_equations * log modulus).

Out: intersect(a1,d1,a2,d2) returns the first common value at least max(a1,a2) and the new period.

```

using T = __int128;

T norm(T a, T m){
    a %= m;
    if (a < 0) a += m;
    return a;
}

T exgcd(T a, T b, T &x, T &y){
    T xx = y = 0;
    T yy = x = 1;
    while (b){
        T q = a / b, t = b;
        b = a % b; a = t;
        t = xx;
        xx = x - q * xx;
        x = t; t = yy;
        yy = y - q * yy;
        y = t;
    }
    return a;
}

pair<T, T> CRT(T a1, T m1, T a2, T m2){
    a1 = norm(a1, m1);
    a2 = norm(a2, m2);
    T p, q;
    T g = exgcd(m1, m2, p, q);
    if (a1 % g != a2 % g) return {0, -1};
    T m = m1 / g * m2;
    p = norm(p, m);
    q = norm(q, m);
    T x = (p * a2 % m * (m1 / g) % m + q * a1 % m * (m2 / g) % m) % m;
    return {x, m};
}

pair<T, T> CRT(vector<pair<T, T>> eq){
    pair<T, T> ret = {0, 1};
    for (auto [a, m] : eq){
        ret = CRT(ret.first, ret.second, a, m);
        if (ret.second == -1) return ret;
    }
    return ret;
}

pair<T, T> intersect(T a1, T d1, T a2, T d2){
    auto x = CRT(a1, d1, a2, d2);
    T a = x.first, d = x.second;
}

```

```

    if (d == -1) return {0, 0};
    T st = max(a1, a2);
    if (a < st) a += (st - a + d - 1) / d * d;
    return {a, d};
}

template<typename T>
T mod_inv_in_range(T a, T m){
    T x = a, y = m;
    T vx = 1, vy = 0;
    while (x){
        T k = y / x;
        y %= x;
        vy -= k * vx;
        swap(x, y);
        swap(vx, vy);
    }
    assert(y == 1);
    return vy < 0 ? m + vy : vy;
}

template<typename T>
struct extended_gcd_result{
    T gcd, coeff_a, coeff_b;
};

template<typename T>
extended_gcd_result<T> extended_gcd(T a, T b){
    T x = a, y = b;
    T ax = 1, ay = 0;
    T bx = 0, by = 1;
    while (x){
        T k = y / x;
        y %= x;
        ay -= k * ax;
        by -= k * bx;
        swap(x, y);
        swap(ax, ay);
        swap(bx, by);
    }
    return {y, ay, by};
}

template<typename T>
T mod_inv(T a, T m){
    a %= m;
    if (a < 0) a += m;
    return mod_inv_in_range(a, m);
}

```

Discrete log.cpp

Use: discrete_log(a,b,mod) returns the minimum $x \geq 0$ with $a^x = b \bmod \text{mod}$, or -1 if none.

Care: Handles gcd(a,mod)>1 cases; unordered_map memory is $O(\sqrt{\text{mod}})$, so avoid it for many huge calls.

Complexity: Baby-step giant-step uses $O(\sqrt{\text{mod}})$ time and memory.

Pitfall: For non-coprime cases the function reduces the modulus; test a=0 or b=0 cases separately.

```

ll mul_mod(ll a, ll b, ll mod){
    return (__int128)a * b % mod;
}

ll fp(ll a, ll b, ll mod){
    a %= mod;

```

```

    ll res = 1 % mod;
    while (b){
        if (b & 1) res = mul_mod(res, a, mod);
        a = mul_mod(a, a, mod);
        b >>= 1;
    }
    return res;
}

// returns minimum x >= 0 with a^x = b (mod mod), or -1 if none
ll discrete_log(ll a, ll b, ll mod){
    a %= mod;
    b %= mod;
    if (mod == 1) return 0;
    ll add = 0, k = 1, g;
    while ((g = gcd(a, mod)) > 1){
        if (b == k) return add;
        if (b % g) return -1;
        b /= g;
        mod /= g;
        add++;
        k = mul_mod(k, a / g, mod);
    }
    if (b == k) return add;
    ll n = sqrtl(mod) + 1;
    unordered_map<ll, ll> vals;
    ll cur = b;
    for (ll q = 0; q <= n; q++){
        vals[cur] = q;
        cur = mul_mod(cur, a, mod);
    }
    ll an = fp(a, n, mod);
    cur = k;
    for (ll p = 1; p <= n + 1; p++){
        cur = mul_mod(cur, an, mod);
        auto it = vals.find(cur);
        if (it != vals.end()){
            ll ans = n * p - it->second + add;
            if (ans >= 0) return ans;
        }
    }
    return -1;
}

```

Fast prime sieve.cpp

Use: sieve(N) returns all primes $\leq N$ quickly; tune Q/L only if memory or speed needs it.

Care: Designed for large single limits; for small repeated limits, the simpler sieve in Number_theory may be easier.

Complexity: Optimized for huge N with wheel/blocking; memory is roughly $O(\sqrt{\text{N}} + \text{block})$.

Out: Returns a vector of primes in increasing order.

```

// credit: min_25, about 0.5s for n = 1e9
vector<int> sieve(const int N, const int Q = 17, const int L = 1 << 15) {
    if (N < 2) return {};
    if (N < 7){
        vector<int> primes;
        for (int p : {2, 3, 5}){
            if (p <= N) primes.push_back(p);
        }
    }

```

```

    return primes;
}
static const int rs[] = {1, 7, 11, 13, 17, 19, 23, 29};
auto approx_prime_count = [] (const int N) -> int {
    return N > 60184 ? N / (log(N) - 1.1)
        : max(1., N / (log(N) - 1.11)) + 1;
};

const int v = sqrt(N), vv = sqrt(v);
vector<bool> isp(v + 1, true);
if (v >= 0) isp[0] = false;
if (v >= 1) isp[1] = false;
for (int i = 2; i <= vv; ++i) if (isp[i]) {
    for (int j = i * i; j <= v; j += i) isp[j] = false;
}

const int rsize = approx_prime_count(N + 30);
vector<int> primes = {2, 3, 5};
int psize = 3;
primes.resize(rsize);

vector<int> sprimes_p;
vector<array<int, 8>> sprimes_pos;
size_t pbeg = 0;
int prod = 1;
for (int p = 7; p <= v; ++p) {
    if (!isp[p]) continue;
    if (p <= Q) prod *= p, ++pbeg, primes[psize++] = p;
    array<int, 8> pos;
    for (int t = 0; t < 8; ++t) {
        int j = (p <= Q) ? p : p * p;
        while (j % 30 != rs[t]) j += p << 1;
        pos[t] = j / 30;
    }
    sprimes_p.push_back(p);
    sprimes_pos.push_back(pos);
}

vector<unsigned char> pre(prod, 0xFF);
for (size_t pi = 0; pi < pbeg; ++pi) {
    const int p = sprimes_p[pi];
    for (int t = 0; t < 8; ++t) {
        const unsigned char m = ~(1 << t);
        for (int i = sprimes_pos[pi][t]; i < prod; i += p) pre[i] &= m;
    }
}

const int block_size = (L + prod - 1) / prod * prod;
vector<unsigned char> block(block_size);
unsigned char* pblock = block.data();
const int M = (N + 29) / 30;

for (int beg = 0; beg < M; beg += block_size, pblock -= block_size) {
    int end = min(M, beg + block_size);
    for (int i = beg; i < end; i += prod) {
        copy(pre.begin(), pre.end(), pblock + i);
    }
    if (beg == 0) pblock[0] &= 0xFE;
    for (size_t pi = pbeg; pi < sprimes_p.size(); ++pi) {

```

```

        const int p = sprimes_p[pi];
        for (int t = 0; t < 8; ++t) {
            int i = sprimes_pos[pi][t];
            const unsigned char m = ~(1 << t);
            for (; i < end; i += p) pblock[i] &= m;
            sprimes_pos[pi][t] = i;
        }
    }
    for (int i = beg; i < end; ++i) {
        for (int m = pblock[i]; m > 0; m &= m - 1) {
            primes[psize++] = i * 30 + rs[__builtin_ctz(m)];
        }
    }
}
assert(psize <= rsize);
while (psize > 0 && primes[psize - 1] > N) --psize;
primes.resize(psize);
return primes;
}

```

FFT string_matching.cpp

Use: conv(a,b) multiplies integer vectors; wildcard_matching(s,t) returns starts where pattern t matches text s with '?' wildcards.

Out: wildcard_matching(text,pattern) returns all starts where every non-'?' aligned character matches.

Complexity: Convolutions cost $O(n \log n)$, with three convolutions for wildcard matching.

Pitfall: The matching formula assumes lowercase letters mapped to positive values and '?' as wildcard.

```

using cd = complex<double>;
const double PI = acos(-1);

int next_power_two(int n){
    int p = 1;
    while (p < n) p <= 1;
    return p;
}

vector<cd> fft_roots{0, 1};
void prepare_roots(int n){
    if ((int)fft_roots.size() >= n) return;
    int old_size = fft_roots.size();
    fft_roots.resize(n);
    for (int len = old_size; len < n; len <= 1){
        cd w = polar(1.0, PI / len);
        for (int i = len / 2; i < len; i++){
            fft_roots[2 * i] = fft_roots[i];
            fft_roots[2 * i + 1] = fft_roots[i] * w;
        }
    }
}

void fft(vector<cd> &a){
    int n = a.size();
    if (n <= 1) return;
    assert((n & (n - 1)) == 0);
    static vector<int> cached_rev;
    if ((int)cached_rev.size() != n){
        int lg = __builtin_ctz(n);
        cached_rev.assign(n, 0);
    }
}

```

```

        for (int i = 0; i < n; i++){
            cached_rev[i] = (cached_rev[i >> 1] >> 1) | ((i & 1) << (lg - 1));
        }
    }
    prepare_roots(n);
    for (int i = 0; i < n; i++){
        if (i < cached_rev[i]) swap(a[i], a[cached_rev[i]]);
    }
    for (int len = 1; len < n; len <= 1){
        for (int i = 0; i < n; i += 2 * len){
            for (int j = 0; j < len; j++){
                cd z = fft_roots[len + j] * a[i + j + len], &ai = a[i + j];
                a[i + j + len] = ai - z;
                ai += z;
            }
        }
    }
}

void ifft(vector<cd> &a){
    if ((int)a.size() <= 1) return;
    reverse(a.begin() + 1, a.end());
    fft(a);
    for (cd &x : a) x /= (double)a.size();
}

void dft(vector<cd> &a){
    fft(a);
}

void idft(vector<cd> &a){
    ifft(a);
}

vector<ll> conv(const vector<ll> &a, const vector<ll> &b){
    if (a.empty() || b.empty()) return {};
    int s = a.size() + b.size() - 1;
    int n = next_power_two(s);
    vector<cd> L(n), R(n);
    for (int i = 0; i < (int)a.size(); i++) L[i] = a[i];
    for (int i = 0; i < (int)b.size(); i++) R[i] = b[i];
    dft(L);
    dft(R);
    for (int i = 0; i < n; i++) L[i] *= R[i];
    idft(L);
    vector<ll> ret(s);
    for (int i = 0; i < s; i++){
        ret[i] = llround(real(L[i]));
    }
    return ret;
}

// match iff sum s*t*(s-t)^2 over aligned non-'?' chars is 0
vector<int> wildcard_matching(const string &s, const string &t){
    int n = s.size(), m = t.size();
    if (m > n) return {};
    vector<ll> s1(n), s2(n), s3(n), t1(m), t2(m), t3(m);
    for (int i = 0; i < n; i++){
        s1[i] = s[i] == '?' ? 0 : s[i] - 'a' + 1;

```

```

        s2[i] = s1[i] * s1[i];
        s3[i] = s1[i] * s2[i];
    }
    for (int i = 0; i < m; i++){
        t1[i] = t[i] == '?' ? 0 : t[i] - 'a' + 1;
        t2[i] = t1[i] * t1[i];
        t3[i] = t1[i] * t2[i];
    }
    reverse(t1.begin(), t1.end());
    reverse(t2.begin(), t2.end());
    reverse(t3.begin(), t3.end());
    vector<ll> a = conv(s1, t3);
    vector<ll> b = conv(s2, t2);
    vector<ll> c = conv(s3, t1);
    vector<int> ret;
    for (int i = m - 1; i < n; i++){
        if (a[i] - 2 * b[i] + c[i] == 0) ret.push_back(i - m + 1);
    }
    return ret;
}

```

FFT with MOD.cpp

Use: `conv<M>(a,b)` returns convolution modulo M using FFT splitting; inputs are integer coefficient vectors.
 Care: Use this when target modulus is not NTT-friendly; coefficients are reduced modulo template parameter M.

Complexity: Convolution is $O(n \log n)$, with coefficient splitting to control precision under modulus M.

Pitfall: Still uses doubles, so for extreme lengths or coefficients compare with CRT/NTT if suspicious.

```

using C = complex<double>;
using ld = long double;
const ld PI = acosl(-1.0L);
int nxt_pw(int n){
    int p = 1;
    while (p < n) p <= 1;
    return p;
}

vector<complex<ld>> roots_ld{0, 1};
vector<C> roots{0, 1};
void prepare_roots(int n){
    if ((int)roots.size() >= n) return;
    int old_size = roots.size();
    roots_ld.resize(n);
    roots.resize(n);
    for (int len = old_size; len < n; len <= 1){
        auto w = polar(1.0L, PI / len);
        for (int i = len / 2; i < len; i++){
            roots_ld[2 * i] = roots_ld[i];
            roots_ld[2 * i + 1] = roots_ld[i] * w;
            roots[2 * i] = C((double)roots_ld[2 * i].real(), (double)roots_ld[2 * i]
                ↪ i].imag());
            roots[2 * i + 1] = C((double)roots_ld[2 * i + 1].real(), (double)roots_ld[2 * i
                ↪ i + 1].imag());
        }
    }
}

void fft(vector<C> &a){

```

```

int n = a.size();
if (n <= 1) return;
assert((n & (n - 1)) == 0);
static vector<int> revs;
if ((int)revs.size() != n){
    int lg = __builtin_ctz(n);
    revs.assign(n, 0);
    for (int i = 0; i < n; i++){
        revs[i] = (revs[i >> 1] >> 1) | ((i & 1) << (lg - 1));
    }
}
prepare_roots(n);
for (int i = 0; i < n; i++){
    if (i < revs[i]) swap(a[i], a[revs[i]]);
}
for (int len = 1; len < n; len <= 1){
    for (int i = 0; i < n; i += 2 * len){
        for (int j = 0; j < len; j++){
            auto x = (double *)&roots[j + len], y = (double *)&a[i + j + len];
            C z(x[0] * y[0] - x[1] * y[1], x[0] * y[1] + x[1] * y[0]);
            a[i + j + len] = a[i + j] - z;
            a[i + j] += z;
        }
    }
}
}

void ifft(vector<C> &a){
    if ((int)a.size() <= 1) return;
    reverse(a.begin() + 1, a.end());
    fft(a);
    for (C &x : a) x /= (double)a.size();
}

template<int M>
vector<int> conv(const vector<int> &a, const vector<int> &b){
    if (a.empty() || b.empty()) return {};
    vector<int> res(a.size() + b.size() - 1);
    int n = nxt_pw(res.size()), cut = sqrt(M);
    vector<C> L(n), R(n), outl(n), outs(n);
    for (int i = 0; i < (int)a.size(); i++){
        ll x = a[i] % M;
        if (x < 0) x += M;
        L[i] = C((int)x / cut, (int)x % cut);
    }
    for (int i = 0; i < (int)b.size(); i++){
        ll x = b[i] % M;
        if (x < 0) x += M;
        R[i] = C((int)x / cut, (int)x % cut);
    }
    fft(L); fft(R);
    C I(0, 1);
    for (int i = 0; i < n; i++){
        int j = -i & (n - 1);
        outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
        outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / I;
    }
    fft(outl); fft(outs);
    for (int i = 0; i < (int)res.size(); i++){
        ll av = ll(real(outl[i]) + 0.5), cv = ll(imag(outs[i]) + 0.5);
        ll bv = ll(imag(outl[i]) + 0.5) + ll(real(outs[i]) + 0.5);

```

```

        res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
    }
    return res;
}

```

FFT.cpp

Use: conv(a,b) returns integer convolution of vectors a and b using doubles, best when coefficients/results fit rounding safety.

Care: Rounding is the danger point; split coefficients or use NTT/CRT when products can exceed double precision safety.

Complexity: Convolution is $O(n \log n)$ after padding to the next power of two.

Pitfall: Negative coefficients are fine, but very large absolute values can break llround accuracy.

```

using cd = complex<double>;
const double PI = acos(-1);

int nxt_pw(int n){
    int p = 1;
    while (p < n) p <= 1;
    return p;
}

vector<cd> roots{0, 1};
void prepare_roots(int n){
    if ((int)roots.size() >= n) return;
    int old_size = roots.size();
    roots.resize(n);
    for (int len = old_size; len < n; len <= 1){
        cd w = polar(1.0, PI / len);
        for (int i = len / 2; i < len; i++){
            roots[2 * i] = roots[i];
            roots[2 * i + 1] = roots[i] * w;
        }
    }
}

void fft(vector<cd> &a){
    int n = a.size();
    if (n <= 1) return;
    assert((n & (n - 1)) == 0);
    static vector<int> revs;
    if ((int)revs.size() != n){
        int lg = __builtin_ctz(n);
        revs.assign(n, 0);
        for (int i = 0; i < n; i++){
            revs[i] = (revs[i >> 1] >> 1) | ((i & 1) << (lg - 1));
        }
    }
    prepare_roots(n);
    for (int i = 0; i < n; i++){
        if (i < revs[i]) swap(a[i], a[revs[i]]);
    }
    for (int len = 1; len < n; len <= 1){
        for (int i = 0; i < n; i += 2 * len){
            for (int j = 0; j < len; j++){
                cd z = roots[len + j] * a[i + j + len], &ai = a[i + j];
                a[i + j + len] = ai - z;
                ai += z;
            }
        }
    }
}

```

```

    }
}

void ifft(vector<cd> &a){
    if ((int)a.size() <= 1) return;
    reverse(a.begin() + 1, a.end());
    fft(a);
    for (cd &x : a) x /= (double)a.size();
}

vector<ll> conv(const vector<ll> &a, const vector<ll> &b){
    if (a.empty() || b.empty()) return {};
    int s = a.size() + b.size() - 1;
    int n = nxt_pw(s);
    vector<cd> L(n), R(n);
    for (int i = 0; i < (int)a.size(); i++) L[i] = a[i];
    for (int i = 0; i < (int)b.size(); i++) R[i] = b[i];
    fft(L); fft(R);
    for (int i = 0; i < n; i++) L[i] *= R[i];
    ifft(L);
    vector<ll> ret(s);
    for (int i = 0; i < s; i++){
        ret[i] = llround(real(L[i]));
    }
    return ret;
}

```

FWHT.cpp

Use: Vectors must have power-of-two size; transforms mutate in place and convolution helpers return a new vector.

Apps: xor_convolution/and_convolution/or_convolution combine by bitwise operation; subset_convolution combines masks by disjoint submasks.

Care: Transforms mutate arrays in place; pass copies if you still need the original vectors after convolution.

Complexity: xor/and/or transforms are $O(n \log n)$; subset_convolution is $O(n \log^2 n)$ by mask layers.

Pitfall: For modular arithmetic, T must support +, -, *, and division by n for inverse xor.

```

bool is_power_two(int n){
    return n > 0 && (n & (n - 1)) == 0;
}

template<typename T>
void xor_transform(vector<T> &a, bool inv = false){
    int n = a.size();
    assert(is_power_two(n));
    for (int len = 1; len < n; len <= 1){
        for (int i = 0; i < n; i += 2 * len){
            for (int j = 0; j < len; j++){
                T x = a[i + j], y = a[i + j + len];
                a[i + j] = x + y;
                a[i + j + len] = x - y;
            }
        }
    }
    if (inv){
        T iv = T(n);
        for (auto &x : a) x /= iv;
    }
}

template<typename T>

```

```

void and_transform(vector<T> &a, bool inv = false){
    int n = a.size();
    assert(is_power_two(n));
    for (int len = 1; len < n; len <= 1){
        for (int i = 0; i < n; i += 2 * len){
            for (int j = 0; j < len; j++){
                if (inv) a[i + j] -= a[i + j + len];
                else a[i + j] += a[i + j + len];
            }
        }
    }
}

template<typename T>
void or_transform(vector<T> &a, bool inv = false){
    int n = a.size();
    assert(is_power_two(n));
    for (int len = 1; len < n; len <= 1){
        for (int i = 0; i < n; i += 2 * len){
            for (int j = 0; j < len; j++){
                if (inv) a[i + j + len] -= a[i + j];
                else a[i + j + len] += a[i + j];
            }
        }
    }
}

template<typename T>
vector<T> xor_convolution(vector<T> a, vector<T> b){
    assert(a.size() == b.size());
    xor_transform(a);
    xor_transform(b);
    for (int i = 0; i < (int)a.size(); i++) a[i] *= b[i];
    xor_transform(a, true);
    return a;
}

template<typename T>
vector<T> and_convolution(vector<T> a, vector<T> b){
    assert(a.size() == b.size());
    and_transform(a);
    and_transform(b);
    for (int i = 0; i < (int)a.size(); i++) a[i] *= b[i];
    and_transform(a, true);
    return a;
}

template<typename T>
vector<T> or_convolution(vector<T> a, vector<T> b){
    assert(a.size() == b.size());
    or_transform(a);
    or_transform(b);
    for (int i = 0; i < (int)a.size(); i++) a[i] *= b[i];
    or_transform(a, true);
    return a;
}

template<typename T>
vector<T> subset_convolution(vector<T> a, vector<T> b){
    assert(a.size() == b.size() && is_power_two(a.size()));
    int n = a.size(), lg = __builtin_ctz(n);
    vector<vector<T>> A(lg + 1, vector<T>(n)), B(lg + 1, vector<T>(n));
    for (int mask = 0; mask < n; mask++){
        A[__builtin_popcount(mask)][mask] = a[mask];

```

```

        B[__builtin_popcount(mask)][mask] = b[mask];
    }
    for (int i = 0; i <= lg; i++){
        or_transform(A[i]);
        or_transform(B[i]);
    }
    vector<T> tmp(n), ret(n);
    for (int k = 0; k <= lg; k++){
        fill(tmp.begin(), tmp.end(), T(0));
        for (int i = 0; i <= k; i++){
            for (int mask = 0; mask < n; mask++){
                tmp[mask] += A[i][mask] * B[k - i][mask];
            }
        }
        or_transform(tmp, true);
        for (int mask = 0; mask < n; mask++){
            if (__builtin_popcount(mask) == k) ret[mask] = tmp[mask];
        }
    }
    return ret;
}

```

Gaussian with Doubles.cpp

Use: gauss(a,ans) solves linear equations over doubles and returns 0 none, 1 unique, or INF many solutions.

Care: Matrix a is augmented with b as the last column; tune EPS for geometry or near-singular systems.

Complexity: Gaussian elimination is $O(\text{rows} * \text{cols}^2)$ for an augmented matrix.

Pitfall: Floating comparisons depend on EPS; too small or too large EPS changes rank decisions.

```

const double EPS = 1e-9;
const int INF = 2;
int gauss(vector<vector<double>> a, vector<double> &ans) {
    int n = a.size(), m = (int)a[0].size() - 1;
    vector<int> where(m, -1);
    for (int col = 0, row = 0; col < m && row < n; col++) {
        int sel = row;
        for (int i = row; i < n; i++) {
            if (abs(a[i][col]) > abs(a[sel][col])) sel = i;
        }
        if (abs(a[sel][col]) < EPS) continue;
        for (int j = col; j <= m; j++) swap(a[sel][j], a[row][j]);
        where[col] = row;

        for (int i = 0; i < n; i++) {
            if (i != row) {
                double c = a[i][col] / a[row][col];
                for (int j = col; j <= m; j++) {
                    a[i][j] -= a[row][j] * c;
                }
            }
        }
        row++;
    }
    ans.assign(m, 0);
    for (int i = 0; i < m; i++) {
        if (where[i] != -1) ans[i] = a[where[i]][m] / a[where[i]][i];
    }
    for (int i = 0; i < n; i++) {
        double sum = 0;

```

```

        for (int j = 0; j < m; j++) sum += ans[j] * a[i][j];
        if (abs(sum - a[i][m]) > EPS) return 0;
    }
    for (int i = 0; i < m; i++) {
        if (where[i] == -1) return INF;
    }
    return 1;
}

```

Gaussian with MOD.cpp

Use: Set MOD; Gauss(a,ans) solves $Ax=b$, determinant(a) returns det, and matrix_rank(a) returns rank modulo MOD.

Out: Gauss returns -1 for no solution, otherwise the number of free variables and one solution in ans.

Complexity: Gauss, determinant, and matrix_rank are $O(n*m*\min(n,m))$ style elimination.

Pitfall: MOD should be prime because inverses use binpow(x, MOD-2).

```

const ll MOD = 1e9 + 7;
int norm(ll x){
    x %= MOD;
    if (x < 0) x += MOD;
    return x;
}
ll binpow(ll a, ll b){
    ll res = 1;
    while(b){
        if(b & 1) res = (res * a) % MOD;
        a = (a * a) % MOD;
        b >>= 1;
    }
    return res;
}
int Gauss(vector<vector<int>> a, vector<int> &ans){
    int n = a.size(), m = (int)a[0].size() - 1;
    vector<int> pos(m, -1);
    int free_var = 0;
    const ll MODSQ = (ll)MOD * MOD;
    int det = 1, rank = 0;
    for (int col = 0, row = 0; col < m && row < n; col++) {
        int mx = row;
        for (int k = row; k < n; k++) if (a[k][col] > a[mx][col]) mx = k;
        if (a[mx][col] == 0) {det = 0; continue;}
        for (int j = col; j <= m; j++) swap(a[mx][j], a[row][j]);
        if (row != mx) det = det == 0 ? 0 : MOD - det;
        det = 1LL * det * a[row][col] % MOD;
        pos[col] = row;
        int inv = binpow(a[row][col], MOD - 2);
        for (int i = 0; i < n && inv; i++){
            if (i != row && a[i][col]) {
                int x = ((ll)a[i][col] * inv) % MOD;
                for (int j = col; j <= m && x; j++){
                    if (a[row][j]) a[i][j] = (MODSQ + a[i][j] - ((ll)a[row][j] * x)) % MOD;
                }
            }
        }
        row++; ++rank;
    }
    ans.assign(m, 0);

```

```

    for (int i = 0; i < m; i++){
        if (pos[i] == -1) free_var++;
        else ans[i] = ((ll)a[pos[i]][m] * binpow(a[pos[i]][i], MOD - 2)) % MOD;
    }
    for (int i = 0; i < n; i++) {
        ll val = 0;
        for (int j = 0; j < m; j++) val = (val + ((ll)ans[j] * a[i][j])) % MOD;
        if (val != a[i][m]) return -1; //no solution
    }
    return free_var; //has solution
}

int determinant(vector<vector<int>> a){
    int n = a.size();
    if (n == 0) return 1;
    assert((int)a[0].size() == n);
    int det = 1;
    for (int i = 0; i < n; i++){
        for (int j = 0; j < n; j++) a[i][j] = norm(a[i][j]);
    }
    for (int col = 0, row = 0; col < n; col++, row++){
        int sel = row;
        while (sel < n && a[sel][col] == 0) sel++;
        if (sel == n) return 0;
        if (sel != row){
            swap(a[sel], a[row]);
            det = norm(-det);
        }
        det = 1LL * det * a[row][col] % MOD;
        int inv = binpow(a[row][col], MOD - 2);
        for (int i = row + 1; i < n; i++){
            int c = 1LL * a[i][col] * inv % MOD;
            for (int j = col; j < n; j++){
                a[i][j] = norm(a[i][j] - 1LL * a[row][j] * c);
            }
        }
    }
    return det;
}

int matrix_rank(vector<vector<int>> a){
    int n = a.size();
    if (n == 0) return 0;
    int m = a[0].size(), rank = 0;
    for (int i = 0; i < n; i++){
        for (int j = 0; j < m; j++) a[i][j] = norm(a[i][j]);
    }
    for (int col = 0, row = 0; col < m && row < n; col++){
        int sel = row;
        while (sel < n && a[sel][col] == 0) sel++;
        if (sel == n) continue;
        swap(a[sel], a[row]);
        int inv = binpow(a[row][col], MOD - 2);
        for (int i = row + 1; i < n; i++){
            int c = 1LL * a[i][col] * inv % MOD;
            for (int j = col; j < m; j++){
                a[i][j] = norm(a[i][j] - 1LL * a[row][j] * c);
            }
        }
        row++;
        rank++;
    }
}

```

```

    }
    return rank;
}

```

Kth root modulo.cpp

Use: kth_root_mod(a,k,p) returns one x with $x^k = a \pmod{p}$, or -1; kth_roots_mod returns all roots sorted.

Care: This version assumes p is prime; for $a=0$ the only returned root is 0, and many roots are possible when $\gcd(k, p-1) > 1$.

Complexity: primitive_root factors $p-1$ by trial division; discrete log is $O(\sqrt{p})$ memory/time for one query.

Before: Use only when the modulus is prime. For many queries under the same p , cache the primitive root or adapt the file.

Out: kth_roots_mod solves $kz = \log_g(a) \pmod{p-1}$, then maps exponents back by g^z and sorts unique roots.

```

ll normalize(ll x, ll mod){
    x %= mod;
    if (x < 0) x += mod;
    return x;
}

ll mult(ll a, ll b, ll mod){
    return (__int128)normalize(a, mod) * normalize(b, mod) % mod;
}

ll fp(ll a, ll b, ll mod){
    a = normalize(a, mod);
    ll res = 1 % mod;
    while (b){
        if (b & 1) res = mult(res, a, mod);
        a = mult(a, a, mod);
        b >>= 1;
    }
    return res;
}

ll exgcd(ll a, ll b, ll &x, ll &y){
    if (b == 0){
        x = 1;
        y = 0;
        return a;
    }
    ll x1, y1;
    ll g = exgcd(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return g;
}

ll inv_mod(ll a, ll mod){
    ll x, y;
    ll g = exgcd(a, mod, x, y);
    assert(abs(g) == 1);
    return normalize(x, mod);
}

// assumes p is prime
ll primitive_root(ll p){
    if (p == 2) return 1;
    ll phi = p - 1, x = phi;
    vector<ll> fact;
    for (ll d = 2; d <= x / d; d++){

```

```

    if (x % d == 0){
        fact.push_back(d);
        while (x % d == 0) x /= d;
    }
}
if (x > 1) fact.push_back(x);
for (ll g = 2; ; g++){
    bool ok = true;
    for (ll q : fact){
        if (fp(g, phi / q, p) == 1){
            ok = false;
            break;
        }
    }
    if (ok) return g;
}
}

// assumes gcd(a, mod) = 1; returns minimum x >= 0 with a^x = b mod mod, or -1
ll discrete_log_coprime(ll a, ll b, ll mod){
    a = normalize(a, mod);
    b = normalize(b, mod);
    if (mod == 1) return 0;
    ll n = sqrtl(mod - 1) + 1;
    unordered_map<ll, ll> baby;
    baby.reserve(n * 2 + 1);
    ll cur = 1 % mod;
    for (ll j = 0; j < n; j++){
        if (!baby.count(cur)) baby[cur] = j;
        cur = mult(cur, a, mod);
    }
    ll factor = inv_mod(fp(a, n, mod), mod);
    cur = b;
    for (ll i = 0; i <= n; i++){
        auto it = baby.find(cur);
        if (it != baby.end()){
            ll ans = i * n + it->second;
            if (ans < mod) return ans;
        }
        cur = mult(cur, factor, mod);
    }
    return -1;
}

// returns all x with x^k = a mod prime p; may be large if gcd(k, p - 1) is large
vector<ll> kth_roots_mod(ll a, ll k, ll p){
    assert(k >= 1 && p >= 2);
    a = normalize(a, p);
    if (p == 2) return {a};
    if (a == 0) return {0};
    ll g = primitive_root(p);
    ll y = discrete_log_coprime(g, a, p);
    if (y == -1) return {};
    ll order = p - 1, d = gcd(k, order);
    if (y % d != 0) return {};
    ll kk = k / d, yy = y / d, mod = order / d;
    ll z = 0;
    if (mod > 1) z = mult(yy, inv_mod(kk % mod, mod), mod);
    vector<ll> roots;

```

```

    roots.reserve(d);
    for (ll i = 0; i < d; i++){
        roots.push_back(fp(g, z + i * mod, p));
    }
    sort(roots.begin(), roots.end());
    roots.erase(unique(roots.begin(), roots.end()), roots.end());
    return roots;
}

// returns one root, or -1 if no root exists
ll kth_root_mod(ll a, ll k, ll p){
    auto roots = kth_roots_mod(a, k, p);
    return roots.empty() ? -1 : roots[0];
}

```

lagrange interpolation.cpp

Use: interpolate(y,x,first,step) evaluates equally spaced points, interpolate_brute(points,x) arbitrary points, sum_pows(n,k) gives $\sum i^k$.

Care: interpolate(y,x,first,step) assumes $y[i]=f(\text{first}+i*\text{step})$; step must be nonzero modulo MOD.

Complexity: Equally spaced interpolation is $O(n)$; brute interpolation is $O(n^2)$.

When: Use sum_pows when you need $1^k + \dots + n^k$ for huge n under MOD.

Needs: Mint / modnum helper above.

```

const int MOD = 998244353;
using mint = modnum<MOD>;
template<typename T>
mint binpow(T aa, ll b){
    assert(b >= 0);
    mint a = aa, r = 1;
    while (b){
        if (b & 1) r *= a; a *= a; b >>= 1;
    }
    return r;
}

// y[i] = f(first + i * step)
mint interpolate(const vector<mint> &y, ll x, mint first = 0, mint step = 1){
    int n = y.size() - 1;
    assert(n >= 0 && step != 0);
    if (first == 0 && step == 1 && 0 <= x && x <= n) return y[x];
    vector<mint> pref(n + 1, 1), suff(n + 1, 1), fact(n + 1, 1), finv(n + 1, 1);
    for (int i = 1; i <= n; i++){
        pref[i] = pref[i - 1] * (mint(x) - (first + mint(i - 1) * step));
        fact[i] = fact[i - 1] * step * i;
    }
    finv[n] = fact[n].inv();
    for (int i = n - 1; i >= 0; i--){
        suff[i] = suff[i + 1] * (mint(x) - (first + mint(i + 1) * step));
        finv[i] = finv[i + 1] * (i + 1) * step;
    }
    mint ans = 0;
    for (int i = 0; i <= n; i++){
        mint cur = y[i] * pref[i] * suff[i] * finv[i] * finv[n - i];
        if ((n - i) & 1) ans -= cur;
        else ans += cur;
    }
    return ans;
}

```

```

mint interpolate_brute(const vector<pair<mint, mint>> &p, mint x){
    int n = p.size();
    mint ans = 0;
    for (int i = 0; i < n; i++){
        mint num = 1, den = 1;
        for (int j = 0; j < n; j++){
            if (i == j) continue;
            num *= x - p[j].first;
            den *= p[i].first - p[j].first;
        }
        ans += p[i].second * num / den;
    }
    return ans;
}

mint sum_pows(ll n, int k){
    vector<mint> y(k + 2);
    for (int i = 1; i <= k + 1; i++){
        y[i] = y[i - 1] + binpow(i, k);
    }
    return interpolate(y, n);
}

```

matrix_expo.cpp

Use: Mat<mint> A(r,c); set A.a, then mul(A,B) or matpow(A,e) for matrix multiplication/exponentiation.

Care: matpow requires a square matrix; mul_vec applies a matrix to a state vector after exponentiation.

Complexity: matpow is $O(\text{dim}^3 \log \text{exponent})$ with the provided dense multiplication.

Pitfall: Use sparse/custom multiplication for very large states; this helper is for small to medium matrices.

Needs: Mint / modnum helper above.

```

const int MOD = 998244353;
using mint = modnum<MOD>;
template<typename T>
mint binpow(T aa, ll b){
    assert(b >= 0);
    mint a = aa, r = 1;
    while (b){
        if (b & 1) r *= a; a *= a; b >>= 1;
    }
    return r;
}

template<typename T>
struct Mat{
    int r, c;
    vector<vector<T>> a;
    Mat(int r = 0, int c = 0) : r(r), c(c), a(r, vector<T>(c)) {}
    static Mat iden(int n){
        Mat res(n, n);
        for (int i = 0; i < n; i++) res.a[i][i] = 1;
        return res;
    }
    vector<T> mul_vec(const vector<T> &v) const {
        assert(c == (int)v.size());
        vector<T> res(r);
        for (int i = 0; i < r; i++){
            for (int j = 0; j < c; j++){
                res[i] += a[i][j] * v[j];
            }
        }
    }
}

```

```

    }
    return res;
}

};

template<typename T>
Mat<T> mul(const Mat<T> &a, const Mat<T> &b){
    assert(a.c == b.r);
    Mat<T> res(a.r, b.c);
    for (int i = 0; i < a.r; i++){
        for (int k = 0; k < a.c; k++){
            if (a.a[i][k] == T(0)) continue;
            for (int j = 0; j < b.c; j++){
                res.a[i][j] += a.a[i][k] * b.a[k][j];
            }
        }
    }
    return res;
}

template<typename T>
Mat<T> matpow(Mat<T> a, ll b){
    assert(a.r == a.c && b >= 0);
    Mat<T> res = Mat<T>::iden(a.r);
    while (b){
        if (b & 1) res = mul(res, a);
        a = mul(a, a);
        b >>= 1;
    }
    return res;
}

```

NTT.cpp

Use: conv(a,b) pads and multiplies modulo 998244353; dft(a)/idft(a) mutate power-of-two arrays, and all poly_* return trimmed prefixes.

Apps: poly_inv needs $F[0] \neq 0$; poly_ln needs $F[0] = 1$; poly_exp needs $A[0] = 0$; poly_pow/pow2 exponentiate; poly_sqrt returns empty if no sqrt.

Care: dft normalizes inputs and idft performs the reverse/scale step; raw dft/idft still require power-of-two length.

Complexity: dft/idft are $O(n \log n)$; conv is $O(s \log s)$ after padding $s = a.size() + b.size() - 1$ to a power of two.

Before: Call conv for normal multiplication; use dft/idft only when you manually keep vectors already padded to power-of-two length.

Pitfall: Roots and bit-reversal are cached; modulus/root are fixed to 998244353 and 3 unless you rebuild the whole setup.

```

const int MOD = 998244353;
const int root = 3;

int normalize(ll x){ x %= MOD; if (x < 0) x += MOD; return x; }
int add(int a, int b){ a += b; if (a >= MOD) a -= MOD; return a; }
int sub(int a, int b){ a -= b; if (a < 0) a += MOD; return a; }
int mult(int a, int b){ return 1LL * a * b % MOD; }
int fp(int b, ll p){
    int res = 1;
    while (p){
        if (p & 1) res = mult(res, b);
        b = mult(b, b);
        p >>= 1;
    }
}

```

```

    return res;
}
int inv(int x){ return fp(x, MOD - 2); }
int nxt_pw(int n){
    int p = 1;
    while (p < n) p <<= 1;
    return p;
}
vector<int> roots{0, 1};
void prepare_roots(int n){
    if ((int)roots.size() >= n) return;
    int sz = roots.size();
    roots.resize(n);
    for (int len = sz; len < n; len <<= 1){
        int w = fp(root, (MOD - 1) / (len << 1));
        for (int i = len / 2; i < len; i++){
            roots[2 * i] = roots[i];
            roots[2 * i + 1] = mult(roots[i], w);
        }
    }
}
void ntt(vector<int> &a){
    int n = a.size();
    if (n <= 1) return;
    assert((n & (n - 1)) == 0);
    static vector<int> revs;
    if ((int)revs.size() != n){
        int lg = __builtin_ctz(n);
        revs.assign(n, 0);
        for(int i = 0; i < n; i++){
            revs[i] = (revs[i >> 1] >> 1) | ((i & 1) << (lg - 1));
        }
    }
    prepare_roots(n);
    for(int i = 0; i < n; i++) if (i < revs[i]) swap(a[i], a[revs[i]]);
    for(int len = 1; len < n; len <<= 1){
        for (int i = 0; i < n; i += 2 * len){
            for (int j = 0; j < len; j++){
                int u = a[i + j], v = mult(a[i + j + len], roots[len + j]);
                a[i + j] = add(u, v);
                a[i + j + len] = sub(u, v);
            }
        }
    }
}
void intt(vector<int> &a){
    if ((int)a.size() <= 1) return;
    reverse(a.begin() + 1, a.end());
    ntt(a);
    int iv = inv(a.size());
    for(int &x : a) x = mult(x, iv);
}
void dft(vector<int> &a){
    for(int &x : a) x = normalize(x);
    ntt(a);
}
void idft(vector<int> &a){
    intt(a);
}

```

```

}
vector<int> conv(const vector<int> &a, const vector<int> &b){
    if (a.empty() || b.empty()) return {};
    int s = a.size() + b.size() - 1;
    int n = nxt_pw(s);
    vector<int> L(a), R(b);
    L.resize(n); R.resize(n);
    dft(L); dft(R);
    for (int i = 0; i < n; i++) L[i] = mult(L[i], R[i]);
    idft(L);
    return {L.begin(), L.begin() + s};
}
vector<int> poly_trim(const vector<int> &a, int sz){
    int n = min<int>(a.size(), sz);
    return vector<int>(a.begin(), a.begin() + n);
}
vector<int> poly_add(const vector<int> &a, const vector<int> &b, int sz = -1){
    if (sz == -1) sz = max<int>(a.size(), b.size());
    vector<int> r(sz);
    for (int i = 0; i < sz; i++){
        if (i < (int)a.size()) r[i] = add(r[i], normalize(a[i]));
        if (i < (int)b.size()) r[i] = add(r[i], normalize(b[i]));
    }
    return r;
}
vector<int> poly_sub(const vector<int> &a, const vector<int> &b, int sz = -1){
    if (sz == -1) sz = max<int>(a.size(), b.size());
    vector<int> r(sz);
    for (int i = 0; i < sz; i++){
        if (i < (int)a.size()) r[i] = add(r[i], normalize(a[i]));
        if (i < (int)b.size()) r[i] = sub(r[i], normalize(b[i]));
    }
    return r;
}
vector<int> poly_scalar_mul(vector<int> a, int scalar){
    scalar = normalize(scalar);
    for (auto &x : a) x = mult(x, scalar);
    return a;
}
vector<int> poly_derivative(const vector<int> &a){
    if (a.size() <= 1) return vector<int>{0};
    vector<int> d(a.size() - 1);
    for (int i = 1; i < (int)a.size(); i++) d[i - 1] = mult(normalize(a[i]), i);
    return d;
}
vector<int> poly_integral(const vector<int> &a){
    int n = a.size();
    vector<int> res(n + 1);
    for (int i = 0; i < n; i++) res[i + 1] = mult(normalize(a[i]), inv(i + 1));
    return res;
}
vector<int> poly_inv(const vector<int> &F, int m){
    assert(!F.empty() && normalize(F[0]) != 0);
    vector<int> G(1, inv(normalize(F[0])));
    int cur = 1;
    while (cur < m){
        cur <<= 1;
        vector<int> Fcut = poly_trim(F, cur);
    }
}

```

```

        Fcut.resize(cur);
        vector<int> GG = conv(conv(G, G), Fcut);
        G.resize(cur);
        for (int i = 0; i < cur; i++){
            int u = i < (int)GG.size() ? GG[i] : 0;
            G[i] = sub(add(G[i], G[i]), u);
        }
        G.resize(m);
        return G;
    }

vector<int> poly_ln(const vector<int> &F, int n){
    assert(!F.empty() && normalize(F[0]) == 1);
    vector<int> prod = conv(poly_derivative(F), poly_inv(F, n));
    prod.resize(max(0, n - 1));
    vector<int> ret = poly_integral(prod);
    ret.resize(n);
    return ret;
}

vector<int> poly_exp(const vector<int> &A, int n){
    assert(!A.empty() && normalize(A[0]) == 0);
    vector<int> G(1, 1);
    int cur = 1;
    while (cur < n){
        cur <= 1;
        vector<int> Gcut = poly_trim(G, cur);
        Gcut.resize(cur);
        vector<int> diff = poly_sub(poly_trim(A, cur), poly_ln(Gcut, cur), cur);
        diff[0] = add(diff[0], 1);
        G = conv(G, diff);
        G.resize(cur);
    }
    G.resize(n);
    return G;
}

vector<int> poly_pow(const vector<int> &F, ll k, int n){
    if (n == 0) return {};
    if (k == 0){
        vector<int> r(n);
        r[0] = 1;
        return r;
    }

    int shift = 0;
    while (shift < (int)F.size() && normalize(F[shift]) == 0) shift++;
    if (shift == (int)F.size()) return vector<int>(n, 0);
    if (1LL * shift * k >= n) return vector<int>(n, 0);
    int lead = normalize(F[shift]), ilead = inv(lead);
    vector<int> G;
    for (int i = shift; i < (int)F.size(); i++) G.push_back(mult(normalize(F[i]), ilead));
    G.resize(n);
    vector<int> lnG = poly_ln(G, n);
    for (auto &x : lnG) x = mult(x, normalize(k));
    vector<int> H = poly_exp(lnG, n);
    int coef = fp(lead, k);
    for (auto &x : H) x = mult(x, coef);
    vector<int> res(n);
    for (int i = 0; i + shift * k < n && i < (int)H.size(); i++){
        res[i + shift * k] = H[i];
    }
}

```

```

        return res;
    }

vector<int> poly_pow2(vector<int> poly, ll p, int limit = 1e5 + 5){
    vector<int> ans{1};
    while (p){
        if (p & 1) ans = conv(ans, poly);
        poly = conv(poly, poly);
        ans.resize(limit);
        poly.resize(limit);
        p >>= 1;
    }
    return ans;
}

int mod_sqrt(int a){
    a = normalize(a);
    if (a == 0) return 0;
    if (MOD == 2) return a;
    if (fp(a, (MOD - 1) / 2) != 1) return -1;
    if (MOD % 4 == 3) return fp(a, (MOD + 1) / 4);
    int q = MOD - 1, m = 0;
    while ((q & 1) == 0){
        q >>= 1;
        m++;
    }
    int z = 2;
    while (fp(z, (MOD - 1) / 2) != MOD - 1) z++;
    int c = fp(z, q), x = fp(a, (q + 1) / 2), t = fp(a, q);
    while (t != 1){
        int i = 1, tt = mult(t, t);
        while (tt != 1){
            tt = mult(tt, tt);
            i++;
            if (i == m) return -1;
        }
        int b = fp(c, 1LL << (m - i - 1));
        x = mult(x, b);
        c = mult(b, b);
        t = mult(t, c);
        m = i;
    }
    return min(x, MOD - x);
}

vector<int> poly_sqrt(const vector<int> &F, int n){
    if (F.empty() || n == 0) return {};
    int shift = 0;
    while (shift < (int)F.size() && normalize(F[shift]) == 0) shift++;
    if (shift == (int)F.size()) return vector<int>(n, 0);
    if (shift & 1) return {};
    int c0 = mod_sqrt(F[shift]);
    if (c0 == -1) return {};
    int target = n - shift / 2;
    vector<int> shifted_F(F.begin() + shift, F.end());
    vector<int> G(1, c0);
    int inv2 = inv(2);
    int cur = 1;
    while (cur < target){
        cur <= 1;
        vector<int> Fcut = poly_trim(shifted_F, cur);
        Fcut.resize(cur);
    }
}

```

```

vector<int> term = conv(Fcut, poly_inv(G, cur));
term.resize(cur);
G.resize(cur);
for (int i = 0; i < cur; i++) G[i] = mult(add(G[i], term[i]), inv2);
}
vector<int> res(n);
for (int i = 0; i < target && i + shift / 2 < n; i++){
    res[i + shift / 2] = G[i];
}
return res;
}

```

NTT_CRT.cpp

Use: conv_crt(a,b,mod) multiplies under arbitrary target_mod (or exact CRT value if mod=0); conv3 returns residues under three NTT primes.

Care: conv_crt with target_mod=0 returns exact reconstructed values while they fit signed 64-bit.

Complexity: Runs three NTT convolutions, so it is about three times normal NTT cost.

When: Use it for arbitrary-mod convolution or exact integer convolution beyond single-mod precision.

```

constexpr int MOD1 = 998244353;
constexpr int MOD2 = 1004535809;
constexpr int MOD3 = 469762049;

template<int MOD, int root>
struct NTT{
    static int normalize(ll x){ x %= MOD; if (x < 0) x += MOD; return x; }
    static int add(int a, int b){ a += b; if (a >= MOD) a -= MOD; return a; }
    static int sub(int a, int b){ a -= b; if (a < 0) a += MOD; return a; }
    static int mult(int a, int b){ return 1LL * a * b % MOD; }
    static int fp(int b, ll p){
        int res = 1;
        while (p){
            if (p & 1) res = mult(res, b);
            b = mult(b, b);
            p >>= 1;
        }
        return res;
    }
    static int inv(int x){ return fp(x, MOD - 2); }
    static int nxt_pw(int n){
        int p = 1;
        while (p < n) p <<= 1;
        return p;
    }
    static vector<int>& root_values(){
        static vector<int> roots{0, 1};
        return roots;
    }
    static void prepare_roots(int n){
        auto &roots = root_values();
        if ((int)roots.size() >= n) return;
        int sz = roots.size();
        roots.resize(n);
        for (int len = sz; len < n; len <<= 1){
            int w = fp(root, (MOD - 1) / (len << 1));
            for (int i = len / 2; i < len; i++){
                roots[2 * i] = roots[i];
                roots[2 * i + 1] = mult(roots[i], w);
            }
        }
    }
};

```

```

    }
}

static void ntt(vector<int> &a){
    int n = a.size();
    if (n <= 1) return;
    assert((n & (n - 1)) == 0);
    static vector<int> revs;
    if ((int)revs.size() != n){
        int lg = __builtin_ctz(n);
        revs.assign(n, 0);
        for (int i = 0; i < n; i++){
            revs[i] = (revs[i >> 1] >> 1) | ((i & 1) << (lg - 1));
        }
    }
    prepare_roots(n);
    auto &roots = root_values();
    for (int i = 0; i < n; i++){
        if (i < revs[i]) swap(a[i], a[revs[i]]);
    }
    for (int len = 1; len < n; len <<= 1){
        for (int i = 0; i < n; i += 2 * len){
            for (int j = 0; j < len; j++){
                int u = a[i + j], v = mult(a[i + j + len], roots[len + j]);
                a[i + j] = add(u, v);
                a[i + j + len] = sub(u, v);
            }
        }
    }
}

static void intt(vector<int> &a){
    if ((int)a.size() <= 1) return;
    reverse(a.begin() + 1, a.end());
    ntt(a);
    int iv = inv(a.size());
    for (int &x : a) x = mult(x, iv);
}

static void dft(vector<int> &a){
    for (int &x : a) x = normalize(x);
    ntt(a);
}

static void idft(vector<int> &a){
    intt(a);
}

static vector<int> conv(const vector<int> &a, const vector<int> &b){
    if (a.empty() || b.empty()) return {};
    int s = a.size() + b.size() - 1;
    int n = nxt_pw(s);
    vector<int> L(a), R(b);
    L.resize(n); R.resize(n);
    dft(L); dft(R);
    for (int i = 0; i < n; i++) L[i] = mult(L[i], R[i]);
    idft(L);
    return {L.begin(), L.begin() + s};
}

};

using NTT1 = NTT<MOD1, 3>;
using NTT2 = NTT<MOD2, 3>;

```

```

using NTT3 = NTT<MOD3, 3>;

ll extGCD(ll a, ll b, ll &x, ll &y){
    if (b == 0){
        x = 1; y = 0;
        return a;
    }
    ll x1, y1;
    ll d = extGCD(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return d;
}

ll inv(ll a, ll m){
    ll x, y;
    extGCD(a, m, x, y);
    return (x % m + m) % m;
}

vector<array<ll, 3>> conv3(const vector<ll> &a, const vector<ll> &b){
    vector<int> a1(a.size()), a2(a.size()), a3(a.size());
    vector<int> b1(b.size()), b2(b.size()), b3(b.size());
    for (int i = 0; i < (int)a.size(); i++){
        a1[i] = NTT1::normalize(a[i]);
        a2[i] = NTT2::normalize(a[i]);
        a3[i] = NTT3::normalize(a[i]);
    }
    for (int i = 0; i < (int)b.size(); i++){
        b1[i] = NTT1::normalize(b[i]);
        b2[i] = NTT2::normalize(b[i]);
        b3[i] = NTT3::normalize(b[i]);
    }
    auto c1 = NTT1::conv(a1, b1);
    auto c2 = NTT2::conv(a2, b2);
    auto c3 = NTT3::conv(a3, b3);
    vector<array<ll, 3>> ret(c1.size());
    for (int i = 0; i < (int)c1.size(); i++){
        ret[i] = {c1[i], c2[i], c3[i]};
    }
    return ret;
}

vector<ll> conv_crt(const vector<ll> &a, const vector<ll> &b, ll target_mod = 0){
    if (a.empty() || b.empty()) return {};
    auto c = conv3(a, b);
    ll m1m2 = (ll)MOD1 * MOD2;
    ll i1 = inv(MOD1, MOD2);
    ll i2 = inv(m1m2 % MOD3, MOD3);
    __int128_t all_mods = (__int128_t)MOD1 * MOD2 * MOD3;
    vector<ll> res(c.size());
    for (int i = 0; i < (int)c.size(); i++){
        ll v1 = c[i][0], v2 = c[i][1], v3 = c[i][2];
        ll x = NTT2::sub((int)v2, (int)v1);
        x = NTT2::mult((int)x, (int)i1);
        ll val12 = v1 + x * MOD1;
        ll y = NTT3::sub((int)v3, (int)(val12 % MOD3));
        y = NTT3::mult((int)y, (int)i2);
        __int128_t ans = val12 + (__int128_t)y * m1m2;
        if (ans > all_mods / 2) ans -= all_mods;
        if (target_mod){
            ans %= target_mod;

```

```

        if (ans < 0) ans += target_mod;
    }
    res[i] = (ll)ans;
}
return res;
}

```

Number_theory.cpp

Use: Use `fp/inv_mod/is_prime/factorize/divisors/phi_single` directly; call `sieve_*` before reading `spf`, `phi`, `mobius`, `divs`, or `pfact` arrays.

Care: Trial factorization is fine for moderate n ; use Pollard Rho for large 64-bit composites.

Complexity: Single-number factorization here is trial division; sieve helpers are $O(n \log \log n)$ or similar.

Pitfall: `inv_mod` asserts `gcd(a, mod) = 1`; use CRT/`exgcd` handling when inverse may not exist.

```

ll fp(ll a, ll b, ll mod){
    a %= mod;
    if (a < 0) a += mod;
    ll res = 1 % mod;
    while (b){
        if (b & 1) res = (__int128)res * a % mod;
        a = (__int128)a * a % mod;
        b >>= 1;
    }
    return res;
}

ll exgcd(ll a, ll b, ll &x, ll &y){
    if (b == 0){
        x = 1;
        y = 0;
        return a;
    }
    ll x1, y1;
    ll g = exgcd(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return g;
}

ll inv_mod(ll a, ll mod){
    ll x, y;
    ll g = exgcd(a, mod, x, y);
    assert(abs(g) == 1);
    x %= mod;
    if (x < 0) x += mod;
    return x;
}

bool is_prime(ll n){
    if (n <= 1) return false;
    if (n <= 3) return true;
    if (n % 2 == 0 || n % 3 == 0) return false;
    for (ll i = 5; i <= n / i; i += 6){
        if (n % i == 0 || n % (i + 2) == 0) return false;
    }
    return true;
}

vector<pair<ll, int>> factorize_pairs(ll n){
    vector<pair<ll, int>> ret;
    for (ll p : {2, 3, 5}){
        if (n % p == 0){

```

```

        int cnt = 0;
        while (n % p == 0){
            n /= p;
            cnt++;
        }
        ret.push_back({p, cnt});
    }
}

static int inc[] = {4, 2, 4, 2, 4, 6, 2, 6};
for (ll p = 7, i = 0; p <= n / p; p += inc[i++ & 7]){
    if (n % p == 0){
        int cnt = 0;
        while (n % p == 0){
            n /= p;
            cnt++;
        }
        ret.push_back({p, cnt});
    }
}
if (n > 1) ret.push_back({n, 1});
return ret;
}

vector<ll> factorize(ll n){
    vector<ll> ret;
    for (auto [p, cnt] : factorize_pairs(n)){
        while (cnt-- > 0) ret.push_back(p);
    }
    return ret;
}

vector<ll> divisors(ll n){
    vector<ll> ret;
    for (ll i = 1; i <= n / i; i++){
        if (n % i == 0){
            ret.push_back(i);
            if (i != n / i) ret.push_back(n / i);
        }
    }
    return ret;
}

ll phi_single(ll n){
    ll ret = n;
    for (auto [p, cnt] : factorize_pairs(n)){
        ret -= ret / p;
    }
    return ret;
}

ll num_divisors(ll n){
    ll ret = 1;
    for (auto [p, cnt] : factorize_pairs(n)){
        ret *= cnt + 1;
    }
    return ret;
}

ll sum_divisors(ll n){
    ll ret = 1;
    for (auto [p, cnt] : factorize_pairs(n)){
        ll cur = 1, pw = 1;
        for (int i = 0; i < cnt; i++){
            pw *= p;

```

```

            cur += pw;
        }
        ret *= cur;
    }
    return ret;
}

vector<int> spf, primes;
void sieve_spf(int n){
    spf.assign(n + 1, 0);
    primes.clear();
    if (n >= 1) spf[1] = 1;
    for (int i = 2; i <= n; i++){
        if (!spf[i]){
            spf[i] = i;
            primes.push_back(i);
        }
        for (int p : primes){
            ll x = 1LL * i * p;
            if (x > n) break;
            spf[x] = p;
            if (p == spf[i]) break;
        }
    }
}

vector<int> phi;
void sieve_phi(int n){
    phi.resize(n + 1);
    iota(phi.begin(), phi.end(), 0);
    for (int i = 2; i <= n; i++){
        if (phi[i] == i){
            for (int j = i; j <= n; j += i){
                phi[j] -= phi[j] / i;
            }
        }
    }
}

vector<int> mobius;
void sieve_mobius(int n){
    mobius.assign(n + 1, 0);
    vector<int> lp(n + 1), ps;
    if (n >= 1) mobius[1] = 1;
    for (int i = 2; i <= n; i++){
        if (!lp[i]){
            lp[i] = i;
            ps.push_back(i);
            mobius[i] = -1;
        }
        for (int p : ps){
            ll x = 1LL * i * p;
            if (x > n) break;
            lp[x] = p;
            if (p == lp[i]){
                mobius[x] = 0;
                break;
            }
            mobius[x] = -mobius[i];
        }
    }
}

```

```

    }
}

vector<vector<int>> divs;
void sieve_divisors(int n){
    divs.assign(n + 1, {});
    for (int i = 1; i <= n; i++){
        for (int j = i; j <= n; j += i){
            divs[j].push_back(i);
        }
    }
}

vector<vector<int>> pfact;
void sieve_prime_factors(int n){
    pfact.assign(n + 1, {});
    for (int i = 2; i <= n; i++){
        if (pfact[i].empty()){
            for (int j = i; j <= n; j += i){
                pfact[j].push_back(i);
            }
        }
    }
}
}

```

Pollard Rho Miller Rabin.cpp

Use: miller_rabin(n) tests 64-bit primality; factorize(n) returns prime factors and factorize_pairs(n) returns {prime, exponent}.

Care: factorize returns unsorted prime factors; sort or call factorize_pairs when you need grouped exponents.

Complexity: Miller-Rabin is fast for 64-bit; Pollard Rho is randomized and usually near $n^{1/4}$.

Pitfall: Because factorization is randomized, retrying can fix rare unlucky splits.

```

using u64 = uint64_t;
using u128 = __uint128_t;

u64 mul_mod(u64 a, u64 b, u64 mod){
    return (u128)a * b % mod;
}

u64 add_mod(u64 a, u64 b, u64 mod){
    a += b;
    if (a >= mod || a < b) a -= mod;
    return a;
}

u64 fp(u64 a, u64 b, u64 mod){
    u64 res = 1 % mod;
    while (b){
        if (b & 1) res = mul_mod(res, a, mod);
        a = mul_mod(a, a, mod);
        b >>= 1;
    }
    return res;
}

bool miller_rabin(u64 n){
    if (n < 2) return false;
    for (u64 p : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37}){
        if (n % p == 0) return n == p;
    }
}

```

```

u64 d = n - 1, s = 0;
while ((d & 1) == 0){
    d >>= 1;
    s++;
}

for (u64 a : {2ULL, 325ULL, 9375ULL, 28178ULL, 450775ULL, 9780504ULL, 1795265022ULL}){
    if (a % n == 0) continue;
    u64 x = fp(a, d, n);
    if (x == 1 || x == n - 1) continue;
    bool comp = true;
    for (u64 r = 1; r < s; r++){
        x = mul_mod(x, x, n);
        if (x == n - 1){
            comp = false;
            break;
        }
    }
    if (comp) return false;
}
return true;
}

mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());
u64 pollard_rho(u64 n){
    if (n % 2 == 0) return 2;
    if (n % 3 == 0) return 3;
    while (true){
        u64 c = uniform_int_distribution<u64>(1, n - 1)(rng);
        u64 y = uniform_int_distribution<u64>(0, n - 1)(rng);
        u64 x = 0, ys = 0, q = 1, g = 1;
        auto f = [&](u64 v){ return add_mod(mul_mod(v, v, n), c, n); };
        for (u64 r = 1; g == 1; r <= 1){
            x = y;
            for (u64 i = 0; i < r; i++) y = f(y);
            for (u64 k = 0; k < r && g == 1; k += 128){
                ys = y;
                for (u64 i = 0; i < min<u64>(128, r - k); i++){
                    y = f(y);
                    u64 dif = x > y ? x - y : y - x;
                    if (dif) q = mul_mod(q, dif, n);
                }
                g = gcd(q, n);
            }
        }
        if (g == n){
            do{
                ys = f(ys);
                u64 dif = x > ys ? x - ys : ys - x;
                g = gcd(dif, n);
            }while (g == 1);
        }
        if (g != n) return g;
    }
}

void factor(u64 n, vector<u64> &ret){
    if (n == 1) return;
    if (miller_rabin(n)){
        ret.push_back(n);
        return;
    }
}

```

```

    u64 d = pollard_rho(n);
    factor(d, ret);
    factor(n / d, ret);
}
vector<u64> factorize(u64 n){
    vector<u64> ret;
    factor(n, ret);
    sort(ret.begin(), ret.end());
    return ret;
}
vector<pair<u64, int>> factorize_pairs(u64 n){
    vector<u64> f = factorize(n);
    vector<pair<u64, int>> ret;
    for (auto p : f){
        if (ret.empty() || ret.back().first != p) ret.push_back({p, 1});
        else ret.back().second++;
    }
    return ret;
}

```

Primitive root.cpp

Use: `primitive_root(p)` returns a primitive root modulo prime p .

Care: The modulus p must be prime; it factors $p-1$ by trial division before testing candidates.

Complexity: Factors $p-1$ by trial division, then tests candidates against each prime factor.

Pitfall: There is no primitive root modulo arbitrary composite moduli in this helper.

```

ll fp(ll a, ll b, ll mod){
    a %= mod;
    ll res = 1 % mod;
    while (b){
        if (b & 1) res = (__int128)res * a % mod;
        a = (__int128)a * a % mod;
        b >>= 1;
    }
    return res;
}
// primitive root modulo prime p
ll primitive_root(ll p){
    if (p == 2) return 1;
    ll phi = p - 1, x = phi;
    vector<ll> fact;
    for (ll i = 2; i <= x / i; i++){
        if (x % i == 0){
            fact.push_back(i);
            while (x % i == 0) x /= i;
        }
    }
    if (x > 1) fact.push_back(x);
    for (ll g = 2; ; g++){
        bool ok = true;
        for (ll q : fact){
            if (fp(g, phi / q, p) == 1){
                ok = false;
                break;
            }
        }
        if (ok) return g;
    }
}

```

```

}

```

rand.cpp

Use: `rnd(l,r)`, `rndll(l,r)`, and `shuffle_vec(v)` give random integers and shuffles from seeded `rng/rng64`.

Care: The seed uses `steady_clock` plus heap address noise; keep one `rng` if you need reproducible local debugging.

Out: `rnd` and `rndll` are inclusive on both l and r .

Pitfall: Do not use this RNG for cryptography; it is for randomized algorithms and testing.

```

mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
mt19937_64 rng64(chrono::steady_clock::now().time_since_epoch().count());

```

```

int rnd(int l, int r){
    return uniform_int_distribution<int>(l, r)(rng);
}
ll rndll(ll l, ll r){
    return uniform_int_distribution<ll>(l, r)(rng64);
}
template<typename T>
void shuffle_vec(vector<T> &v){
    shuffle(v.begin(), v.end(), rng);
}

```

Segmented sieve.cpp

Use: `segmented_sieve(l,r)` returns primes in $[l,r]$, and `segmented_sieve_mask(l,r)` returns primality mask for the interval.

Care: The interval $[l,r]$ is inclusive; memory is $O(r-l+1)$, so keep the segment width reasonable.

Complexity: Runs in about $O((r-l+1) \log \log r + \sqrt{r} \log \log \sqrt{r})$.

Pitfall: When $l \leq 1$ the mask explicitly marks 0 and 1 as non-prime.

```

vector<int> simple_sieve(int n){
    vector<bool> is_prime(n + 1, true);
    vector<int> primes;
    if (n >= 0) is_prime[0] = false;
    if (n >= 1) is_prime[1] = false;
    for (int i = 2; i <= n; i++){
        if (!is_prime[i]) continue;
        primes.push_back(i);
        if (1LL * i * i <= n){
            for (ll j = 1LL * i * i; j <= n; j += i){
                is_prime[j] = false;
            }
        }
    }
    return primes;
}
vector<bool> segmented_sieve_mask(ll l, ll r){
    if (l > r) return {};
    int root = sqrtl(r);
    while (1LL * (root + 1) * (root + 1) <= r) root++;
    while (1LL * root * root > r) root--;
    vector<int> primes = simple_sieve(root);
    vector<bool> is_prime(r - l + 1, true);
    for (ll p : primes){
        ll start = max(p * p, ((l + p - 1) / p) * p);
        for (ll j = start; j <= r; j += p){
            is_prime[j - l] = false;
        }
    }
}

```

```

    }
}
for (ll x = 1; x <= min(r, 1LL); x++){
    is_prime[x - 1] = false;
}
return is_prime;
}
// returns all primes in [l, r]
vector<ll> segmented_sieve(ll l, ll r){
    vector<bool> is_prime = segmented_sieve_mask(l, r);
    vector<ll> primes;
    for (ll i = 0; i < (ll)is_prime.size(); i++){
        if (is_prime[i]) primes.push_back(l + i);
    }
    return primes;
}
}

```

SOS dp.cpp

Use: For subset/superset transforms vector size must be a power of two; divisor/multiple transforms work on index arrays by value.

Apps: zeta/mobius transforms mutate sums, while gcd_convolution(a,b) and lcm_convolution(a,b) return value-indexed convolutions.

Care: Subset transforms index by bitmask size 2^n ; divisor/multiple transforms expect arrays indexed from 1 to max value.

Complexity: Subset/superset transforms are $O(n \log n)$ where n is number of masks.

Pitfall: gcd/lcm convolution arrays are value-indexed, so index 0 is not part of the number-theory domain.

```

bool is_power_two(int n){
    return n > 0 && (n & (n - 1)) == 0;
}
template<typename T>
void subset_zeta(vector<T> &a){
    int n = a.size();
    assert(is_power_two(n));
    for (int bit = 1; bit < n; bit <= 1){
        for (int mask = 0; mask < n; mask++){
            if (mask & bit) a[mask] += a[mask ^ bit];
        }
    }
}
template<typename T>
void subset_mobius(vector<T> &a){
    int n = a.size();
    assert(is_power_two(n));
    for (int bit = 1; bit < n; bit <= 1){
        for (int mask = 0; mask < n; mask++){
            if (mask & bit) a[mask] -= a[mask ^ bit];
        }
    }
}
template<typename T>
void superset_zeta(vector<T> &a){
    int n = a.size();
    assert(is_power_two(n));
    for (int bit = 1; bit < n; bit <= 1){
        for (int mask = 0; mask < n; mask++){
            if (mask & bit) a[mask ^ bit] += a[mask];
        }
    }
}

```

```

    }
}
template<typename T>
void superset_mobius(vector<T> &a){
    int n = a.size();
    assert(is_power_two(n));
    for (int bit = 1; bit < n; bit <= 1){
        for (int mask = 0; mask < n; mask++){
            if (mask & bit) a[mask ^ bit] -= a[mask];
        }
    }
}
vector<int> spf, primes;
void sieve_primes(int n){
    spf.assign(n + 1, 0);
    primes.clear();
    if (n >= 1) spf[1] = 1;
    for (int i = 2; i <= n; i++){
        if (!spf[i]){
            spf[i] = i;
            primes.push_back(i);
        }
        for (int p : primes){
            ll x = 1LL * i * p;
            if (x > n) break;
            spf[x] = p;
            if (p == spf[i]) break;
        }
    }
}
void ensure_primes(int n){
    if ((int)spf.size() <= n) sieve_primes(n);
}
template<typename T>
void divisor_zeta(vector<T> &a){
    int n = (int)a.size() - 1;
    ensure_primes(n);
    for (int p : primes){
        if (p > n) break;
        for (int i = 1; i * p <= n; i++){
            a[i * p] += a[i];
        }
    }
}
template<typename T>
void divisor_mobius(vector<T> &a){
    int n = (int)a.size() - 1;
    ensure_primes(n);
    for (int p : primes){
        if (p > n) break;
        for (int i = n / p; i >= 1; i--){
            a[i * p] -= a[i];
        }
    }
}
template<typename T>
void multiple_zeta(vector<T> &a){
}

```

```

    int n = (int)a.size() - 1;
    ensure_primes(n);
    for (int p : primes){
        if (p > n) break;
        for (int i = n / p; i >= 1; i--){
            a[i] += a[i * p];
        }
    }
}

template<typename T>
void multiple_mobius(vector<T> &a){
    int n = (int)a.size() - 1;
    ensure_primes(n);
    for (int p : primes){
        if (p > n) break;
        for (int i = 1; i * p <= n; i++){
            a[i] -= a[i * p];
        }
    }
}

template<typename T>
vector<T> gcd_convolution(vector<T> a, vector<T> b){
    assert(a.size() == b.size());
    multiple_zeta(a);
    multiple_zeta(b);
    for (int i = 0; i < (int)a.size(); i++) a[i] *= b[i];
    multiple_mobius(a);
    return a;
}

template<typename T>
vector<T> lcm_convolution(vector<T> a, vector<T> b){
    assert(a.size() == b.size());
    divisor_zeta(a);
    divisor_zeta(b);
    for (int i = 0; i < (int)a.size(); i++) a[i] *= b[i];
    divisor_mobius(a);
    return a;
}

```

xor basis.cpp

Use: XorBasis<LG> xb; insert(x), can(x), max_xor(x), min_xor(x), kth(k), merge(other), and rank()/size() for linear basis queries.

Care: Call normalize() before kth/count_le if you inserted new vectors; LG must cover the highest bit used.

Complexity: insert/can/max/min are $O(LG)$; kth and count_le need normalized basis.

Pitfall: kth is over representable xor values; account for zero and duplicate subset counts separately.

```

template<int LG = 60>
struct XorBasis{
    array<ll, LG> basis{};
    vector<ll> rref;
    ll cnt = 0;
    int sz = 0;

    void clear(){
        basis.fill(0);
        rref.clear();
        cnt = 0;
        sz = 0;
    }
}

```

```

}

bool insert(ll x, bool count = true){
    if (count) cnt++;
    for (int i = LG - 1; i >= 0; i--){
        if (!(x >> i & 1)) continue;
        if (!basis[i]){
            basis[i] = x;
            sz++;
            rref.clear();
            return true;
        }
        x ^= basis[i];
    }
    return false;
}

void merge(const XorBasis &o){
    cnt += o.cnt;
    for (int i = 0; i < LG; i++){
        if (o.basis[i]) insert(o.basis[i], false);
    }
}

void anding(ll val){
    vector<ll> v;
    for (int i = 0; i < LG; i++){
        if (basis[i] & val) v.push_back(basis[i] & val);
    }
    basis.fill(0);
    rref.clear();
    sz = 0;
    for (ll x : v) insert(x, false);
}

int rank() const {
    return sz;
}

ll count_distinct() const {
    return sz >= 63 ? LLONG_MAX : (1LL << sz);
}

ll count_zero_subsets() const {
    return cnt - sz >= 63 ? LLONG_MAX : (1LL << (cnt - sz));
}

bool can(ll x) const {
    for (int i = LG - 1; i >= 0; i--){
        if (x >> i & 1){
            if (!basis[i]) return false;
            x ^= basis[i];
        }
    }
    return true;
}

ll max_xor(ll x = 0) const {
    for (int i = LG - 1; i >= 0; i--){
        if (basis[i] && (x ^ basis[i]) > x) x ^= basis[i];
    }
    return x;
}

ll min_xor(ll x) const {
    for (int i = LG - 1; i >= 0; i--){
        if (basis[i] && (x ^ basis[i]) < x) x ^= basis[i];
    }
}

```

```

    return x;
}
ll min_xor() const {
    if (sz == 0) return 0;
    XorBasis tmp = *this;
    tmp.normalize();
    ll ans = LLONG_MAX;
    for (ll x : tmp.rref){
        ans = min(ans, x);
    }
    return ans;
}
void normalize(){
    rref.clear();
    for (int i = 0; i < LG; i++){
        if (!basis[i]) continue;
        ll x = basis[i];
        for (int j = i - 1; j >= 0; j--){
            if (x >> j & 1) x ^= basis[j];
        }
        rref.push_back(x);
    }
    sort(rref.rbegin(), rref.rend());
}
ll kth(ll k){
    if (rref.empty() && sz) normalize();
    if (sz >= 63 || k >= (1LL << sz)) return -1;
    ll ans = 0;
    for (int i = 0; i < sz; i++){
        if (k >> (sz - i - 1) & 1) ans ^= rref[i];
    }
    return ans;
}
ll count_le(ll val){
    if (val < 0) return 0;
    if (rref.empty() && sz) normalize();
    if (LG < 63 && val >= (1LL << LG)) return sz >= 63 ? LLONG_MAX : (1LL << sz);
    ll ans = 0, mask = 0;
    int j = 0;
    for (int b = LG - 1; b >= 0; b--){
        bool piv = j < sz && (63 - __builtin_clzll(rref[j])) == b;
        int bit = (val >> b) & 1;
        if (piv){
            if (bit){
                if (sz - 1 - j >= 63) return LLONG_MAX;
                ans += 1LL << (sz - 1 - j);
                mask ^= rref[j];
            }
            j++;
        }else{
            int cur = (mask >> b) & 1;
            if (cur < bit){
                if (sz - j >= 63) return LLONG_MAX;
                return ans + (1LL << (sz - j));
            }
            if (cur > bit) return ans;
        }
    }
    return ans + 1;
}

```

```

    }
};

```

5 strings

Aho.cpp

Use: AC ac; add_pattern(p) returns id, build(), then search(text) returns start positions for each pattern id.

Care: Alphabet is lowercase a-z; build() must be called after all patterns and before search/advance use.

Complexity: build is $O(\text{total pattern length} * \text{alphabet})$, search is $O(\text{text length} + \text{reported matches})$.

Pitfall: The while over out_link reports all suffix patterns; many matches can make output large.

```
struct AC{
    int N, P;
    static const int A = 26;
    vector<array<int, A>> next;
    vector<int> link, out_link, pat_len;
    vector<vector<int>> out, rev;
    AC(): N(0), P(0) {node();}
    int node() {
        next.push_back(array<int, A>{0});
        rev.push_back({});
        link.emplace_back(0);
        out_link.emplace_back(0);
        out.emplace_back();
        return N++;
    }
    inline int get (char c){ return c - 'a'; }
    int add_pattern(const string & T){
        int u = 0;
        for(auto c : T){
            if(!next[u][get(c)]) next[u][get(c)] = node();
            u = next[u][get(c)];
        }
        out[u].push_back(P);
        pat_len.push_back(T.length());
        return P++;
    }
    void build(){
        queue<int> q; q.push(0);
        while(!q.empty()){
            int u = q.front(); q.pop();
            for(int c = 0; c < A; c++){
                int v = next[u][c];
                if(!v) next[u][c] = next[link[u]][c];
                else{
                    link[v] = u ? next[link[u]][c] : 0;
                    rev[link[v]].push_back(v);
                    out_link[v] = out[link[v]].empty() ? out_link[link[v]] : link[v];
                    q.push(v);
                }
            }
        }
    }
    int advance(int u, char c){
        return next[u][get(c)];
    }
    vector<vector<int>> search(const string & s){
        int n = (int)s.length(), m = (int)pat_len.size();
        int u = 0;
        vector<vector<int>> ret(m);
        for(int i = 0; i < n; i++){
            char c = s[i];
```

```
            u = advance(u, c);
            int x = u;
            while(x != 0){
                for(int v : out[x]) ret[v].push_back(i - pat_len[v] + 1);
                x = out_link[x];
            }
        }
        return ret;
    }
};
```

Bitset string matching.cpp

Use: BitsetMatcher bm(text); occurrences(p) returns 0-indexed starts, count(p) counts matches, and set_char(i,c) updates text.

Care: N is the maximum text length and SIG=256 handles bytes; memory is SIG*N bits, so reduce SIG for small alphabets.

Complexity: Each query costs $O(|\text{pattern}| * N / \text{word_size})$; listing occurrences also costs output scan time.

When: Best for many exact pattern queries on one mostly static text; use KMP/Z if there is only one pattern.

Pitfall: match_bits(p)[i] means p starts at i; only inspect i with $i+|p| \leq \text{text length}$.

```
const int N = 3e5 + 5;
const int SIG = 256;

// Many pattern queries on one text:  $O(|p| * N / \text{word\_size})$ , memory  $O(\text{SIG} * N / \text{word\_size})$ .
// match_bits(p)[i] = 1 iff p occurs starting at 0-indexed position i.
struct BitsetMatcher{
    string s;
    vector<bitset<N>> at;

    BitsetMatcher() : at(SIG) {}
    BitsetMatcher(const string &s) : at(SIG) {
        build(s);
    }
    void build(const string &s){
        s = s;
        assert((int)s.size() < N);
        for (auto &b : at) b.reset();
        for (int i = 0; i < (int)s.size(); i++){
            at[(unsigned char)s[i]].set(i);
        }
    }
    void set_char(int i, char c){
        assert(0 <= i && i < (int)s.size());
        if (s[i] == c) return;
        at[(unsigned char)s[i]].reset(i);
        s[i] = c;
        at[(unsigned char)s[i]].set(i);
    }
    bitset<N> match_bits(const string &p) const{
        int n = s.size(), m = p.size();
        bitset<N> ret;
        if (m > n) return ret;
        if (m == 0){
            for (int i = 0; i <= n; i++) ret.set(i);
            return ret;
        }
        ret.set();
        for (int i = 0; i < m; i++){
```

```

        ret &= at[(unsigned char)p[i]] >> i;
    }
    return ret;
}
int count(const string &p) const{
    return (int)match_bits(p).count();
}
bool exists(const string &p) const{
    return match_bits(p).any();
}
vector<int> occurrences(const string &p) const{
    int n = s.size(), m = p.size();
    vector<int> ret;
    if (m > n) return ret;
    bitset<N> good = match_bits(p);
    for (int i = 0; i + m <= n; i++){
        if (good[i]) ret.push_back(i);
    }
    return ret;
}
};

```

dynamic_aho.cpp

Use: dynamic_AC ac; insert(pattern), then cnt(text) returns total occurrences of all inserted patterns in text.

Care: Good for online insertions without deletions; cnt(text) sums matches across the logarithmic collection of automata.

Complexity: Each insert rebuilds $O(\log \text{number_of_patterns})$ automata by binary-counter merging.

Pitfall: No deletion support; rebuild from scratch if patterns need to be removed.

```

struct AC{
    int N, P;
    static const int A = 26;
    vector<array<int, A>> next;
    vector<int> link, out_link, cnt;
    vector<vector<int>> out, rev;
    AC(): N(0), P(0) {node();}
    int node() {
        next.push_back(array<int, A>{0});
        rev.push_back({});
        cnt.emplace_back(0);
        link.emplace_back(0);
        out_link.emplace_back(0);
        out.emplace_back();
        return N++;
    }
    inline int get (char c){ return c - 'a'; }
    int add_pattern(const string & T){
        int u = 0;
        for(auto c : T){
            if(!next[u][get(c)]) next[u][get(c)] = node();
            u = next[u][get(c)];
        }
        out[u].push_back(P);
        cnt[u]++;
        return P++;
    }
    void build(){
        queue<int> q; q.push(0);

```

```

        while(!q.empty()){
            int u = q.front(); q.pop();
            for(int c = 0; c < A; c++){
                int v = next[u][c];
                if(!v) next[u][c] = next[link[u]][c];
                else{
                    link[v] = u ? next[link[u]][c] : 0;
                    rev[link[v]].push_back(v);
                    cnt[v] += cnt[link[v]];
                    out_link[v] = out[link[v]].empty() ? out_link[link[v]] : link[v];
                    q.push(v);
                }
            }
        }
    }
    int advance(int u, char c){
        return next[u][get(c)];
    }
    vector<vector<int>> search(const string & s, vector<int>& pat){
        int n = (int)s.length(), m = (int)pat.size();
        int u = 0;
        vector<vector<int>> ret(m);
        for(int i = 0; i < n; i++){
            char c = s[i];
            u = advance(u, c);
            int x = u;
            while(x != 0){
                for(int v : out[x]) ret[v].push_back(i - pat[v] + 1);
                x = out_link[x];
            }
        }
        return ret;
    }
};

```

```

struct dynamic_AC{
    int lg;
    vector<vector<string>> a;
    vector<AC> aho;
    dynamic_AC(int LOG = 20){
        lg = LOG;
        a = vector<vector<string>>(lg);
        aho = vector<AC>(lg);
    }

    void insert(const string & s){
        vector<string> have = {s};
        for(int i = 0; i < lg; i++){
            if(i == lg - 1){
                lg++;
                a.push_back({});
                aho.push_back(AC());
            }
            if(a[i].empty()){
                swap(have, a[i]);
                for(auto & v : a[i]) aho[i].add_pattern(v);
                aho[i].build();
                break;
            }else{

```

```

        have.insert(end(have), begin(a[i]), end(a[i]));
        vector<string>().swap(a[i]);
        aho[i] = AC();
    }
}

ll cnt(const string & s){
    ll answer = 0;
    for(int i = 0; i < lg; i++){
        int state = 0;
        for(auto & v : s){
            state = aho[i].advance(state, v);
            answer += aho[i].cnt[state];
        }
    }
    return answer;
}
};

```

Generalized suffix automaton.cpp

Use: GSAM g(total_len, m); g.init(total_len,m), g.build(strings); lowercase strings and m <= 63 for mask-based methods.

Apps: get_lcs(K) LCS length for first K strings; distinct_substrings() count; match(P) total occurrences; get_kth(K) kth distinct substring.

Apps: shortest_distinguishing() gives min unique length.

Apps: get_all_occurrences(P) gives {string_id,start}; string_specific_occ(P,i) counts in strings[i].

Care: Call build_tree() through build(); mask-based LCS/distinguishing routines need at most 63 strings.

Complexity: Build is linear in total length; occurrence-heavy routines may traverse suffix-link subtrees.

Pitfall: Masks use ll, so K beyond 63 strings needs a different bitset representation.

```

struct GSAM{
    static const int A = 26;
    struct node{
        int len, link;
        bool is_cloned;
        int nxt[A];
        node(){
            memset(nxt, -1, sizeof nxt);
            len = 0;
            link = -1;
            is_cloned = false;
        }
    };
    int sz;
    vector<node> t;
    vector<int> cnt;
    vector<vector<int>> inv, g;
    vector<ll> freq;
    GSAM(){}
    GSAM(int n, int m){
        init(n, m);
    }
    void init(int n, int m){
        t.assign(2 * n + 5, node());
        cnt.assign(2 * n + 5, 0);
        freq.assign(2 * n + 5, -1);
        g.assign(2 * n + 5, {});
    }
};

```

```

        inv.assign(m, {});
        sz = 1;
    }
    int get(char c){
        return c - 'a';
    }
    int extend(char c, int last){
        int c_idx = get(c);
        if (t[last].nxt[c_idx] != -1){
            int q = t[last].nxt[c_idx];
            if (t[last].len + 1 == t[q].len) return q;
            int clone = sz++;
            t[clone] = t[q];
            t[clone].len = t[last].len + 1;
            t[clone].is_cloned = true;
            int p = last;
            while (p != -1 && t[p].nxt[c_idx] == q){
                t[p].nxt[c_idx] = clone;
                p = t[p].link;
            }
            t[q].link = clone;
            return clone;
        }
        int cur = sz++;
        t[cur].len = t[last].len + 1;
        int p = last;
        while (p != -1 && t[p].nxt[c_idx] == -1){
            t[p].nxt[c_idx] = cur;
            p = t[p].link;
        }
        if (p == -1){
            t[cur].link = 0;
        }else{
            int q = t[p].nxt[c_idx];
            if (t[p].len + 1 == t[q].len){
                t[cur].link = q;
            }else{
                int clone = sz++;
                t[clone] = t[q];
                t[clone].len = t[p].len + 1;
                while (p != -1 && t[p].nxt[c_idx] == q){
                    t[p].nxt[c_idx] = clone;
                    p = t[p].link;
                }
                t[q].link = t[cur].link = clone;
                t[clone].is_cloned = true;
            }
        }
        return cur;
    }
    void build_tree(){
        for (int i = 1; i < sz; i++){
            g[t[i].link].push_back(i);
        }
    }
    void build(const vector<string> &strings){
        for (int i = 0; i < (int)strings.size(); i++){
            int lst = 0;
            inv[i].resize(strings[i].size());

```

```

        for (int j = 0; j < (int)strings[i].size(); j++){
            lst = extend(strings[i][j], lst);
            cnt[lst]++;
            inv[i][j] = lst;
        }
    }
    build_tree();
}
ll occ(int i){
    if (~freq[i]) return freq[i];
    ll ret = cnt[i];
    for (auto x : g[i]){
        ret += occ(x);
    }
    return freq[i] = ret;
}
void getMask(int u, vector<ll> &mask){
    for (int v : g[u]){
        getMask(v, mask);
        mask[u] |= mask[v];
    }
}
int get_lcs(int K){
    vector<ll> mask(sz, 0);
    for (int i = 0; i < (int)inv.size(); i++){
        for (int j = 0; j < (int)inv[i].size(); j++){
            mask[inv[i][j]] |= (1LL << i);
        }
    }
    getMask(0, mask);
    int max_len = 0;
    ll target = (1LL << K) - 1;
    for (int i = 1; i < sz; i++){
        if ((mask[i] & target) == target){
            max_len = max(max_len, t[i].len);
        }
    }
    return max_len;
}
ll distinct_substrings(){
    ll total = 0;
    for (int i = 1; i < sz; i++){
        total += t[i].len - t[t[i].link].len;
    }
    return total;
}
ll match(const string &P){
    int u = 0;
    for (char c : P){
        if (t[u].nxt[get(c)] == -1) return 0;
        u = t[u].nxt[get(c)];
    }
    return occ(u);
}
ll calc(int u, vector<ll> &dp){
    if (dp[u] != -1) return dp[u];
    dp[u] = 1;
    for (int c = 0; c < A; c++){
        if (t[u].nxt[c] != -1){

```

```

            dp[u] += calc(t[u].nxt[c], dp);
        }
    }
    return dp[u];
}
string get_kth(ll K){
    vector<ll> dp(sz, -1);
    calc(0, dp);
    string res = "";
    int u = 0;
    while (K > 0){
        for (int c = 0; c < A; c++){
            if (t[u].nxt[c] != -1){
                int v = t[u].nxt[c];
                if (dp[v] >= K){
                    res += (char)('a' + c);
                    u = v;
                    K--;
                    break;
                }else{
                    K -= dp[v];
                }
            }
        }
    }
    return res;
}
int shortest_distinguishing(){
    vector<ll> mask(sz, 0);
    for (int i = 0; i < (int)inv.size(); i++){
        for (int j = 0; j < (int)inv[i].size(); j++){
            mask[inv[i][j]] |= (1LL << i);
        }
    }
    getMask(0, mask);
    int min_len = 1e9;
    for (int i = 1; i < sz; i++){
        if (__builtin_popcountll(mask[i]) == 1){
            min_len = min(min_len, t[t[i].link].len + 1);
        }
    }
    return min_len == 1e9 ? -1 : min_len;
}
vector<pair<int, int>> get_all_occurrences(const string &P){
    int u = 0;
    for (char c : P){
        if (t[u].nxt[get(c)] == -1) return {};
        u = t[u].nxt[get(c)];
    }
    vector<vector<pair<int, int>>> state_endpoints(sz);
    for (int i = 0; i < (int)inv.size(); i++){
        for (int j = 0; j < (int)inv[i].size(); j++){
            state_endpoints[inv[i][j]].push_back({i, j});
        }
    }
    vector<pair<int, int>> res;
    collect(u, (int)P.length(), state_endpoints, res);
    return res;
}

```

```

void collect(int u, int len, vector<vector<pair<int, int>>> &ends, vector<pair<int,
↳ int>> &res){
    for (auto ep : ends[u]){
        res.push_back({ep.first, ep.second - len + 1});
    }
    for (int v : g[u]){
        collect(v, len, ends, res);
    }
}

ll dfsSp(int u, vector<int> &cnt2, vector<ll> &freq2){
    if (freq2[u] != -1) return freq2[u];
    ll ret = cnt2[u];
    for (int v : g[u]){
        ret += dfsSp(v, cnt2, freq2);
    }
    return freq2[u] = ret;
}

ll string_specific_occ(const string &P, int str_idx){
    int u = 0;
    for (char c : P){
        if (t[u].nxt[get(c)] == -1) return 0;
        u = t[u].nxt[get(c)];
    }
    vector<int> cnt2(sz, 0);
    for (int j = 0; j < (int)inv[str_idx].size(); j++){
        cnt2[inv[str_idx][j]]++;
    }
    vector<ll> freq2(sz, -1);
    return dfsSp(u, cnt2, freq2);
}
};

```

KMP.cpp

Use: KMP(s) builds pi; find_occurrences(pattern,text) returns all 0-indexed starts; automaton(pattern) builds transitions.

Care: find_occurrences concatenates pattern + '#' + text, so choose a separator not present if alphabet is not lowercase.

Complexity: KMP and find_occurrences are $O(n+m)$; automaton is $O(\text{pattern_length} * 26)$.

Pitfall: The automaton assumes lowercase a-z transitions.

```

vector<int> KMP(const string &s){
    int n = s.length();
    vector<int> pi(n);
    for (int i = 1; i < n; i++){
        int j = pi[i - 1];
        while (j > 0 && s[i] != s[j]){
            j = pi[j - 1];
        }
        if (s[i] == s[j]) j++;
        pi[i] = j;
    }
    return pi;
}

vector<int> find_occurrences(const string &pat, const string &text){
    string s = pat + '#' + text;
    vector<int> pi = KMP(s), ret;
    int m = pat.size();
    for (int i = m + 1; i < (int)s.size(); i++){

```

```

        if (pi[i] == m) ret.push_back(i - 2 * m);
    }
    return ret;
}

vector<vector<int>> automaton(string s){ // aut[s.length() + 1][26]
    s += '#';
    int n = s.length();
    vector<int> pi = KMP(s);
    vector<vector<int>> aut(n, vector<int>(26));
    for (int i = 0; i < n; i++){
        for (int j = 0; j < 26; j++){
            if (i > 0 && 'a' + j != s[i]){
                aut[i][j] = aut[pi[i - 1]][j];
            }else{
                aut[i][j] = i + ('a' + j == s[i]);
            }
        }
    }
    return aut;
}

```

manacher.cpp

Use: manacher m(s); odd/even radii, ispal(l,r), longest_palindrome(), count_palindromes(), starts, and ends are built.

Out: starts[i] and ends[i] count palindromes starting/ending at i after build(), useful for palindrome DP counts.

Complexity: build is $O(n)$; palindrome checks and longest radii queries are $O(1)$.

Pitfall: ispal(l,r) expects inclusive 0-indexed bounds.

```

struct manacher{
    vector<int> p, odd, even;
    vector<ll> starts, ends;
    manacher(){
        build(S);
    }
    void run_manacher(const string &s){
        int n = s.length();
        p.assign(n, 1);
        int l = 1, r = 1;
        for (int i = 1; i < n; i++){
            p[i] = i < r ? min(r - i, p[l + r - i]) : 1;
            while (i - p[i] >= 0 && i + p[i] < n && s[i + p[i]] == s[i - p[i]]){
                p[i]++;
            }
            if (r < i + p[i]){
                r = i + p[i];
                l = i - p[i];
            }
        }
    }
    void build(const string &s){
        string t = "#";
        t.reserve(s.length() * 2 + 1);
        for (char c : s){
            t += c;
            t += '#';
        }
        run_manacher(t);
    }

```

```

    build_radix(s.length());
    build_counts(s.length());
}

void build_radix(int n){
    odd.assign(n, 0);
    even.assign(n, 0);
    for(int i = 0; i < n; i++){
        odd[i] = get_longest(i, false) / 2 + 1;
        even[i] = get_longest(i, true) / 2;
    }
}

int get_longest(int cen, bool even){
    return p[2 * cen + 1 + even] - 1;
}

bool ispal(int l, int r){
    return (r - l + 1) <= get_longest((l + r) >> 1, l % 2 != r % 2);
}

pair<int, int> longest_palindrome(){
    int best = 0, pos = 0;
    for (int i = 0; i < (int)p.size(); i++){
        if (p[i] - 1 > best){
            best = p[i] - 1;
            pos = i;
        }
    }
    int l = (pos - best) / 2;
    return {l, l + best - 1};
}

ll count_palindromes(){
    ll ret = 0;
    for (int x : p){
        ret += x / 2;
    }
    return ret;
}

void build_counts(int n){
    starts.assign(n + 1, 0);
    ends.assign(n + 1, 0);
    for (int i = 0; i < (int)p.size(); i++){
        int len = p[i] - 1;
        if (!len) continue;
        int l = (i - len + 1) / 2;
        int r = (i - 1) / 2;
        starts[l]++;
        starts[r + 1]--;
        int el = i - r - 1;
        int er = i - l - 1;
        ends[el]++;
        ends[er + 1]--;
    }
    for (int i = 1; i < n; i++){
        starts[i] += starts[i - 1];
        ends[i] += ends[i - 1];
    }
    starts.pop_back();
    ends.pop_back();
}
};

```

palindromic tree.cpp

Use: PalindromicTree pt; build(s) or init()+extend(c); rollback() removes the last added char.

Apps: cnt_dis() distinct count; pal_cnt() total palindrome substrings; call propagate_occurrences() before final get_all(); get_longest() returns longest palindrome; get_all() returns {palindrome, occurrences}.

Care: rollback() undoes the last extend; call propagate_occurrences() once after final build before occurrence reports.

Complexity: Each extend is amortized $O(1)$ for fixed alphabet; build is $O(n)$.

Pitfall: Alphabet is lowercase a-z; change A/get mapping before using other characters.

```

struct PalindromicTree{
    static const int A = 26;
    struct node{
        int nxt[A], len, st, en, link, cnt, oc, smart_link[A];
        node(){
            memset(nxt, 0, sizeof nxt);
            for (int i = 0; i < A; i++) smart_link[i] = 1;
            oc = cnt = 0;
        }
    };
    string s;
    vector<node> t;
    vector<array<int, 4>> changes;
    int sz, last;
    vector<int> pref;
    ll tot_pals;
    vector<ll> tot_pals_hist;
    PalindromicTree(){
        init();
    }
    void init(){
        s = "";
        t.assign(3, node());
        sz = 2;
        last = 2;
        t[1].len = -1;
        t[1].link = 1;
        t[2].len = 0;
        t[2].link = 1;
        changes.clear();
        pref.clear();
        tot_pals = 0;
        tot_pals_hist.clear();
    }
    void build(const string &str){
        init();
        for (char c : str){
            extend(c);
        }
    }
    int extend(char c){
        tot_pals_hist.push_back(tot_pals);
        int cur = last, pos = s.size();
        pref.push_back(0);
        if (pos) pref[pos] = pref[pos - 1];
        s += c;
        int ch = c - 'a';
        if (pos - t[cur].len - 1 < 0 || s[pos - t[cur].len - 1] != c){
            cur = t[cur].smart_link[ch];

```

```

}
if (t[cur].nxt[ch]){
    changes.push_back({last, t[cur].nxt[ch], -1, -1});
    last = t[cur].nxt[ch];
    t[last].oc++;
    tot_pals += t[last].cnt;
    return 0;
}
changes.push_back({last, -1, cur, ch});
sz++;
last = sz;
t.emplace_back();
t[sz].oc = 1;
t[sz].len = t[cur].len + 2;
pref[pos] = max(pref[pos], t[sz].len);
t[cur].nxt[ch] = sz;
t[sz].en = pos;
t[sz].st = pos - t[sz].len + 1;
if (t[sz].len == 1){
    t[sz].link = 2;
    t[sz].cnt = 1;
    t[sz].smart_link[ch] = 2;
    tot_pals += t[sz].cnt;
    return 1;
}
t[sz].link = t[t[cur].smart_link[ch]].nxt[ch];
for (int i = 0; i < A; i++){
    int x = t[sz].link;
    if (pos - t[x].len >= 0 && s[pos - t[x].len] - 'a' == i){
        t[sz].smart_link[i] = x;
    }else{
        t[sz].smart_link[i] = t[x].smart_link[i];
    }
}
t[sz].cnt = 1 + t[t[sz].link].cnt;
tot_pals += t[sz].cnt;
return 1;
}
}

void rollback(){
    if (s.empty()) return;
    s.pop_back();
    pref.pop_back();
    auto x = changes.back();
    int prvlst = x[0], oc = x[1], c = x[2], ch = x[3];
    changes.pop_back();
    last = prvlst;
    if (oc == -1){
        t[c].nxt[ch] = 0;
        t.pop_back();
        sz--;
    }else{
        t[oc].oc--;
    }
    tot_pals = tot_pals_hist.back();
    tot_pals_hist.pop_back();
}

int cnt_dis() const{ return sz - 2; }
ll pal_cnt() const{ return tot_pals; }
void propagate_occurrences(){

```

```

    for (int i = sz; i >= 3; i--){
        t[t[i].link].oc += t[i].oc;
    }
}

string get_longest() const{
    int max_len = 0, best_st = -1;
    for (int i = 3; i <= sz; i++){
        if (t[i].len > max_len){
            max_len = t[i].len;
            best_st = t[i].st;
        }
    }
    if (best_st == -1) return "";
    return s.substr(best_st, max_len);
}

vector<pair<string, int>> get_all() const{
    vector<pair<string, int>> res;
    for (int i = 3; i <= sz; i++){
        res.push_back({s.substr(t[i].st, t[i].len), t[i].oc});
    }
    return res;
}
};

```

String_Hashing.cpp

Use: Call `init_hash()` once before Hashing; Hashing `h(s)`, `get(l,r)` returns forward hash, `inv(l,r)` reverse hash, and `ispal(l,r)` checks palindrome.

Care: Call `init_hash()` once; `get(l,r)`, `inv(l,r)`, and `ispal(l,r)` are inclusive and 0-indexed.

Complexity: After `init_hash`, build is $O(n)$, push/pop are $O(1)$, and substring hashes are $O(1)$.

Pitfall: N controls maximum precomputed length; increase it before hashing longer strings.

```

using u64 = uint64_t;

const int N = 2e5 + 5;
const u64 md = (1ULL << 61) - 1;
mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count() ^
    (uintptr_t)make_unique<char>().get());
const u64 B = (md >> 2) + rng() % (md >> 1);
inline u64 add(u64 a, u64 b){
    return a + b >= md ? a + b - md : a + b;
}
inline u64 mul(u64 a, u64 b){
    __uint128_t l = (__uint128_t)a * b;
    u64 ret = (l & md) + (l >> 61);
    return ret >= md ? ret - md : ret;
}
u64 binpow(u64 a, u64 b){
    u64 ret = 1;
    while(b){
        if(b & 1) ret = mul(ret, a);
        a = mul(a, a);
        b >>= 1;
    }
    return ret;
}

const u64 IB = binpow(B, md - 2);
vector<u64> PB(N), PIB(N);
void init_hash(){

```

```

PB[0] = PIB[0] = 1;
for (int i = 1; i < N; i++){
    PB[i] = mul(PB[i - 1], B);
    PIB[i] = mul(PIB[i - 1], IB);
}
}
struct Hash : pair<u64, int>{
    Hash(){}
    Hash(const pair<u64, int> &x) : pair(x){}
    Hash(int x) : pair(x, 1){}
    Hash(const string &x){
        for (char c : x) *this = *this + c;
    }
    friend Hash operator+(const Hash &f, const Hash &o){
        return pair{add(mul(f.first, PB[o.second]), o.first), f.second + o.second};
    }
};
struct Hashing{
    vector<Hash> p, rev;
    Hashing(){}
    Hashing(const string &t){
        build(t);
    }
    void build(const string &t){
        clear();
        for (char c : t) push_back(c);
    }
    void push_back(char c){
        if (p.empty()){
            p.push_back(Hash(c));
            rev.push_back(Hash(c));
        }else{
            p.push_back(p.back() + Hash(c));
            rev.push_back(Hash(c) + rev.back());
        }
    }
    void pop_back(){
        assert(!p.empty());
        p.pop_back();
        rev.pop_back();
    }
    Hash get(int l, int r){
        if (l > r) return {};
        if (!l) return p[r];
        return pair{add(p[r].first, md - mul(p[l - 1].first, PB[r - l + 1]), r - l + 1};
    }
    Hash inv(int l, int r){
        if (l > r) return {};
        if (!l) return rev[r];
        u64 val = add(rev[r].first, md - rev[l - 1].first);
        val = mul(val, PIB[l]);
        return pair{val, r - l + 1};
    }
    bool ispal(int l, int r){
        return get(l, r) == inv(l, r);
    }
    int size() const{
        return p.size();
    }
}

```

```

void clear(){
    p.clear();
    rev.clear();
}
};

```

Suffixautomaton.cpp

Use: SAM sam; sam.build(s); methods assume lowercase letters and 0-indexed positions.

Apps: distinct_substrings() count; get_kth(K) kth distinct substring; get_kth_all(K) kth substring counting occurrences; match(P) occurrence count; first_occurrence(P) first start or -1; all_occurrences(P) sorted starts; lcs(T) longest common substring length.

Care: build() also builds suffix-link tree data needed by occurrence methods; alphabet is lowercase a-z.

Complexity: Build is O(n); many query methods are O(length of query) plus output size.

Pitfall: get_kth uses lexicographic lowercase order and assumes K is within the number of available substrings.

```

struct SAM{
    static const int A = 26;
    struct node{
        int len, link, firstpos, lstpos;
        bool is_cloned;
        int nxt[A];
        node(){
            memset(nxt, -1, sizeof nxt);
            len = 0;
            link = -1;
            firstpos = 0;
            lstpos = 0;
            is_cloned = false;
        }
    };
    int sz, last;
    vector<node> t;
    vector<int> terminal, inv;
    vector<ll> freq;
    vector<vector<int>> g;
    SAM(){}
    SAM(int n){
        init(n);
    }
    void init(int n){
        t.assign(2 * n + 5, node());
        terminal.assign(2 * n + 5, 0);
        inv.assign(n + 5, 0);
        freq.assign(2 * n + 5, -1);
        g.assign(2 * n + 5, {});
        sz = 1;
        last = 0;
    }
    int get(char c){
        return c - 'a';
    }
    void extend(char c){
        int cur = sz++;
        t[cur].len = t[last].len + 1;
        t[cur].firstpos = t[cur].lstpos = t[cur].len - 1;
        int p = last;
        while (p != -1 && t[p].nxt[get(c)] == -1){
            t[p].nxt[get(c)] = cur;

```

```

        p = t[p].link;
    }
    if (p == -1){
        t[cur].link = 0;
    }else{
        int q = t[p].nxt[get(c)];
        if (t[p].len + 1 == t[q].len){
            t[cur].link = q;
        }else{
            int clone = sz++;
            t[clone] = t[q];
            t[clone].len = t[p].len + 1;
            while (p != -1 && t[p].nxt[get(c)] == q){
                t[p].nxt[get(c)] = clone;
                p = t[p].link;
            }
            t[q].link = t[cur].link = clone;
            t[clone].is_cloned = true;
        }
    }
    last = cur;
}

void build_tree(){
    for (int i = 1; i < sz; i++){
        g[t[i].link].push_back(i);
    }
}

void build(const string &s){
    init(s.size());
    int j = 0;
    for (auto x : s){
        extend(x);
        terminal[last] = j + 1;
        inv[j++] = last;
    }
    build_tree();
}

ll occ(int i){
    if (~freq[i]) return freq[i];
    ll ret = terminal[i] > 0;
    for (auto x : g[i]){
        ret += occ(x);
    }
    return freq[i] = ret;
}

ll distinct_substrings(){
    ll total = 0;
    for (int i = 1; i < sz; i++){
        total += t[i].len - t[t[i].link].len;
    }
    return total;
}

ll calc(int u, vector<ll> &dp){
    if (dp[u] != -1) return dp[u];
    dp[u] = 1;
    for (int c = 0; c < A; c++){
        if (t[u].nxt[c] != -1){
            dp[u] += calc(t[u].nxt[c], dp);
        }
    }
}

```

```

    }
    return dp[u];
}

string get_kth(ll K){
    vector<ll> dp(sz, -1);
    calc(0, dp);
    if (K <= 0 || K >= dp[0]) return "";
    string res = "";
    int u = 0;
    while (K > 0){
        for (int c = 0; c < A; c++){
            if (t[u].nxt[c] == -1) continue;
            int v = t[u].nxt[c];
            if (dp[v] >= K){
                res += (char)('a' + c);
                u = v;
                K--;
                break;
            }
            K -= dp[v];
        }
    }
    return res;
}

ll calcAll(int u, vector<ll> &dp){
    if (dp[u] != -1) return dp[u];
    dp[u] = occ(u);
    for (int c = 0; c < A; c++){
        if (t[u].nxt[c] != -1){
            dp[u] += calcAll(t[u].nxt[c], dp);
        }
    }
    return dp[u];
}

string get_kth_all(ll K){
    vector<ll> dp(sz, -1);
    calcAll(0, dp);
    if (K <= 0 || K > dp[0]) return "";
    string res = "";
    int u = 0;
    while (K > 0){
        bool moved = false;
        for (int c = 0; c < A; c++){
            if (t[u].nxt[c] == -1) continue;
            int v = t[u].nxt[c];
            if (dp[v] >= K){
                res += (char)('a' + c);
                u = v;
                K -= occ(v);
                moved = true;
                break;
            }
        }
        K -= dp[v];
    }
    if (!moved || K <= 0) break;
}
return res;
}

ll match(const string &P){

```

```

int u = 0;
for (char c : P){
    if (t[u].nxt[get(c)] == -1) return 0;
    u = t[u].nxt[get(c)];
}
return occ(u);
}

int first_occurrence(const string &P){
    int u = 0;
    for (char c : P){
        if (t[u].nxt[get(c)] == -1) return -1;
        u = t[u].nxt[get(c)];
    }
    return t[u].firstpos - (int)P.length() + 1;
}

void getOcc(int u, int len, vector<int> &res){
    if (terminal[u] > 0){
        res.push_back(terminal[u] - len);
    }
    for (int v : g[u]){
        getOcc(v, len, res);
    }
}

vector<int> all_occurrences(const string &P){
    int u = 0;
    for (char c : P){
        if (t[u].nxt[get(c)] == -1) return {};
        u = t[u].nxt[get(c)];
    }
    vector<int> res;
    getOcc(u, (int)P.length(), res);
    sort(res.begin(), res.end());
    return res;
}

int lcs(const string &T){
    int u = 0, l = 0, max_len = 0;
    for (char c : T){
        int idx = get(c);
        while (u && t[u].nxt[idx] == -1){
            u = t[u].link;
            l = t[u].len;
        }
        if (t[u].nxt[idx] != -1){
            u = t[u].nxt[idx];
            l++;
        }
        max_len = max(max_len, l);
    }
    return max_len;
}
};

```

Trie_strings.cpp

Use: Trie tr; insert(s), erase(s), count(s), count_prefix(s), and contains(s) for lowercase strings.

Care: erase(s) returns false if the word is absent; counts keep duplicate insertions, not just boolean presence.

Complexity: insert, erase, count, and prefix queries are $O(\text{length of string})$.

Pitfall: The implementation assumes lowercase letters; adapt child size and get mapping for other alphabets.

```

struct Trie{
    vector<array<int, 26>> nxt;
    vector<int> cnt, ending;
    Trie(){
        init();
    }
    void init(){
        nxt.clear();
        cnt.clear();
        ending.clear();
        node();
    }
    int node(){
        nxt.push_back(array<int, 26>{0});
        cnt.push_back(0);
        ending.push_back(0);
        return (int)nxt.size() - 1;
    }
    void insert(const string &s){
        int u = 0;
        cnt[u]++;
        for (char c : s){
            int idx = c - 'a';
            if (!nxt[u][idx]) nxt[u][idx] = node();
            u = nxt[u][idx];
            cnt[u]++;
        }
        ending[u]++;
    }
    bool erase(const string &s){
        if (!count(s)) return false;
        int u = 0;
        cnt[u]--;
        for (char c : s){
            int idx = c - 'a';
            u = nxt[u][idx];
            cnt[u]--;
        }
        ending[u]--;
        return true;
    }
    int count(const string &s){
        int u = 0;
        for (char c : s){
            int idx = c - 'a';
            if (!nxt[u][idx]) return 0;
            u = nxt[u][idx];
        }
        return ending[u];
    }
    int count_prefix(const string &s){
        int u = 0;
        for (char c : s){
            int idx = c - 'a';
            if (!nxt[u][idx]) return 0;
            u = nxt[u][idx];
        }
        return cnt[u];
    }
};

```

```

    }
    bool contains(const string &s){
        return count(s) > 0;
    }
    void clear(){
        init();
    }
}
};

```

Z_algorithm.cpp

Use: zfunction(s) returns z[i], the longest prefix of s matching s starting at i.

Care: For pattern matching, run zfunction(pattern + '#' + text) and test z positions at least pattern length.

Complexity: zfunction runs in $O(n)$.

Out: z[0] is left as 0 in this implementation, so do not expect z[0]=n.

```

vector<int> zfunction(const string& s){
    int n = s.size();
    vector<int> z(n);
    for(int i = 1, l = 0, r = 0; i < n; i++){
        if(i < r) z[i] = min(r - i, z[i - l]);
        while(i + z[i] < n && s[z[i]] == s[i + z[i]]) z[i]++;
        if(i + z[i] > r){
            r = i + z[i];
            l = i;
        }
    }
    return z;
}

```

6 DP Optimizations and Misc

divide and conquer dp.cpp

Use: Set n and k , edit $\text{add}(i)/\text{rem}(i)$ so $\text{cur}=\text{cost}(L,R)$, then $\text{compute}()$; $\text{answer}()$ returns $\text{dp}[n]$ after k layers.

Care: This maintains cost by moving $[L,R]$; add/rem may use current L/R exactly like Mo-style pointer movement.

Complexity: Each layer is $O(n \log n)$ divide-and-conquer states plus the amortized moving-window cost changes.

Recurrence: The template computes $\text{dp}[\text{mid}][\text{layer}] = \min \text{dp}[\text{split}-1][\text{layer}-1] + \text{cost}(\text{split}, \text{mid})$.

Pitfall: The optimization is only valid when optimal split points are monotone; otherwise use plain DP or another optimization.

```
const int N = 1e5 + 5;
const ll inf = (1LL << 62);

// dp[i][layer] = minimum cost to cover prefix [1..i] using this many layers.
// Transition shape: dp[mid][it] = min_{split <= mid} dp[split-1][it ^ 1] + cost(split,
//   mid).
// Valid only when the optimal split point is monotone as mid increases.
int n, k, it = 1, L = 1, R = 0;
ll dp[N][2], cur = 0;

void add(int i){
    (void)i;
    // i has just entered the current interval [L, R].
    // Update cur so it equals cost(L, R).
}

void rem(int i){
    (void)i;
    // i is still inside [L, R] and is about to leave.
    // Update cur so it equals the interval after removing i.
}

void reset_window(){
    L = 1;
    R = 0;
    cur = 0;
}

void move(int l, int r){
    while(R < r) add(++R);
    while(L > l) add(--L);
    while(R > r) rem(R--);
    while(L < l) rem(L++);
}

void compute_layer(){
    vector<array<int, 4>> q;
    q.reserve(2 * n);
    q.push_back({1, n, 1, n}); // {answer range l,r, allowed opt range lx,rx}
    while(!q.empty()){
        auto [l, r, lx, rx] = q.back();
        q.pop_back();
        int mid = (l + r) >> 1, opt = lx;
        dp[mid][it] = inf;
        for(int split = lx; split <= min(rx, mid); split++){
            move(split, mid);
            ll val = dp[split-1][it ^ 1] + cur;
            if(val < dp[mid][it]){
                dp[mid][it] = val;
                opt = split;
            }
        }
    }
}
```

```
        if(l <= mid-1) q.push_back({l, mid-1, lx, opt});
        if(mid+1 <= r) q.push_back({mid+1, r, opt, rx});
    }
}

void compute(){
    for(int i = 1; i <= n; i++){
        dp[i][0] = dp[i][1] = inf;
    }
    dp[0][0] = dp[0][1] = 0;
    it = 1;
    for(int layer = 1; layer <= k; layer++){
        for(int i = 1; i <= n; i++) dp[i][it] = inf;
        reset_window();
        compute_layer();
        it ^= 1;
    }
}

ll answer(){
    return dp[n][it ^ 1];
}
```

Custom hashing.cpp

Use: Use `unordered_map<K,V,custom_hash>` for integers or `unordered_map<pair<ll,ll>,V,pair_hash>` for pairs to resist hacks.

Care: Use these hashes only in unordered containers; ordered maps/sets do not need them and cannot use them directly.

When: Use it for adversarial `unordered_map/unordered_set` tests with integer or pair keys.

Pitfall: Custom hash changes bucket distribution, not equality; pair equality still uses normal pair comparison.

```
struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }

    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM =
            chrono::steady_clock::now().time_since_epoch().count();
        return (size_t) splitmix64(x + FIXED_RANDOM);
    }
};

struct pair_hash {
    size_t operator()(const pair<ll,ll>& p) const noexcept {
        uint64_t a = (uint64_t) p.first;
        uint64_t b = (uint64_t) p.second;
        uint64_t z = a + 0x9e3779b97f4a7c15ULL + (b << 6) + (b >> 2);
        return (size_t) custom_hash::splitmix64(z);
    }
};
```

Josephus.cpp

Use: `josephus_kth_removed(n,k,m)` returns the m -th removed 1-indexed label; survivor/order helpers cover common variants.

Care: $k=1$ is handled directly; `josephus_order(n,k)` uses Fenwick and returns the whole deletion order in 1-indexed labels.

Complexity: `kth_removed/survivor` use jump compression and are fast for large n ; `josephus_order` is $O(n \log n)$.

Out: survivor1 returns the 1-indexed survivor; survivor0_slow is the classic $O(n)$ 0-indexed recurrence.

```
// Returns the m-th removed person, 1-indexed labels,  $1 \leq m \leq n$ .  
// People are removed by counting k alive people starting from person 1.
```

```
11 josephus_kth_removed(11 n, 11 k, 11 m){  
    assert(n >= 1 && k >= 1 && 1 <= m && m <= n);  
    m = n - m;  
    if (k == 1) return n - m;  
    11 alive = m;  
    while (alive < n){  
        11 jump = (11)((__int128)alive - m + k - 2) / (k - 1));  
        if ((__int128)alive + jump > n) jump = n - alive;  
        else if (jump == 0) jump = 1;  
        alive += jump;  
        m = (11)((m + (__int128)jump * k) % alive);  
    }  
    return m + 1;  
}  
11 josephus_survivor1(11 n, 11 k){  
    return josephus_kth_removed(n, k, n);  
}  
11 josephus_survivor0_slow(11 n, 11 k){  
    assert(n >= 1 && k >= 1);  
    11 ans = 0;  
    for (11 i = 1; i <= n; i++){  
        ans = (ans + k) % i;  
    }  
    return ans;  
}
```

```
struct FenwickJosephus{  
    int n;  
    vector<int> bit;  
    FenwickJosephus(){}  
    FenwickJosephus(int _n){ init(_n); }  
    void init(int _n){  
        n = _n;  
        bit.assign(n + 1, 0);  
        for (int i = 1; i <= n; i++) add(i, 1);  
    }  
    void add(int i, int v){  
        for (; i <= n; i += i & -i) bit[i] += v;  
    }  
    int kth(int k){  
        int pos = 0;  
        for (int pw = 1 << __lg(n); pw; pw >>= 1){  
            int nxt = pos + pw;  
            if (nxt <= n && bit[nxt] < k){  
                pos = nxt;  
                k -= bit[nxt];  
            }  
        }  
        return pos + 1;  
    }  
};  
vector<int> josephus_order(int n, 11 k){  
    assert(n >= 1 && k >= 1);  
    FenwickJosephus fw(n);  
    vector<int> order;
```

```
    order.reserve(n);  
    11 cur = 0;  
    for (int alive = n; alive >= 1; alive--){  
        cur = (cur + k - 1) % alive;  
        int pos = fw.kth((int)cur + 1);  
        order.push_back(pos);  
        fw.add(pos, -1);  
    }  
    return order;  
}
```

Permutation cycles.cpp

Use: Helpers use 0-indexed permutations with $p[i]$ =image of i ; cycle order represents $v \rightarrow \text{next}(v)$.

decompose: perm_decompose(p) gets a permutation p and returns its disjoint cycles in orbit order; $O(n)$.

restore: perm_restore(n,cycles) gets cycle lists and returns p with each element mapped to the next element; $O(\text{total length})$.

power: perm_power_cycles(p,k) gets p and $k \geq 0$, then returns the cycle decomposition of p^k ; $O(n)$.

root: perm_root_cycles(p,k,toggle) gets p and $k > 0$, returns cycles of q with $q^k = p$ or empty if impossible.

swap list: perm_transpositions(cycles) returns swaps (c0,ci) for each cycle, count n-number_of_cycles; $O(\text{total length})$.

Root detail: For a length L cycle in p^k , preimages are formed by merging equal-length cycles according to $\text{gcd}(L,k)$.

Toggle: toggle tries one legal 2^* need merge to change transposition parity when the problem asks for a parity target.

Pitfall: The helpers do not validate that p is a permutation; duplicates or out-of-range images break the orbit assumptions.

```
vector<vector<int>> perm_decompose(const vector<int> &p){  
    int n = p.size();  
    vector<vector<int>> cycles;  
    vector<bool> vis(n);  
    for (int i = 0; i < n; i++){  
        if (vis[i]) continue;  
        vector<int> cycle;  
        int u = i;  
        while (!vis[u]){  
            cycle.push_back(u);  
            vis[u] = true;  
            u = p[u];  
        }  
        cycles.push_back(cycle);  
    }  
    return cycles;  
}  
vector<int> perm_restore(int n, const vector<vector<int>> &cycles){  
    vector<int> p(n);  
    for (const auto &cycle : cycles){  
        int m = cycle.size();  
        for (int i = 0; i < m; i++){  
            p[cycle[i]] = cycle[(i + 1) % m];  
        }  
    }  
    return p;  
}  
vector<vector<int>> perm_power_cycles(const vector<int> &p, 11 k){  
    assert(k >= 0);  
    auto cycles = perm_decompose(p);
```

```

vector<vector<int>>> ans;
for (const auto &cycle : cycles){
    int len = cycle.size();
    int step = k % len;
    int g = gcd(step, len);
    for (int i = 0; i < g; i++){
        vector<int> cur;
        for (int j = i, cnt = 0; cnt < len / g; cnt++, j = (j + step) % len){
            cur.push_back(cycle[j]);
        }
        ans.push_back(cur);
    }
}
return ans;
}

vector<vector<int>>> perm_root_cycles(const vector<int> &p, ll k, bool toggle = false){
    assert(k > 0);
    int n = p.size();
    vector<vector<vector<int>>>> by_len(n + 1);
    for (auto &cycle : perm_decompose(p)){
        by_len[(int)cycle.size()].push_back(cycle);
    }
    vector<vector<int>>> ans;
    for (int len = 1; len <= n; len++){
        if (by_len[len].empty()) continue;
        ll tmp = k, need = 1;
        int g = gcd((ll)len, tmp);
        while (g != 1){
            need *= g;
            tmp /= g;
            g = gcd((ll)len, tmp);
        }
        if ((ll)by_len[len].size() % need != 0) return {};
        int id = 0, block = need;
        if (toggle && tmp % 2 == 0 && (ll)by_len[len].size() >= 2 * need){
            int merged = 2 * block;
            int sz = merged * len;
            vector<int> cycle(sz);
            for (int x = 0; x < merged; x++){
                for (int y = 0; y < len; y++){
                    cycle[(x + 1LL * y * k) % sz] = by_len[len][x][y];
                }
            }
            ans.push_back(cycle);
            id = merged;
            toggle = false;
        }
        for (int i = id; i < (int)by_len[len].size(); i += block){
            int sz = block * len;
            vector<int> cycle(sz);
            for (int x = 0; x < block; x++){
                for (int y = 0; y < len; y++){
                    cycle[(x + 1LL * y * k) % sz] = by_len[len][i + x][y];
                }
            }
            ans.push_back(cycle);
        }
    }
}
return ans;

```

```

}
vector<pair<int, int>> perm_transpositions(const vector<vector<int>>> &cycles){
    vector<pair<int, int>> ans;
    for (const auto &cycle : cycles){
        for (int i = (int)cycle.size() - 1; i > 0; i--){
            ans.push_back({cycle[0], cycle[i]});
        }
    }
    return ans;
}

```